Class 10

Biology

(Solutions)

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1. Life Process- Digestion and Absorption Solutions

LEVEL-I

1. Correct Option: (C) iii & iv

Cuscuta, ticks, and leeches are parasites that feed by living on or inside another organism's body. Cuscuta species are plant parasites. Since they lack chlorophyll, they cannot use photosynthesis to produce their sustenance. To grow, they depend on other plants for nutrients, which weakens the host plant. Most leeches are sanguivorous, that is they feed as blood sucking parasites on preferred hosts.

2. Correct Option: (B) humidity

There are many factors that can alter the rate of photosynthesis. However, one that does not affect this process is Humidity. Studies are indicative of the fact that humidity plays an almost negligible role in the process of photosynthesis.

3. Correct Option: (B) Remove starch from leaves

The process is known as destarching. This is done so because the result of the experiment is not affected.

4. Correct Option: (A) Bile

Bile is a digestive juice that has no enzymes. Bile is an alkaline liquid that helps in emulsification of fat i.e., it helps in breakdown of large fat globules into smaller droplets that increases its surface area and accelerates digestion.

5. Correct Option: (A) The entire body surfaces

In single celled animals like amoeba, the food is taken up by the whole body. The process by which the single celled animals feed is known as Phagocytosis. In this process, the single celled organisms bring the food into the inner part of the cell for digestion.

6. Correct Option: (A) C.B. Van Neil

By studying purple Sulphur bacteria and green Sulphur bacteria he was the first scientist to demonstrate, in 1931, that photosynthesis is a light-dependent redox reaction in which hydrogen from an oxidizable compound reduces carbon dioxide to cellular materials.

7. Correct Option: (C) Starch

In the presence of starch, iodine changes its color from brown to blue-black or purple. So, iodine can be used to test the presence of starch.

8. Correct Option: (C) Joseph priestly

In 1770 he performed a series of experiments that showed the essential role of air in the growth of green plants. Priestley hypothesis: Plants restore to the air whatever breathing animals and burning candles.

9. Correct Option: (B) Jan Ingenhousz

Oxygen is evolved during photosynthesis as demonstrated by Jan Ingenhousz through an experiment on the aquatic plant -Hydrilla. He showed that in bright sunlight, small bubbles formed around the green parts of the plant while in dark, no bubbles were formed. He identified those bubbles as oxygen.

10. Correct Option: (C) Engelman

Engelmann used oxygen loving bacteria to grow on the algae. The bacteria congregating in the region of the red and blue light indicated that there was more oxygen being produced in these areas, indicating there was more photosynthetic activity in these areas.

11. Correct Option: (A) Pelletier & Caventou

Chlorophyll was first extracted by Pelletier and Caventou in 1817. At the time of their work, scientists know that green plants consume carbon dioxide and release oxygen.

12. Correct Option: (B) 40-100

Chloroplasts are present in large numbers in the cells (around 40 – 100) of plant tissue like the stomatal guard cells and ground tissues.

13. Correct Option: (A) Daniel I. Arnon

In the 1950s, Arnon performed research with Mary Belle Allen and F. Robert Whatley on chloroplasts and their role in photosynthesis, identifying a process which they named "photosynthetic phosphorylation."

14. Correct Option: (B) Photolysis

The splitting of water molecules is called photolysis of water. During the light reaction of photosynthesis, the absorption of light by the chlorophyll results in the loss of the electron from it. To fulfill the electron requirement, the water molecule splits and donates the electron to the chlorophyll and at that time oxygen is released.

15. Correct Option: (B) ATP, NADPH

The assimilatory powers produced during light reaction of photosynthesis are ATP and NADPH. These are utilized to form carbohydrates (sugars) during light independent reactions of photosynthesis.

16. Correct Option: (A) Less than even 1/1000 of a second

1/1000th seconds because there is A server-containing system.

17. Correct Option: (C) Blue & Red

Chlorophyll is a photosynthetic pigment. Violet-blue and Orange-red light wavelengths are absorbed by chlorophyll A. The rate of photosynthesis is more in blue and red light.

18. Correct Option: (B) $C_6H_{12}O_6$

The end product of this reaction is that the Co2 is converted into glucose (Carbohydrates form).

19. Correct Option: (A) Amylase

Salivary amylase (Ptyalin) is a glucose-polymer cleavage enzyme that is produced by the salivary glands.

20. Correct Option: (A) Bile juice

Emulsification is done by the action of bile secretes from the liver. Bile contains bile pigment and bile salts. Bile salts break down the fats into smaller globules and result in a milky emulsion. These emulsified fats are again digested by pancreatic lipase into fatty acid and glycerol.

21. Correct Option: (D) Both a & b

Pancreatic juice consists of alkaline (chiefly bicarbonate) fluid and enzymes; 200–800 ml is produced each day. Enzymes, such as trypsin, chymotrypsin, lipase, and amylase, are essential for the digestion of most of the protein, fat, and carbohydrate in the meal.

22. Correct Option: (C) Hydrochloric acid

Gastric juice is a unique combination of hydrochloric acid (HCl), lipase, and pepsin.

23. Correct Option: (D) All the above

We can avoid indigestion by having a simple well-balanced diet, eating food in a leisurely manner, and thoroughly masticating food and taking violent exercise after eating food.

24. Correct Option: (A) Kwashiorkor

Kwashiorkor is a type of malnutrition characterized by severe protein deficiency. It causes fluid retention and a swollen, distended abdomen. Kwashiorkor most commonly affects children, particularly in developing countries with high levels of poverty and food insecurity.

25. Correct Option: (A) Riboflavin (Vitamin B2)

Vitamin B2 deficiency (ariboflavinosis) can cause glossitis, along with angular cheilitis, cheilosis, peripheral neuropathy and other signs and symptoms. The glossitis in vitamin B2 deficiency is described as magenta.

26.Correct Option: (C) Vitamin D

This vitamin should be a part of your diet to help your fracture heal. It helps your blood take in and use calcium and build up the minerals in your bones.

27.Correct Option: (B) Vitamin K

Vitamin K is necessary for blood clotting, and it prevents excessive bleeding. This is because vitamin K helps to produce proteins required for blood coagulation. If a person has a deficiency of vitamin K it increases the time taken by the blood to clot. Severe deficiency of vitamin K causes death due to excessive blood loss in case of a cut or an injury.

28.Correct Option: (B) Folic Acid

Folic acid is an essential nutrient necessary for protein and nucleic acid synthesis (DNA and RNA). The production and regulation of these biological macromolecules are essential for survival and replication of organisms.

29.Correct Option: (C) Vitamin A, D, E, K

Vitamins A, D, E, and K are called fat-soluble vitamins, because they are soluble in organic solvents and are absorbed and transported in a manner similar to that of fats.

30.Correct Option: (C) Ingestion →Digestion → Absorption →Assimilation →Egestion

1) Ingestion - It includes the processes that take place while the food is in the mouth also known as a buccal cavity (such as chewing and grinding) and swallowing of the food - which sends it onward down the digestive tract.

2) Digestion - It is the process of breaking down food into simpler substances that can then be absorbed and assimilated into the tissues of the body.

3) Absorption - The digested food is absorbed into the bodily fluids from the alimentary canal. The process of absorption takes place in the small intestine with the help of villi (they are minute finger-like processes lined against the walls of the small intestine).

4) Assimilation - the process by which chemicals from food are taken into the cells of the body after digestion and absorption. The food absorbed by blood is transported to various parts of the body by the blood vessels.

5) Egestion - the process of removing undigested food material from the body through the anus.

Match the Following

1. Correct Matching is: a - R, b - Q, c - T, d - P, e – S

a) Light reaction – (R) Grana
b) Dark reaction – (Q) Stroma
c) Photolysis – (T) Release of Oxygen
d) Photon – (P) Smallest energy of light
e) Stomata – (S) Exchange of Gases

2. Correct Matching is: a - R, b - Q, c - P, d - S, e - T

a) C. B. Van Neil – (R) Formulated equation of photosynthesis
b) Jan Ingenhousz – (Q) Release of oxygen from Hydrilla plants
c) Engelmann – (P) Detected the point of maximum rate of photosynthesis
d) Pelletier & Caventou – (S) Observed chlorophyll in chloroplast
e) Daniel I. Arnon – (T) Extracted chloroplast

3.Correct Matching is: a - R, b - P, c - S, d - T, e – Q

a) Salivary glands – (R) Ptyalin
b) Stomach – (P) Gastric Juice
c) Liver – (S) Bile Juice
d) Pancreas – (T) Trypsin, Amylase, Lipase
e) Small Intestine – (Q) Intestinal juice

4.Correct Matching is: a - R, b - Q, c - S, d - P, e - T

a) Cyanocobalamin (B12) – (R) Pernicious anemia
b) Ascorbic Acid (C) – (Q) Scurvy
c) PhylloQuinone (K) – (S) Blood Clotting problems
d) Riboflavin (B2) – (P) glossitis
e) Niacin (B3) – (T) Pellagra

LEVEL-I

1. Correct option: (A) Ptyalin

It breaks starch into maltose and dextrin. It will break large, insoluble starch molecules into soluble starches (amylodextrin, erythrodextrin, and achrodextrin) to produce successively smaller starches and ultimately maltose.

2. Correct option: (C) Protein

Trypsin is an enzyme that helps us digest protein. In the small intestine, trypsin breaks down proteins, continuing the process of digestion that began in the stomach. It may also be referred to as a proteolytic enzyme, or proteinase.

3. Correct option: (D) Glucose, Glucose

Maltose, also known as malt, is a disaccharide made up of two alpha D-glucose units. The two units of glucose are linked with an alpha 1,4 glycosidic bond.

4. Correct option: (B) Lipase

Lipases are digestive enzymes that catalyse the hydrolysis of fat (lipid, triglyceride) into free fatty acids (FFA) and monoglycerides, molecules that can be absorbed into the bloodstream.

5. Correct option: (B) Insulin, Glucagon

The production of pancreatic hormones, including insulin, somatostatin, gastrin, and glucagon, play an important role in maintaining sugar and salt balance in our bodies.

6. Correct option: (A) Mouth

Saliva contains enzyme Ptyalin or Salivary amylase which helps digest starch present in food into dimeric maltose which is made up of glucose units.

7. Correct option: (B) Liver

The liver is the largest gland in the human body. It is also the largest (internal)organ in our body and can weigh up to 1.5 kg for a human adult. That is, about 1/50th of the body weight is because of the liver.

8. Correct option: (A) protein into peptones

Trypsin is an enzyme produced by the pancreas which breaks down protein molecules into simple and small protein molecules called proteoses and peptones.

9. Correct option: (D) Milk

Rennin is a proteolytic enzyme found in the gastric juice in the stomach of infants which helps in the digestion of Casein (milk protein). Rennin is also known as Chymosin synthesized by the chief cells.

10. Correct option: (A) Peristalsis

Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract.

11. Correct option: (D) 2123/2123

The dental formula of adult human teeth is 2123/2123. It means that there are 2 incisors, 1 canine, 2 premolars and 3 molars in each half of the upper and lower jaw.

12. Correct option: (C) Epiglottis

Epiglottis is a flap-like structure present in the larynx which covers the respiratory tract when food enters the oesophagus. It is the function of epiglottis to prevent the entry of food into the respiratory tract.

13. Correct option: (C) Enamel

Enamel is the hardest substance of the human body, and it is the outer layer of the crown. Dentine is a layer that lies beneath the enamel of the tooth.

14. Correct option: (C) Mouth

The digestion of starch starts in the mouth and then continues in the intestine. The food in the mouth is chewed and mixed with salivary amylase to help in the easy movement of the food in the esophagus.

15. Correct option: (C) Production of Insulin

Insulin is a hormone, produced by beta cells of islets of Langerhans of pancreas.

16. Correct option: (C) Conversion of Glucose into Glycogen

Glycogenesis, the formation of glycogen, the primary carbohydrate stored in the liver and muscle cells of animals, from glucose. Glycogenesis takes place when blood glucose levels are sufficiently high to allow excess glucose to be stored in liver and muscle cells.

17. Correct option: (D) Sodium glycolate and taurocholate

In humans, taurocholic acid and glycocholic acid (derivatives of cholic acid) and Tauro chenodeoxycholic acid and glycochenodeoxycholic acid (derivatives of chenodeoxycholic acid) are the major bile salts.

LEVEL-II

1. Correct option: Correct option: (B) Stenson's duct

A duct, called Stensen's duct, drains saliva from the parotid gland into the mouth, at the area of the upper cheeks.

2. Correct option: (B) Peptic ulcers

In humans, excessive stimulation of the Vagus nerve disrupts the mechanism of hydrochloric acid secretion. Excessive secretion of gastric acid causes peptic ulcers. So, excessive stimulation of the Vagus nerve in humans may lead to peptic ulcers.

3. Correct option: (D) Bile Salts

Bile is a watery greenish fluid containing bile salts, bile pigments, cholestrol and phospholipid. Blue salts play an important role in digestion of fats. Therefore, in their absence emulsification of fat cannot take place.

4. Correct option: (D) Number and type of teeth in one half of both jaws

It showcases the peculiar pattern of arrangement, the kind of teeth present, and the number of a particular tooth in the mouth, at a given age. The dental formula constitutes the number of teeth in their respective position as they are present in the jaws. It's 2123/2123 per half of the jaw.

5. Correct option: (D) K

It is Vitamin K2, a component of vitamin K, synthesized by bacteria present in the intestine and it helps to treat osteoporosis.

6. Correct option: (C) Premolar and Molar

The premolar and molars are called cheek teeth. These are also called lophodont teeth. Their free ends are flattened and bear transverse ridges to crush and grind the food.

7. Correct option: (C) Night Blindness – Retinol – Carrot and milk

Night blindness (in which it is difficult or impossible to see in relatively low light) is one of the clinical signs of vitamin A deficiency and is common during pregnancy in developing countries. Retinol is the main circulating form of vitamin A in blood and plasma, and is present in carrots, green vegetables, and dairy products.

8. Correct option: (A) HCl

Ptyalin is a form of amylase found in the saliva of humans and some other animals. It converts starch into dextrin and maltose. HCl of the gastric juice inactivates or destroys salivary amylase or ptyalin in the stomach is well known to physiologists and physiological chemists.

9. Correct option: (C) Third molar, two in number

Humans have 12 molars, six in each jaw. Four of those are wisdom teeth, which is also called the third molar, which come in between the ages of 17 to 25.

10. Correct option: (A) Watery stool with blood and mucus

Patients with colitis present with watery diarrhea, abdominal pain, tenesmus, urgency, fever, tiredness, and blood in the stool.

11. Correct option: (C) 2 ATP

In anaerobic respiration, the ATP is only produced from Glycolysis as the NADH produced is used again in fermentative pathways. So only 2 ATPs are produced during Substrate level phosphorylation.

12. Correct option: (B) Ileum

Ileum is the part of the small intestine. Its mucosa is raised into numerous microscopic projections called the villi. The villi contain minute blind ended lymph vessel lacteals, which help in the absorption of fats.

13. Correct option: (B) Cholecystokinin

Cholecystokinin (in fact it is cholecystokinin pancreozymin) is a hormone which is secreted by the small intestine and stimulates pancreas to secrete and release digestive enzymes in pancreatic juice. Secretin causes the release of bicarbonates in the pancreatic juice.

14. Correct option: (A) Thiamine (B1)

Vitamin-B1 (thiamine) deficiency is most observed in alcoholic person. Prolonged deficiency of thiamine may lead to decreased mental function double vision reduced muscular contraction, lacking in decision capability. These all the ill effect results in a disorder known as Wernicke's syndrome.

15. Correct option: (C) Intestine

Duocrinin is a gastrointestinal hormone that is secreted within the intestine and stimulates the secretory activity of the duodenal glands (Brunner glands). Cholecystokinin is synthesized by the cells of the mucosal epithelium of the small intestine and secreted in the duodenum. It causes the release of digestive enzymes and bile from the pancreas and gallbladder, respectively.

16. Correct option: (B) Vitamin K

Hemorrhagic disease is a bleeding problem that occurs in a baby during the first few days of life. Babies are normally born with low levels of vitamin K, an essential factor in blood clotting. A deficiency in vitamin K is the main cause of hemorrhagic disease in newborn babies.

17. Correct option: (D) 38 ATP

In aerobic respiration, the full oxidation of one glucose molecule can theoretically yield 38 ATP molecules. But the net gain is always 36 ATP as 2 ATP has been previously used during glycolysis.

18. Correct option: (B) Milk changing casein into calcium para caseinate at pH 1-3

Rennin is found in gastric juice of infants which causes coagulation of milk and is the main component for the digestive process of infants. In the presence of calcium, this enzyme converts casein of milk into paracasein which is then treated by pepsin. Hence, rennin acts on milk proteins and converts casein into paracasein. Rennin has an optimum pH of 3 for proteolysis.

19. Correct option: (D) Maltose – Maltase

Maltase is an enzyme that hydrolyzes Maltose into glucose monomers in the intestine of humans. Rennin is a protein digesting enzyme found in infants that helps in digestion of milk protein casein. Starch is digested by amylase in the mouth. Whereas steapsins/ lipases are responsible for fat breakdown. Chymotrypsin is also a proteases that digests proteins in peptides and peptones.

20. Correct option: (C) A is true but R is false

Numerous tube-like glands are present in the wall of the small intestine which secrete intestinal juice into the intestinal lumen. This juice contains several enzymes like enterokinase, amino peptidases, peptidases, maltase, lactase, lipase etc. for digesting various types of food. On the other hand, the enzymes RNase and DNase are present in the pancreatic juice.

21. Correct option: (D) A is false but R is true

The large intestine is responsible for the absorption of water and minerals. The mucosa has many goblet cells but does not have any villi.

22. Correct option: (A) Both A and R are true, and R is correct explanation of A

The peristaltic movement of the gut is aided by the activity of muscles. The food material is mixed with the enzymes in the gut when the muscle contracts and relaxes. Different glands in the alimentary canal release these enzymes. This helps proper food digestion in the small intestine.

23. Correct option: (A) Both A and B are true, and R is the correct explanation of A

Carbohydrates are more suitable to produce energy in the body than proteins and fats because carbohydrate molecules contain relatively more oxygen than the others, and consequently, require less molecules of oxygen for their oxidation. Carbohydrates are also stored in the tissue as glycogen for use in the production of energy, when necessary.

24. Correct option: (C) A is true, but R is false

In the small intestine Pancreatic amylase converts starch and dextrose into maltose and small intestine is main site for digestion of carbohydrates.

25. Correct option: (C) A is true, but R is false

About 30% of starch is hydrolyzed in the oral cavity by the action of salivary amylase into a disaccharide- maltose. Sucrase hydrolyses sucrose into glucose and fructose.

2. Life Process - Breathing and Exchanges of Gas Solutions

LEVEL-I

1. Correct option: (D) b and c

The plain dwellers when move to the higher altitudes certain physiological changes occur in order to cope up with decreased oxygen levels. The important changes are increased breathing rate, increased red blood cells productions with a decreased oxygen binding capacity.

2. Correct option: (1A) bicarbonate

Approximately seventy percent of carbon dioxide absorbed by the blood will be transported to the lungs as bicarbonate ions.

3. Correct option: (A) 1000-1200 mL

Total oxygen that can be carried by blood in 1000-1200 ml.

4. Correct option: (A) Both assertions and reason are true, and the reason is the correct explanation of the assertion

5. Correct option: (D) 1%-2%

Approximately 98% of total oxygen transported in the blood is bound to haemoglobin, while only 2% is dissolved directly in plasma.

6. Correct option: (A) HCO_3 bicarbonate ions

About 70% of CO2 is converted into bicarbonate ions and are transported in plasma.

7. Correct option: (C) Nostrils →Pharynx →Larynx →Trachea →Bronchi→Bronchioles →Alveoli

When you inhale through your nose or mouth, air travels down your pharynx (back of your throat), passes through your larynx (voice box) and into your trachea (windpipe). Your trachea is divided into two air passages called bronchial tubes. One bronchial tube leads to your left lung, the other to your right lung.

8. Correct option: (C) Diaphragm and external intercostal muscles contract

During inhalation, the diaphragm contracts and moves downwards. This increases the space in your chest cavity into which the lungs expand. The intercostal muscles between the ribs also help in enlarging the chest cavity. Whereas during exhalation the diaphragm relaxes and moves upwards into the chest cavity.

9. Correct option: (A) CO

Carbon monoxide combines with hemoglobin to form carboxyhemoglobin at any or all of the oxygen-binding sites of hemoglobin, and also acts to increase the stability of the bond between hemoglobin and oxygen, reducing the ability of the hemoglobin molecule to release oxygen bound to other oxygen-binding sites.

10. Correct option: (A) Sodium bicarbonate

Carbon dioxide can be transported through the blood via three methods. It is dissolved directly in the blood, bound to plasma proteins or hemoglobin, or converted into bicarbonate. The majority of carbon dioxide is transported as part of the bicarbonate system.

11. Correct option: (D) 4.5%

Inhaled air has about 0.04% of carbon dioxide. This is also the amount of carbon dioxide present in the atmosphere. The exhaled air contains all the carbon dioxide that is a by-product of the cellular metabolic processes. Hence, the percentage of carbon dioxide in the exhaled air is 4.5%.

12. Correct option: (B) Alveoli

Both the lungs in humans are filled with tiny balloon-like structures called air sacs or alveoli. Their membrane is highly thin and is surrounded by many capillaries. Alveoli are the actual sites where gaseous exchange takes place between the air in the lungs and blood.

13. Correct option: (C) Epiglottis

The epiglottis is an elastic cartilaginous leaf-shaped flap covering the opening of the larynx.

14. Correct option: (B) Thoracic cavity

Your lungs are located in your chest (your thorax). Your thoracic cavity is the name of the space that contains your lungs and other organs. Your lungs rest on a muscle called your diaphragm.

15. Correct option: (A) As a solution, carbamino compounds, bicarbonates

There are three means by which carbon dioxide is transported in the bloodstream from peripheral tissues and back to the lungs: (1) dissolved gas, (2) bicarbonate, and (3) carbaminohemoglobin bound to hemoglobin (and other proteins).

16. Correct option: (C) Tidal Volume

Tidal volume is the volume of air normally inspired or expired in one breath without any effort. It is about 500 ml in the average human adult male.

17. Correct option: (C) Envelops the lungs

A pleura is a serous membrane that folds back on itself to form a two-layered membranous pleural sac. The outer layer is called the parietal pleura and attaches to the chest wall. The inner layer is called the visceral pleura and covers the lungs, blood vessels, nerves, and bronchi.

18. Correct option: (B) More surface area for diffusion of gases.

There are approximately 600 – 800 million alveoli in the two lungs of a human being. The alveoli are sac-like structures that store the air that one breathes in. The alveoli are stretchable.

19. Correct option: (A)a – r; b – s; c – q; d – p

Column – I		**Column – II**
a) Larynx	-	r) voice box
b) Trachea	-	s) Windpipe
c) Alveoli	-	q) air sacs
d) Epiglottis	-	p) Lid of larynx

LEVEL-II

1. Correct option: (A) Sigmoid

The oxygen-hemoglobin dissociation curve is a graphical representation of the relationship between the amount of oxygen bound to hemoglobin and the partial pressure of oxygen in the blood. The curve is sigmoidal, with a steep slope at low partial pressures of oxygen and a more gradual slope at higher partial pressures.

2. Correct option: (D) Hyaline cartilage

Trachea is lined with incomplete rings of hyaline cartilage.

3. Correct option: (C) Reduces oxygen carrying capacity of blood

CO is not added to tobacco but is formed when tobacco is burned incompletely. This happens when there is too little oxygen present to convert all of the carbon in the tobacco into harmless carbon dioxide. Cigarette smoke can contain large quantities of CO. CO has higher affinity than O2 for hemoglobin.

4. Correct option: (B) Cartilaginous rings

The trachea is supported by c-shaped rings of cartilage which hold them in place when there is less air inside it. Due to this, the walls of trachea do not collapse even when there is a little air. So, the correct option is 'cartilaginous rings.

5. Correct option: (B) Jerky incomplete inspiration

A hiccup can be described as a jerky incomplete inspiration. It is an involuntary contraction of the diaphragm that may repeat several times per minute. In medical terms it is known as synchronous diaphragmatic flutter SDF

6. Correct option: (C) CO_2 concentration in artery blood

Thus, the respiratory centres that are present in the brain are stimulated by the carbon dioxide content in the arterial blood.

7. Correct option: (A) Bicarbonate; 70%

During transport, most of CO 2 is passed into erythrocytes. A major part (nearly 70%) forms carbonic acid on reacting with water in erythrocytes and then form bicarbonate.

8. Correct option: (B) Several hundred grams

In the blood of an adult man the total content of haemoglobin is roughly several hundred grams

9. Correct option: (D) 150 mL

Anatomical dead space is the volume of the conducting airways (from the nose, mouth, and trachea to the terminal bronchioles). These conduct gas to the alveoli, but no gas exchange occurs here.

10. Correct option: (A) 16%

Inhaled air contains approx. 21%oxygen, 0.04% carbon dioxide and 79%nitrogen while exhaled air contains approx. 16% oxygen, 4% carbon dioxide and 79% nitrogen.

11. Correct option: (B) Kept floating

A child was killed through asphyxiation. Postmortem confirmed it because a piece of lung put in water kept floating.

12. Correct option: (B) Cigarette smoking

Emphysema is lung condition that causes shortness of breath due to damage of air sacs in the lungs (alveoli). The main cause of emphysema is a long–term exposure to air–born irritants including tobacco smoke, air pollution, dust.

13. Correct option: (C) Residual Volume.

Residual volume is the amount of air that remains in the lungs after forceful expiration. It is about 1,500 ml.

14. Correct option: (C) Directly proportional to the concentration of CO in the medium

The binding of one CO molecule to haemoglobin increases the affinity of the other binding spots for oxygen, leading to a left shift in the dissociation curve. This shift prevents oxygen unloading in peripheral tissue and therefore the oxygen concentration of the tissue is much lower than normal.

15. Correct option: (C) blood buffer

The bicarbonate buffering system of blood is an important buffer system in the acid-base homeostasis of living things, including humans. As a buffer, it tends to maintain a relatively constant plasma pH and counteract any force that would alter. In this system, carbon dioxide CO2 combines with water H2O to form carbonic acid H2CO3, which in turn rapidly dissociates to form hydrogen ions H+ and bicarbonate HCO3 as shown in the reactions below.

$CO_2 + H_2O \rightleftarrows H_2CO_3 \rightleftarrows HCO_3^- + H^+$

The carbon dioxide - carbonic acid equilibrium is catalysed by the enzyme carbonic anhydrase; the carbonic acid - bicarbonate equilibrium is simple proton dissociation/association and needs no catalyst. Any disturbance of the system will be compensated by a shift in the chemical equilibrium.

16. Correct option: (A) As a solution, carbamino compounds, bicarbonates

There are three means by which carbon dioxide is transported in the bloodstream from peripheral tissues and back to the lungs: (1) dissolved gas, (2) bicarbonate, and (3) carbaminohemoglobin bound to hemoglobin (and other proteins).

17. Correct option: (A) To regulate inhalation

The pneumotaxic center, located in the upper pons, sends inhibitory impulses to the inspiratory center, terminating inspiration, and thereby regulating inspiratory volume and respiratory rate. This center likely is involved in the fine-tuning of breathing.

18. Correct option: (D) Chloride shift

Chloride shift (also known as the Hamburger shift or Hamburger phenomenon, named after Hartog Jakob Hamburger) is a process which occurs in a cardiovascular system and refers to the exchange of bicarbonate ($HCO3^-$) and chloride (Cl^-) across the membrane of red blood cells (RBCs).

19. Correct option: (A) Relaxes and arches.

When you breathe out, or exhale, your diaphragm and rib muscles relax, reducing the space in the chest cavity. As the chest cavity gets smaller, your lungs deflate, similar to how air releases from a balloon.

20. Correct option: (C) Diaphragm and external intercostal muscles contract

During inspiration, the dome shaped diaphragm stretches and become flat and increases the size of the thoracic chamber. As a result, the pressure in the thorax is reduced to less than atmospheric pressure. Air, therefore, enters the lungs.

21. Correct option: (A) Chloride shift

The chloride content of RBCs increases when oxygenated blood becomes deoxygenated. This is temed as chloride shift or hamburger shift.

22. Correct option: (D) Common cold by corona virus

SARS Coronavirus 2, an RNA virus causes severe pulmonary distress conditions infecting the mucosal tissues of lungs, resulting in difficulty in breathing and life-threatening conditions. Coronaviruses accounts for approx. 20% of common cold viral infections.

23. Correct option: (C) nostrils-pharynx-larynx-trachea-bronchi-bronchioles-alveoli

When you inhale through your nose or mouth, air travels down your pharynx (back of your throat), passes through your larynx (voice box) and into your trachea (windpipe). Your trachea is divided into two air passages called bronchial tubes. One bronchial tube leads to your left lung, the other to your right lung.

24. Correct option: (D) a-i, b-ii, c-iii, d-iv

Silicosis is an interstitial lung disease caused by breathing in tiny bits of silica, a common mineral found in many types of rock and soil. Over time, exposure to silica particles causes permanent lung scarring, called pulmonary fibrosis.

Emphysema develops over time and involves the gradual damage of lung tissue, specifically the destruction of the alveoli.

During an asthma attack, the smooth muscle layer goes into spasm, narrowing the airway causing difficulty in breathing.

Bronchitis is a condition that develops when the airways in the lungs, called bronchial tubes, become inflamed and cause coughing, often with mucus production.

25. Correct option: (B) a – ii; b – iii; c – iv; d – i

Vital capacity (VC) is the maximum amount of air a person can expel from the lungs after a maximum inhalation. It is equal to the sum of inspiratory reserve volume, tidal volume, and expiratory reserve volume. The total lung capacity (TLC), about 6,000 mL, is the maximum amount of air that can fill the lungs (TLC = TV + IRV + ERV + RV). The maximum volume of air that can be inspired after reaching the end of a normal, quiet expiration. It is the sum of the tidal volume and the inspiratory reserve volume (TV + IRV). Expiratory Capacity (EC) Total volume of air a person can expire after a normal inspiration. It includes tidal volume and expiratory reserve volume (TV+ERV).

3. Life Process – Transportation

Solutions

LEVEL-I

1. Correct option: (A) Fish

A closed-type circulatory system is seen in fish. Blood circulates via a series of vessels in a closed circulatory system without coming into direct touch with human tissues. An open circulatory system, in contrast, allows blood to circulate more freely and directly bathe bodily tissues.

The circulatory systems of snails, cockroaches, and scorpions are all open. As a result, their blood does not benefit from blood arteries and travels freely through their bodily cavities. Although this circulatory system is not as effective as a closed system, it is enough for these tiny creatures.

2. Correct option: (B) Lymphocytes and monocytes

White blood cells known as agranulocytes are distinguished by the lack of granules in their cytoplasm. About 20–30% of all white blood cells are this smallest subset of white blood cells. Both lymphocytes and monocytes are agranulocytes.

The generation of antibodies and immunological memory are two important functions of lymphocytes in the immune response. By phagocytosing pathogens and presenting antigens to other immune cells, monocytes, on the other hand, contribute to the immune system's defense against infections.

Granulocytes, such as eosinophils and basophils, have granules in their cytoplasm. Inflammation and allergic responses are both mediated by granulocytes.

3. Correct option: (B) Myogenic

In vertebrates, the heartbeat is myogenic, which implies that it develops inside the heart muscle itself. Pacemaker cells are specialized cells within the heart that produce electrical impulses that cause the heart muscles to contract, so starting the heartbeat. The sinoatrial node, or SA node, is found in the right atrium of the heart and contains these pacemaker cells. About 70 times every minute, the SA node produces an electrical impulse that causes the heart to beat. As a result of this contraction, blood is pumped into the ventricles, which contract and send blood to the rest of the body.

Although it does not start the pulse, the nervous system can affect it by accelerating or decelerating it.

4. Correct option: (C) Right atrium and right ventricle

In the right atrium and right ventricle of the heart, there is a valve called the tricuspid. It stops the right ventricle's right atrium's blood from returning there. Three leaflets, or flaps of tissue, make up the tricuspid valve, which opens and closes to let blood flow in one direction.

The left atrium and left ventricle are where you'll find the bicuspid valve. Furthermore, it stops blood from returning from the left ventricle to the left atrium. Between the left ventricle and the aorta is where the aortic valve is situated. It stops the left ventricle's leftward return of blood from the aorta.

5. Correct option: (D) Ca

Coagulation, the process by which blood clots, depends heavily on calcium ions. They are necessary for a number of clotting cascade processes, such as the activation of clotting factors and the production of fibrin, which aids in forming a stable blood clot to stop excessive bleeding.

The remaining ions, sodium (Na), potassium (K), and magnesium (Mg), are not directly involved in blood coagulation. However, they can influence the activity of other ions or proteins involved in the clotting cascade, which can have an indirect effect on blood clotting.

6. Correct option: (A) One minute

Cardiac output measures how much blood the heart pumps in a minute. It is computed by multiplying the heart rate (the number of beats per minute) by the stroke volume (the volume of blood pumped by each heartbeat). In order to quantify cardiac output, liters per minute are used. A healthy adult's resting heart rate is around 5 liters per minute. This indicates that the heart circulates the body with around 5 liters of blood each minute.

The other choices are incorrect. The cardiac index refers to the volume of blood that the heart pumps in a second. The term "stroke volume" refers to the volume of blood that the heart pumps during each beat. The minute ventilation refers to the volume of blood the heart pumps out per 100 heartbeats.

7. Correct option: (D) Pulmonary vein

In the pulmonary veins, oxygenated blood is located. These veins return blood that has been filled with oxygen from the lungs to the left atrium of the heart. To provide the body's tissues and organs access to oxygen, this oxygenated blood is subsequently pushed into the systemic circulation.

From the right ventricle of the heart to the lungs, deoxygenated blood travels by the pulmonary artery. Deoxygenated blood from the body enters the right atrium and right ventricle.

8. Correct option: (C) 0.8 seconds

The cardiac cycle in man takes about 0.8 seconds. The heart needs this amount of time to go through one full cycle of contraction and relaxation. Systole and diastole are the two stages of the heart cycle.

The stage of contraction is called systole. The heart muscle contracts and pumps blood out of the heart during systole. The relaxation phase is known as diastole. The heart muscle relaxes and fills with blood during diastole.

The pacemaker cells in the heart control the cardiac cycle. These cells produce electrical impulses that contract and relax the heart muscle. The sinoatrial node (SA node), which is found in the right atrium of the heart, is where the pacemaker cells are situated.

9. Correct option: (B) bilirubin

Blood coagulation is not something that the word "bilirubin" may be used for. Heme from haemoglobin is broken down to produce bilirubin. It is then broken down by the liver and eliminated as bile, giving faces its distinctive brown color. Blood coagulation and clotting are not directly affected by bilirubin.

The blood coagulation process also involves the other words, fibrin, fibrinogen, and calcium. A clot is created when the protein fibrin creates a web that enfolds platelets and blood cells. The starting point for fibrin is fibrinogen. The activation of numerous proteins involved in the coagulation cascade requires calcium.

10. Correct option: (D) All the above

The thymus, tonsils, and lymph nodes are only a few human organs that contain lymphoid tissue. These organs are crucial parts of the immune system that aid in the development of immune cells as well as the filtration and surveillance of the lymphatic fluid for infections and foreign objects.

T lymphocytes (T cells), which are essential for adaptive immunity, are matured in the thymus. By capturing and recognizing possible infections coming via the mouth and throat, tonsils act as defense systems. Filtering lymphatic fluid is crucial because lymph nodes contain immune cells that hunt out and destroy dangerous chemicals.

The body's defense against infections and disorders is ensured by the lymphoid tissues as a whole, which support immunological responses.

11. Correct option: (A) Both 'A' and 'R' are true, and 'R' is correct explanation of 'A'.

The absence of clotting proteins in blood serum accounts for the fact that it does not have the ability to clot. When blood clots, clotting proteins like fibrinogen cause the blood to solidify and create a mass. Serum is the liquid fraction of blood that is left over after the clotting process has eliminated the clotting factors; as a result, serum lacks the ability to clot. Therefore, the reason (R) correctly explains why the assertion (A) is true.

12. Correct option: (B) Calcium ions and platelet factor

Blood coagulation depends on calcium ions and platelet factors. The activation of clotting factors and the production of fibrin are two phases in the clotting cascade activated by calcium ions. When platelets come into touch with injured tissue, platelet factors are released from the platelets. They cause the coagulation cascade's other proteins to bind with one another, causing a blood clot to develop.

Anticoagulants, such as heparin, oxalates, and citrates, stop blood from clotting. They are thus not necessary for the coagulation of blood.

13. Correct option: (C) Blood group is designated based on the presence of antibodies in the blood plasma.

Instead, then considering the presence of antibodies in the blood plasma, blood groups are really determined by the presence or absence of certain antigens on the surface of red blood cells. Option 3 is wrong because it depicts the methodology used to determine blood groups incorrectly. The presence or absence of antigens like A, B, and Rh(D) on the surface of red blood cells determines a person's blood type, and the compatibility of organ transplants and blood transfusions depends on these antigen-antibody interactions.

The other claims are accurate. Antigens on the surface of red blood cells, not antibodies in blood plasma, are used to determine blood group. A person with type A blood can get blood from a type B donor. A person with type A blood cannot, however, donate blood to someone with type B blood. A person with blood type AB may accept blood from any blood type.

14. Correct option: (B) 1 – c, 2 – a, 3 – d, 4 –b

1)Lymphatic system - c. To drain back the tissue fluid to the circulatory system

The lymphatic system helps to maintain fluid balance by collecting and returning extra tissue fluid (lymph) to the circulatory system as well as moving immune cells.

2) Pulmonary vein - a. Carries oxygenated blood

The pulmonary vein facilitates the delivery of oxygen from the lungs to the rest of the body by transporting oxygenated blood from the lungs to the left atrium of the heart.

3) Thrombocytes - d. Coagulation of blood

Platelets called thrombocytes are necessary for the coagulation of blood. When there is an injury, they are crucial in the formation of blood clots that stop excessive bleeding.

4) Lymphocytes - b. Immune Response

Platelets called thrombocytes are necessary for the coagulation of blood. When there is an injury, they are crucial in the formation of blood clots that stop excessive bleeding.

15. Correct option: (A) Action mentioned in Statement 1 is dependent on action mentioned in Statement 2

According to Statement 1, the atria get blood from every area of the body, which then flows to the ventricles. As noted in Statement 2, the coordinated action potential

produced at the Sino-Atrial (SA) node aids in the flow of blood from the atria to the ventricles. The electrical impulse that causes atrial contraction and assists in blood pumping into the ventricles is started by the SA node. As a result, Statement 1 and Statement 2 both depend on the electrical signal produced at the SA node in order for blood to flow from the atria to the ventricles.

LEVEL-II

1. Correct option: (A) Left atrium → left ventricle → aorta

Left atrium: The oxygenated blood from the lungs enters the left atrium. **Left ventricle**: The left atrium pumps the blood into the left ventricle. **Aorta**: The left ventricle pumps the blood into the aorta, the main artery of the body. **Arteries**: The aorta branches into smaller arteries, which carry the blood to all parts of the body, including the hand.

The other answer choices are incorrect because they do not show the correct path of the blood flow.

2. Correct option: (B) Only (ii) is false

Statement (i) is true: About 6-8 percent of the blood plasma is made up of proteins. Albumin, which serves to regulate blood pressure and transport nutrients and hormones throughout the body, is the protein that makes up the majority of plasma.

Statement (ii) is false: Mineral content is not very high in plasma. In actuality, it only has approximately 0.9% mineral content. In plasma, sodium, chloride, potassium, and bicarbonate are the minerals that are most prevalent.

Statement (iii) is true: Serum is plasma minus the clotting components. Proteins called clotting factors aid in blood clotting. To avoid the blood clotting in the test tube when blood is collected for testing, the clotting factors are eliminated. Serum is the resultant liquid.

Statement (iv) is true: Since these substances are constantly moving throughout the body, they are also present in plasma. These materials are delivered by the plasma to the cells, where they are used for cellular development, energy production, and repair.

3. Correct option: (B) Only ii

An open circulatory system has an unregulated blood flow to many organs. This is due to the fact that there is no method to regulate the blood's flow to certain organs, and blood flows freely through the bodily cavities.

Due to the blood not being restricted to arteries but rather flowing freely through the bodily cavities, the blood pressure in an open circulatory system is significantly lower than in a closed circulatory system. Small blood channels called capillaries link veins and arteries. They are absent from open circulatory systems. In an open circulatory system, the blood returns slowly to the heart. This is due to the heart's inability to pump blood through the body.

4. Correct option: (B) Both assertion and reason are true, but reason is not the correct explanation of assertion

The claim and the justification are both true. White blood cells (WBCs) do travel from the circulation via the endothelium (inner lining) of blood vessels to the site of damage or infection, a process known as diapedesis, to collect at the site of wounds. The justification given, nevertheless, is not the one that adequately explains the claim. White blood cells actively squeeze through endothelial cell gaps during diapedesis, however the phrase "squeezing from the endothelium" may be deceptive. The primary idea is that white blood cells may go to the site of inflammation or infection thanks to diapedesis, which enables them to exit the circulation.

5. Correct option: (D) A – iii, B – iv, C – i, D – ii

A. Plasma - iii. Fluid part of blood

The liquid portion of blood is called plasma. Water, electrolytes, hormones, waste materials, and numerous proteins are all present in it.

B. Red blood cells - iv. Carry oxygen

Erythrocytes, another name for red blood cells, are in charge of returning carbon dioxide to the lungs for expiration and transporting oxygen from the lungs to the body's tissues.

C. White blood cells - i. Eat up the bacteria

Leukocytes, commonly known as white blood cells, are important components of the immune system. They support the body's defence mechanisms against diseases and outside aggressors including germs and viruses.

D. Platelets - ii. Coagulation of blood

The process of blood coagulation or clotting involves platelets. By generating clots at the site of an injury, they aid in controlling bleeding and limiting excessive blood loss.

6. Correct option: (C) Only iv

i. This statement is incorrect. Instead of ventricles, the human heart's upper two chambers are known as atria. The atria are where the heart's blood is received.

ii. This statement is correct. Ventricles are the actual names of the lowest two chambers of the human heart. Blood is pumped from the heart through ventricles.

iii. This statement is correct. The septum, not the pericardium, serves as the barrier between the heart's chambers. A two-layered sac called the pericardium protects the heart by surrounding it.

iv. This statement is incorrect. Deoxygenated blood is transported via the pulmonary artery from the right ventricle to the lungs, where it is oxygenated. After receiving oxygen in the lungs, the blood travels back to the heart through the pulmonary veins.

7. Correct option: (D) A, B, AB and O

Blood group AB is the consequence of a blood group 'A' (genotype AO) person marrying a blood group 'B' (genotype BO) person. This happens because the possible kid can receive the A allele from the first parent and the B allele from the second parent. In addition, the child may receive the O allele from the second parent and the A allele from the first parent, resulting in blood group A. They might also acquire the blood type B allele from their second parent and the blood group O allele from their first parent. Last but not least, if both parents inherit the O gene, the child would have the blood group O.

8. Correct option: (A) Both A and B antigens on RBC but no antibodies in the plasma

Red blood cells (RBCs) from people with blood type AB are coated with both A and B antigens. This implies that they won't experience an immunological reaction to the transfusion if they get blood from donors of any blood group.

They don't have any plasma antibodies to A or B antigens. They won't be able to develop an immune response to entering blood that contains A or B antigens due to this absence of antibodies.

The A or B antigens are not present on the red blood cells of people with blood type O. Additionally, they do not possess either anti-A or anti-B antibodies. However, they are only able to accept blood from O blood types. This is due to the fact that if they receive blood from someone with blood group A or B, their body may create antibodies to the A or B antigens, which may result in a response known as a hemolytic transfusion reaction.

9. Correct option: (D) A – Pulmonary vein-takes impure blood from body parts, PO2 = 60 mm Hg

The correct label and its functions among the given options are Pulmonary vein - takes impure blood from body parts, PO2 = 60 mm Hg

Oxygenated blood is returned to the heart by pulmonary veins from the lungs.

After being oxygenated in the lungs, the blood has a comparatively greater oxygen concentration (PO2 = 60 mm Hg) in the pulmonary veins.

The pulmonary veins contain oxygen-rich blood that has been oxygenated in the lungs, making it relatively pure in terms of oxygen concentration. As a result, the phrase "impure blood" does not apply to them.

10. Correct option: (D) Type O

Blood type O, sometimes referred to as the universal donor blood type, can be safely used to transfuse a person with unknown blood group who has lost a significant amount of blood. People of any ABO blood type can have blood type O because it lacks A or B antigens on the surface of the red blood cells.

Blood from a type O donor can be safely transfused in an emergency when the recipient's blood type is unknown.

11. Correct option: (B) Only i

Blood pressure is the force that the heart's pumping action places on the blood arteries as it travels through them. Systolic blood pressure refers to the force exerted by blood when blood vessels constrict, while diastolic blood pressure refers to the force exerted when blood vessels relax. 120/80 is the typical blood pressure reading.

Statement i is true. This number for blood pressure is extremely high and may harm kidney and brain blood vessels. It can cause significant health issues like renal disease and a stroke if left untreated.

Statement ii is false. The 130/90 mm Hg blood pressure reading is not very high and may be controlled by meditating and drinking some cold liquids. There's no need for therapy for this.

Statement iii is false. Adults should have a blood pressure of 120/80 mm Hg. A measurement of 100/55 mm Hg is seen as being on the low side and might be an indication of low blood pressure.

Statement iv is false. Low blood pressure is defined as having a measurement of 105/50 mm Hg or below. You could experience fatigue, dizzy, and lightheadedness from low blood pressure. Additionally, it may make it challenging to focus and think effectively. It would not necessarily make someone "very active."

12. Correct option: (A) i and iii

Statement i is true because arteries have thicker walls and a smaller diameter than veins. This occurs so that the arteries, which transport oxygenated blood from the heart to the rest of the body, can handle the pressure of the blood. Conversely, veins do not require as much strength since they just transport deoxygenated blood from the rest of the body back to the heart.

Statement iii is accurate since calcium ions are crucial for blood coagulation. A number of clotting factors, which are proteins that aid in stopping bleeding, must first be activated by calcium ions.

Statement ii is incorrect Because only individuals of blood group AB may donate blood to them. This is due to the fact that individuals with blood type AB contain both A and B antigens on their red blood cells. When they receive blood from someone with a different blood type, their immune system will fight the foreign antigens and result in a transfusion response.

13. Correct option: (D) Only D

Adult human red blood cells (RBCs) have a special property called enucleation that gives them extra room for transporting oxygen. RBCs can increase their haemoglobin content, which is in charge of binding and transporting oxygen throughout the body, by eliminating the nucleus and other organelles. RBCs are better able to carry out their main role because to this adaptation, which also improves oxygen transport efficiency. The bone marrow produces RBCs, which are then discharged into the circulation and have a lifespan of roughly 120 days. They are then disassembled and recycled. RBCs are somatic cells, which means they are cells that are part of the tissues that make up the body. Because they lack the organelles and enzymes required for metabolism, RBCs cannot metabolize.

14. Correct option: (B) P – Artery, Q – Vein, R - Capillary

P – Artery - Arteries are blood arteries with strong walls that transport oxygenated blood from the heart. They feature a thick layer of smooth muscle and a limited lumen (the area through which blood flows). This enables them to resist the intense blood pressure caused by the heart's pumping action.

Q – Vein - Veins are blood channels with thin walls that return anaemic blood to the heart. Compared to arteries, they have a broader lumen and a thinner coating of smooth muscle. As a result, the blood may return to the heart more easily.

R - Capillary - The body's tiniest blood vessels are called capillaries. Because of their extremely thin walls, the cells can receive oxygen and nutrients from the blood. Arterioles and veins are linked by capillaries

15. Correct option: (C) Both the statements are correct

Statement 1: When blood clots, serum, liquid rich in proteins, separates off. Red blood cells, white blood cells, platelets, and plasma all make up blood. The platelets and fibrinogen in the plasma create a network known as fibrin during blood clotting. Red and white blood cells become trapped in this net, creating a clot. The liquid that remains after the clot develops is called serum. Proteins including albumin, globulins, and fibrinogen are abundant in it.

Statement 2: An insoluble protein called fibrin is created during blood clotting from fibrinogen. The plasma contains fibrinogen, a soluble protein. When blood clots, fibrin, a mesh made of platelets and plasma fibrinogen, is formed. The white blood cells and red blood cells get caught in this net, creating a clot. Because the fibrin is insoluble, it does not disintegrate in water. This is what gives the clot its hard, solid texture.

16. Correct option: (D) S

Blood that has lost its oxygen content travels from the heart to the lungs through the pulmonary artery. As the blood has not yet been oxygenated in the lungs, the oxygen level in the blood in the pulmonary artery is quite low. Of all the blood arteries in the body, the blood in the pulmonary artery will contain the least amount of oxygen.

The chart shows that sample S has the lowest oxygen content, so it is most likely taken from the pulmonary artery.

17. Correct option: (B) Larger veins

Through the bigger veins, lymph is cleansed and its beneficial components are reintroduced into the circulatory system. Larger bodily veins eventually get the contents of lymphatic channels. This is how the lymphatic system helps to keep the fluid balance in the body while moving trash, bacteria, and other materials out of tissues.

Blood is transported from the heart through arteries. They receive no lymph. The tiniest blood vessels are capillaries. They serve as the point where the blood and tissues

exchange nutrients and oxygen. They receive no lymph. A muscular organ, the heart circulates blood throughout the body. It receives no lymph.

18. Correct option: (C) i , iii and iv

Statement i is true. Platelets are sometimes referred to as thrombocytes.

Statement ii is false. Because they are anucleated, platelets lack a nucleus. Additionally, their blood volume is substantially larger than that of 6000–8000 mm3, with an average of 150,000–400,000 platelets per mm3.

Statement iii is true. They assist in the blood's coagulation. For blood to clot, platelets are required. When a blood artery is damaged, platelets congregate to create a plug that aids in stopping the bleeding.

Statement iv is true. A decrease in their population may result in clotting issues, which may cause the body to lose large amounts of blood. Additionally true is this. Thrombocytopenia is a condition that can result from a decrease in platelet production. Thrombocytopenia may result in severe bleeding, which is potentially fatal.

19. Correct option: (D) i, ii, iii and iv

A set of processes known as the cardiac cycle help the blood to flow through the heart. Systole, which denotes the contraction of the cardiac muscles, and diastole, which denotes their relaxation, are the two primary phases of it (ii). Specialized nodal tissues, namely the sinoatrial node (SA node) and atrioventricular node (AV node), which control the heart's rhythm and coordination, orchestrate the cardiac cycle (iii). Changes in the length of the cardiac cycle have a direct effect on the cardiac output, causing an increase or reduction in the volume of blood the heart pumps per unit of time (iv). As a result, each of the assertions (i, ii, iii, and iv) offered accurately describes a different element of the cardiac cycle.

20. Correct option: (B) A vein starts from an organ and ends up in another organ.

An intricate circulatory system known as a "portal system" involves the passage of blood through two capillary networks before it returns to the heart. In this system, a vein emerges from one organ, where it draws blood from capillaries, before transporting it to a different organ's second set of capillaries and ultimately returning to the heart. Before the blood is returned to the heart, this configuration enables an effective interchange of nutrients, gases, or other things between two distinct organs. The definitions of the other choices (A, C, and D) do not adequately convey the idea of a gateway system.

4. Life Process - Excretory System

Solutions

LEVEL-I

1. Correct option: (B) Nephron

In the human excretory system, the nephron is the excretory unit. It is a tiny organ located in the kidneys and is in charge of filtering blood, eliminating waste, and controlling the fluid balance in the body. Millions of nephrons are found inside each kidney, and they are essential for preserving the body's equilibrium by removing waste materials, extra ions, and water from the blood to make urine. The urine that the nephron produces then travels through the urinary system before being expelled from the body. The body's interior environment is maintained and kept in balance thanks to this effective filtration and waste disposal procedure.

The fundamental element of the nervous system is the neuron. Signals are sent by them throughout the body. Invertebrates excrete their waste through nephridia. Worms, insects, and snails are just a few of the many species that have them. The excretory system of flatworms is the flame cell. They are specific cells that filter blood and expel trash.

2. Correct option: (B) Kidney → ureters → bladder → urethra

After the nephrons in the kidneys filter and remove waste products, excess water, and ions from the blood, the resulting fluid, called urine, is then transported through the ureters, which are muscular tubes connecting each kidney to the bladder. The urine is temporarily stored in the bladder until it reaches a certain level of fullness. When the bladder is filled, the urine is expelled from the body through the urethra, a tube that leads from the bladder to the external opening. This process allows the body to efficiently eliminate waste and maintain the proper balance of fluids and ions.

3. Correct option: (B) Vasopressin

Vasopressin, also known as antidiuretic hormone (ADH), has a direct impact on urination. Produced in the hypothalamus and released by the posterior pituitary gland, vasopressin regulates the water balance in the body by controlling the reabsorption of water in the kidneys. When vasopressin levels are high, the kidneys reabsorb more water from the urine, resulting in concentrated urine and decreased urine output. Conversely, when vasopressin levels are low, more water is excreted in the urine, leading to increased urine output and potentially more diluted urine. This hormone plays a crucial role in maintaining proper fluid balance and preventing excessive dehydration or water retention.

Urination is not directly influenced by estrogen, FSH, or adrenal hormones. Cortisol and adrenaline are only two of the chemicals that the adrenal gland generates. Although this is an indirect consequence, cortisol can cause an increase in urine output. Urine output can also be increased by adrenaline; however, this normally only happens in stressful circumstances. FSH is a hormone that affects how many eggs women produce. Women's ovaries generate the hormone known as estrogen. By influencing the bladder, FSH and estrogen can both indirectly affect urination. They don't, however, directly affect urine.

4. Correct option: (A) Glomerular filtration, Tubular reabsorption, Tubular secretion

Glomerular filtration: In order to create urine, this is the initial step. A network of capillaries in the nephron called the glomerulus is where it takes place. Known as the Bowman's capsule, the glomerulus is encircled by this structure. Blood pressure drives water and other tiny solutes into the Bowman's capsule as blood passes through the glomerulus. Water, glucose, amino acids, salts, and waste materials can be found in this filtrate.

Tubular reabsorption: The second stage in the production of urine is this. The renal tubule, a protracted, winding tube that is a component of the nephron, is where it develops. Cells having unique characteristics that enable them to selectively reabsorb particular compounds from the filtrate line the renal tubule. Reabsorbed substances include salts, water, glucose, and amino acids. The waste is not reabsorbed.
Tubular secretion: Urine formation proceeds via this stage. A renal tubule hosts it. During tubular secretion, certain chemicals are actively transferred from the blood into the filtrate. Potassium ions, certain medicines, and hydrogen ions are some of these compounds.

5. Correct option: (C) Bowman's capsule

Glomerulus and Bowman's capsule make comprise the renal corpuscle. The Bowman's capsule is a component of the nephron that is located in the cortex of the outer zone of the kidney. A system of capillaries called the glomerulus filters blood. It acts as the first site of filtration when blood components are filtered into the renal tubules and encloses the glomerulus, a network of capillaries. By removing waste and water from the blood, this mechanism starts the creation of urine.
The medulla, the kidney's inner region, contains the proximal convoluted tubule (PCT), the distal convoluted tubule (DCT), and the loop of Henle, whereas the cortex, the kidney's outer region, has the bowman's capsule.

6. Correct option: (A) Cortex

The kidney's cortical region is its outermost region. It produces urine and filters blood, and it is dark brown in color. The proximal and distal convoluted tubules, which are involved in the reabsorption and secretion processes, as well as the glomeruli, which filter blood, are among the many structures found in the renal cortex. The nephrons, the kidney's microscopic filtering units, are housed in this area of the kidney, which is crucial in the early stages of urine generation.
The renal pyramids are what make up the medulla. The renal pyramids are radially organized cone-shaped structures. Urine is gathered from the renal tubules via collecting ducts found in the renal pyramids. The calyces, which are cup-shaped structures that collect urine from the renal pyramids, are where the pee travels after leaving the collecting ducts. The ureter, a tube that transports urine from the kidney to the bladder, is where the calyces then empty into.

7. Correct option: (A) Proximal convoluted tubule

In the renal system, most glucose molecules are reabsorbed in the proximal convoluted tubule (PCT). This tubule is situated in the cortex of the kidney and is responsible for the initial reabsorption of various valuable substances, including glucose, amino acids, ions, and water, from the filtrate. Glucose is actively transported from the PCT back into the bloodstream, ensuring that the body retains this essential energy source rather than excreting it in the urine. This process helps maintain proper blood glucose levels and efficiently utilizes nutrients during urine formation.
Reabsorbing water and concentrating the urine are two functions of the distal convoluted tubule, loop of Henle, and collecting ducts. However, for reabsorbing glucose, the PCT is the most crucial component of the nephron.

8. Correct option: (B) Waste products

When the kidneys are unable to eliminate waste materials from the blood, uremia develops. This may occur as a result of acute renal injury, chronic kidney disease, or other kidney-damaging disorders.
Waste materials accumulate in the blood when the kidneys are unable to filter them out. Numerous symptoms, such as weariness, nausea, vomiting, lack of appetite, muscular

cramps, itching, and mental disorientation, may result from this. Uremia can be lethal in extreme circumstances.

Urea, creatinine, uric acid, and potassium are among the waste products that accumulate in the blood in uremia. The liver creates urea as a waste product as a result of protein breakdown. A waste product produced by the muscles is creatinine. The breakdown of purines results in the waste product uric acid. Potassium is a crucial mineral that supports the control of the heartbeat and other physiological processes.

Therefore, uremia is a condition in which the body retains waste materials rather than excess water.

9. Correct option: (B) Vasopressin

Vasopressin, sometimes referred to as antidiuretic hormone (ADH), boosts reabsorption in the nephron's collecting ducts. Vasopressin, which is produced in the hypothalamus and released by the posterior pituitary gland, works on the collecting ducts to increase their water permeability. This concentrates the urine and helps the body conserve water by allowing the body to reabsorb more water from the urine back into the bloodstream. This system is necessary to keep the body's fluid balance in check, avoid dehydration, and modify urine concentration in response to hydration requirements.

Renin, aldosterone, and insulin don't make the collecting ducts reabsorb more. Renin stimulates angiotensin II production, which raises blood pressure. Aldosterone increases sodium reabsorption and potassium excretion, which regulates blood pressure and electrolyte balance. By encouraging the absorption of glucose into cells, insulin aids in the regulation of blood sugar levels.

10. Correct option: (D) Increased water consumption

Urine output rises as a direct result of increased water intake. The human body's fluid equilibrium is upset when you drink more water, therefore the extra water needs to be expelled. An increase in urine production is the consequence of the kidneys being prompted to filter and eliminate the extra water and waste.

Increased amounts of exercise can cause perspiration, which can cool the body and lessen the need to urinate. Sweating and reduced urine output are two additional effects of elevated body temperature. Evidently, less water consumption will result in less urine production.

11. Correct option: (B) Podocytes

The Bowman's capsule is an essential part of the nephron of the kidney, which is in charge of the early filtering of blood to create urine. A single layer of specialized cells known as podocytes lines the inside surface of the Bowman's capsule. Larger molecules, including as proteins, are prevented from entering the first filtrate by the intricate foot-like "pedicels" of these podocytes, which wrap around the glomerular capillaries.

The other options are incorrect. The alternate choices are wrong. Some plants' cells contain crystals in the form of needles called raphides. Red blood cells are known as erythrocytes. Dopocytes are not a class of cell.

12. Correct option: (C) Ultra filtration and selective reabsorption

Ultrafiltration, in which blood is filtered through the glomerulus in Bowman's capsule to generate a filtrate, is a step in the nephron's process of producing urine. Small compounds, ions, and water is all present in this filtrate. Then, selective reabsorption occurs, where waste materials and excess substances are left in the developing urine while necessary chemicals like glucose, ions, and water are reabsorbed from the filtrate back into the circulation. The mechanisms involved in producing urine in the nephron are not correctly described by the other alternatives (A, B, and D).

13. Correct option: (C) Distal convoluted tubule

The distal convoluted tubule (DCT) is in charge of preserving the ideal pH and concentration of urine. It is the section of the renal tubule that links to the collecting ducts following the Henle loop. Further ion reabsorption and secretion of water, other chemicals, and other processes take place in the DCT, allowing for fine-tuning of urine composition.

The majority of the glucose, amino acids, and other crucial nutrients from the filtrate are reabsorbed by the proximal convoluted tubule (PCT). The Henle loop is a U-shaped structure that aids in urine concentration. All of the kidney's nephrons' urine must be collected and sent to the bladder through the collecting duct.

14. Correct option: (A) Vasopressin

Vasopressin, commonly known as antidiuretic hormone (ADH), is the primary factor that causes urine to concentrate in the collecting ducts. When the blood osmolarity varies, the hypothalamus produces vasopressin, which is then released by the posterior pituitary gland. As a result of its action on the collecting ducts, which increases their water permeability, more water may be reabsorbed from the developing urine and returned to the circulation. Concentrated urine is the outcome.

The hormone insulin is in charge of controlling blood sugar levels. Blood pressure and salt levels are controlled by aldosterone. The response of "fight or flight" is brought on by adrenaline.

15. Correct option: (A) 300 – 700 ml

The urinary bladder may hold 300 to 700 ml of fluid on average. The bladder is a muscular hollow organ that may expand to hold various volumes of pee. The neurological system receives a signal from the bladder's stretch receptors when urine fills it, which causes the urge to urinate.

The other options (B, C, and D) do not accurately represent the common range of bladder storage capacity.

16. Correct option: (A) 1.5 to 1.7 liters

Each day, the nephrons produce around 180 liters of glomerular filtrate in total. To keep osmotic equilibrium and preserve vital nutrients, only a small fraction of this filtrate—roughly 188 to 188.5 liters per day—is reabsorbed back into the blood vessels. Urination serves as the body's method of removing the residual nephron residue, which contains nitrogenous waste products. An individual produces around 1.5 liters of urine on average each day as a result. This range of urine production falls within 1.5 to 1.7 liters per day.

17. Correct option: B) Charles Hufnage

In 1954, an American surgeon named Dr. Joseph Murray carried out the first successful kidney transplant on a person. On December 23, 1954, the procedure was carried out at the Peter Bent Brigham Hospital in Boston, Massachusetts. Ronald Herrick, a 23-year-old man with end-stage renal failure, was the receiver. Richard Herrick, his identical twin brother, served as the donor.

William Harvey was an English doctor who made the discovery of blood circulation. American surgeon Charles A. Hufnagel created the first prosthetic heart valve in the early 1950s. American agricultural scientist Norman Borlaug received the Nobel Peace Prize in 1970 for his efforts to boost grain production.

18. Correct option: (A) Bilirubin, biliverdin and urochrome

The breakdown products of hemoglobin from dead red blood cells include bilirubin and biliverdin. The substance bilirubin, which gives bruises their yellow color, is finally broken down in the liver and eliminated in bile. The pigment urochrome, which is produced when heme is broken down, is part of what gives urine its yellow color.

Despite not being formed from hemoglobin, uric acid and urea are metabolic wastes as well. These wastes are likewise carried to the kidneys, where they are eliminated through the urine. Although they are not metabolic wastes, water and mineral salts are nonetheless expelled in the urine.

19. Correct option: (C) Adrenal gland

Each kidney's upper surface is occupied by the endocrine glands known as the adrenal glands The adrenal glands are made up of two parts: the cortex and the medulla. The cortex produces a number of hormones, including cortisol, aldosterone, and androgens. The medulla produces the hormone adrenaline. The body's metabolism, stress response, and electrolyte balance are just a few of the physiological processes that these hormones play a significant part in controlling.

In the front of the neck, the thyroid gland is located. An organ behind the stomach is called the pancreas. Both endocrine and exocrine actions are performed by it. The pituitary gland, which regulates several other endocrine glands, is sometimes referred to as the "master gland." It can be found at the bottom of the brain.

20. Correct option: (B) Vasopressin

The illness known as diabetes insipidus is brought on by a lack of the hormone vasopressin, often referred to as antidiuretic hormone (ADH). The hypothalamus makes ADH, which the pituitary gland then releases. By increasing the water permeability of the collecting ducts, it aids in the kidneys' ability to retain water. Vasopressin regulates the reabsorption of water in the kidneys, which is a key factor in managing the water balance. A lack of vasopressin causes the kidneys to be unable to effectively concentrate urine, which causes excessive thirst and urination.

A hormone called insulin is produced by the pancreas. It facilitates the body's utilization of glucose as a fuel. The adrenal glands create a hormone called adrenaline. In times of stress or peril, it is released. The parathyroid glands create the hormone known as parathormone. It aids in controlling blood calcium levels.

21. Correct option: (B) Ureotelism

The removal of urea, a byproduct of metabolism, is called ureotelism. In animals, including humans, this is the most typical way that nitrogenous waste is excreted. Ammonia, a poisonous gas created when proteins break down, is converted to urea in the liver. The urea is subsequently carried to the kidneys, where it is eliminated as urine. The removal of ammonia-based metabolic waste is known as ammoniotelism. Several aquatic creatures, including fish, excrete nitrogenous waste in this basic form. The removal of uric acid-based metabolic waste is known as uricotelism. This is how birds, reptiles, and insects expel nitrogenous waste. Aminotelism is not a true method of eliminating nitrogenous waste.

22. Correct option: (C) Constant composition of the blood irrespective of the nature of the food or fluid intake

While the juxtaglomerular apparatus releases renin to maintain glomerular blood flow and pressure, the kidney's antidiuretic hormone (ADH) controls blood and fluid volume. The renin-angiotensin-aldosterone pathway, on the other hand, is blocked by the atrial natriuretic factor, which lowers blood pressure through vasodilation. In order to maintain constant blood composition regardless of fluid consumption, the kidneys' osmoregulation system uses ADH to regulate water balance. Together, these activities help to keep the body's internal environment consistent in terms of composition, fluid volume, and blood pressure.

23. Correct option: (A) Nephrons

Nephrons are the smallest functional unit of the kidney. They are responsible for filtering the blood and producing urine. Each kidney contains about 1 million nephrons. The

nephron is the kidney's smallest functioning unit. Filtering blood, reabsorbing necessary nutrients, and excreting waste to generate urine are all tasks carried out by nephrons. The glomerulus, Bowman's capsule, proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting ducts are just a few of the components that make up each nephron. These components all work together to carry out the many kidney activities.
The other options (B, C, and D) are components of the nephron or larger structures associated with kidney function.

24. Correct option: (D) Both A and B

The rate at which blood plasma is filtered via the glomeruli (small filtering units) of the kidneys is known as the glomerular filtration rate (GFR). It's an essential indicator of kidney health since it shows how well the kidneys are removing waste and extra chemicals from the blood to create urine.
The correct GFR reading is around 125 mL/min per kidney, or 180 liters per day when the two kidneys are combined.
So, in this case, if the options are presented as "Both (A) and (B)," the correct answer is D) Both (A) and (B) because both the values 125 mL/min and 180 L/day are accurate representations of the glomerular filtration rate.

25. Correct option: (B) Aldosterone

An enzyme in the adrenal cortex called aldosterone is responsible for its production. When blood potassium levels are high or the sodium level is low, it is discharged. In order to enhance salt and water absorption and reduce potassium excretion, aldosterone operates on the kidneys. The fluid and electrolyte balance of the body is preserved as a result.
ADH, also known as vasopressin, is a hormone that controls salt reabsorption. Enzymes called renin and rennin are part of the renin-angiotensin-aldosterone system (RAAS). The reabsorption of sodium is not directly impacted, though.

26. Correct option: (D) ADH

The posterior pituitary gland houses the hormone ADH, also known as antidiuretic hormone, which is generated by the brain. When the concentration of dissolved chemicals in the blood is high or the blood volume is low, it is discharged. Water is conserved and dehydration is prevented because to ADH's effect on the kidneys, which increases water absorption.
Other options are incorrect. They are not involved in reabsorption of water. Growth hormone, also known as STH, is produced by the pituitary gland. The pituitary gland creates ACTH, also known as adrenocorticotropic hormone. It causes the adrenal cortex to generate cortisol when stimulated. The pituitary gland produces luteinizing hormone (LH).

27. Correct option: (C) Both (A) and (B)

Na+ and Cl- are both removed by the ascending limb of Henle's loop and the proximal convoluted tubule. Na+ and Cl- can pass through the ascending limb of Henle's loop, but water cannot. In contrast to water, Na+ and Cl- are actively carried out of the tubule. As a result, a concentrated salt solution forms in the kidney's medulla, which aids in luring water from the collecting duct and into the circulation.
Additionally permeable to Na+ and Cl-, the proximal convoluted tubule also possesses active transport systems for reabsorbing glucose, amino acids, and other nutrients. Na+ and Cl- are still excreted from the tubule, although not in the same quantities as in the ascending limb of Henle's loop.
The descending limb of Henle's loop is impermeable to Na+ and Cl-, but it is permeable to water. This means that water is drawn out of the tubule into the medulla of the kidney, which helps to concentrate the salt solution.

28. Correct option: (B) Kidney

Columns of Bertini are extensions of the cortex of the kidney that project into the medulla. They are made up of connective tissue and blood vessels. They help to support the medulla and to increase the surface area of the kidney. They help to support the medulla and to increase the surface area of the kidney.

Column of Bertini is not found in liver, ovaries, and testes.

29. Correct option: (D) Na^+ ions

Aldosterone is a hormone that is produced by the adrenal glands. It helps to regulate the balance of sodium ions in the body. It acts on the distal convoluted tubule and collecting ducts of the nephron to increase the reabsorption of sodium ions, which in turn affects water reabsorption and blood pressure regulation.

Keto acids, glucose, and K+ ions are not reabsorbed by aldosterone.

30. Correct option: (A) Cortex

Malpighian corpuscles, also known as renal corpuscles, are located in the renal cortex of the kidney. They consist of Bowman's capsule and the enclosed glomerulus, and they play a crucial role in the initial filtration of blood to form the glomerular filtrate.

The other options are incorrect. The kidney's medulla is situated inside the cortex. The testes are where the germ cells are found.

31. Correct option: (C) Liver

The liver generates urea as a waste product. Amino acids that are no longer required by the organism are broken down by the liver. Ammonia and carbon dioxide are the by-products of this breakdown. The liver turns ammonia into urea because it is harmful. Urine is produced when urea is delivered to the kidneys and discharged there.

The urinary bladder, alimentary canal, and kidneys are all involved in the process of urine formation, but they do not synthesize urea.

LEVEL – II

1. Correct option: (D) Nearly 99 percent of the glomerular filtrate is reabsorbed by the renal tubules

Filtering the blood and getting rid of waste is done by the kidneys. The capillary network that filters blood is known as the glomerulus. The majority of the water and nutrients are then reabsorbed in the renal tubules after the filtrate has been passed through them. Only 1% of the filtrate is eliminated in the urine.

Although it is really impermeable to water, the ascending limb of the Loop of Henle actively transfers electrolytes (sodium, potassium, and chloride) out of the tubule. Water can pass through the descending limb of the Loop of Henle and be reabsorbed by osmosis, which concentrates the filtrate. Bicarbonate ions (HCO3) can be reabsorbed by the distal convoluted tubule to assist control the body's acid-base balance.

2. Correct option: (A) Hydrogen ions are actively secreted into the filtrate

Human urine is usually acidic due to the active secretion of hydrogen ions (protons) into the filtrate within the renal tubules. This production of hydrogen ions aids in maintaining the body's acid-base homeostasis and pH balance. By selectively reabsorbing or secreting hydrogen ions and bicarbonate ions, the kidneys play a critical part in preserving the body's acid-base balance, helping to control blood pH. The urine's overall acidity is a result of the active release of hydrogen ions into the filtrate.

The other options are incorrect. The peritubular capillaries' sodium transporter does not switch hydrogen ions for sodium ions. Instead, it swaps sodium ions for chloride ions. This aids in preserving the blood's osmolarity. Plasma proteins that are excreted are not acidic. In

actuality, they have a mild alkaline pH. Acidity is not produced by the interchange of potassium and sodium. In actuality, it guards against alkalosis and buffers the blood.

3. Correct option: (D) i, ii, iii and iv

i. The outer layer of the kidney is called capsule. This is true. The renal capsule is the name of the kidney's outermost covering. It is a fibrous coating that shields the kidney from harm.
ii. Cortex is divided into outer cortex and inner medulla. This is true. The inner medulla and the outer cortex make up the kidney's structure. The proximal and distal convoluted tubules, together with Bowman's capsules and glomeruli, are located inside the cortex.
iii. Medulla is divided into medullary pyramids. This is true. The kidney's interior layer is called the medulla. The pyramids of the medulla are separated from it. Conical-shaped structures called the medullary pyramids extend into the renal pelvis.
iv. The cortex extends in between the medullary pyramids which is called as columns of Bertini. This is also correct. In between the medullary pyramids, the cortex expands. The columns of Bertini are a protrusion of the cortex. The medullary pyramids are supported in part by the Bertini columns.

4. Correct option: (D) A – iv, B – iii, C – ii, D – i

A. PCT - iv. Reabsorbing about 90% of the important buffer HCO_3 from filtrate
The initial component of the nephron is called the proximal convoluted tubule (PCT). About 60% of the water and 90% of the glucose, amino acids, and other tiny molecules from the filtrate are reabsorbed by it. The PCT is also involved in the hemostasis of Na+ and K+. It does not, however, absorb HCO3.

B. Descending loop of Henle - iii. Permeable to water but not to salt.
The second portion of the nephron is called the ascending Henle loop. Water cannot pass through it, but NaCl can. This enables the NaCl to accumulate in the kidney's medulla, aiding in the formation of a concentration gradient that stimulates water reabsorption in the ascending loop of Henle.

C. Ascending loop of Henle - ii. Permeable to NaCl but impermeable to water.
The third component of the nephron is the ascending Henle loop. Salt cannot pass through it, but water can. As a result, more water may be reabsorbed from the filtrate, increasing the concentration of the filtrate. It does not, however, work in the hemostasis of Na+ and K+.

D. DCT - i. Functions in Na+ & K+ hemostasis.
Fourth in the nephron structure is the distal convoluted tubule (DCT). It is in charge of secreting ions like K+ and H+ as well as reabsorbing roughly 10% of the water. Additionally, the DCT reabsorbs 90% of the crucial buffer HCO3 from filtrate.

5. Correct option: (C)

The concentration of insulin in the bloodstream is uniformly distributed throughout the body, indicating a consistent distribution throughout the whole bloodstream. Insulin continually decreases from the blood when it is metabolized and used, demonstrating a constant rate of elimination. When the blood concentration is graphed over time, the drop is exponential. In contrast, insulin is eliminated from the circulation by urinary excretion, resulting in an initial decrease in urine concentration that progressively rises over time as insulin is filtered and eliminated. As a result, it is anticipated that the graph of urine insulin concentration vs time would follow the opposite pattern of the blood concentration graph, beginning low and gradually rising.

6. Correct option: (D) i and iv

Loop of Henle (i): A vital component of the nephron that greatly aids in the production of concentrated urine is the Loop of Henle. It produces an osmotic gradient in the kidney medulla, allowing for the active transport of ions (such as sodium and chloride) in the

ascending leg and the reabsorption of water from the filtrate in the descending limb. The concentration of the urine depends on this gradient.

Vasa Recta (iv): In the kidney's medulla, parallel to the Loop of Henle, are specialized capillaries known as the vasa recta. The osmotic gradient created by the Loop of Henle is maintained in large part by them. The interchange of water and solutes is possible in the vasa recta without interfering with the concentration gradient, which helps keep the osmotic gradient from dissipating.

Even while Bowman's capsule (iii) and the glomerulus (ii) are both engaged in the early filtering of blood to create the filtrate, neither is directly in charge of the concentration of urine. The vasa recta maintain the concentration process, which is principally driven by the reabsorption and control of water and solutes in the Loop of Henle.

7. Correct option: (B) Only 3

Urine is hypertonic in distal convoluted tubule. This is incorrect. The urine is actually hypotonic in the distal convoluted tubule.

When the urine passes into collecting tubule, it becomes hypertonic. This is also incorrect. The urine becomes more hypotonic as it passes into the collecting tubule.

Urine is isotonic in proximal convoluted tubule. This is correct. The urine is isotonic in the proximal convoluted tubule.

Urine becomes more and more hypotonic as it passes through Henle's loop. This is also correct. The urine becomes more and more hypotonic as it passes through the ascending limb of the loop of Henle.

8. Correct option: (C) A is true and R is false

The assertion is true. Podocytes are specialized cells that line the interior of Bowman's capsule. To create a filtration barrier, the foot processes of podocytes interdigitate with one another. Water and tiny molecules can travel across this barrier without being blocked, but proteins and other big molecules cannot.

The reason is false. The surface area for absorption is not increased by the podocytes' projections. They expand the filtering surface area. The basement membrane and the podocyte foot processes combine to create the filtration barrier. The foot processes are secured to the basement membrane and tightly sealed together thanks to the projections of the podocytes. Larger molecules cannot cross the filtration barrier because of this seal.

9. Correct option: (C) A – Medullary pyramid, B – Calyx, C – Renal artery, D – Renal vein, E - Ureter

A) Medullary pyramid: The kidney's medulla contains triangle-shaped structures called medullary pyramids. They have channels for collecting urine, which are used to move it from the nephrons to the renal pelvis.

B) Calyx: Resembling a cup, the calyx absorbs urine from the medullary pyramids. The renal pelvis is reached by the union of many small calyces into large calyces.

C) Renal artery: From the heart to the kidneys, the renal artery transports oxygenated blood. Through the renal arteries, blood enters the kidneys, giving them with nutrients and oxygen.

D) Renal vein: Deoxygenated blood from the kidneys is transported via the renal vein to the inferior vena cava, which then returns the blood to the heart.

E) Ureter: The ureter is a tube that transports urine from the kidney's renal pelvis to the bladder, where it is temporarily stored until being eliminated.

10. Correct option: (B) Inside the kidney the cortical region extends in between the medullary pyramids as renal pelvis

Inside the kidney the cortical region extends in between the medullary pyramids as renal pelvis is incorrect. The renal pelvis, which is part of the collecting system that collects urine from the medullary pyramids and directs it into the ureter for delivery to the

bladder, is situated in the middle of the kidney. The medullary pyramids are deeper structures in the medullary zone, whereas the cortical area is found on the kidney's outermost section. The renal pelvis is located deeper inside the kidney and is not an extension of the cortical area.

The other given statements are correct.

11. Correct option: (C) Ammoniotelic - Aquatic Amphibia; Ureotelic - Frog, Humans; Uricotelic - Pigeon, Lizards, Cockroach

Ammonia is the main nitrogenous waste that ammonotelic animals emit. Ammonia is extremely harmful, and it takes a lot of water to successfully dilute and expel it. Water is available to aquatic creatures, such as amphibians, allowing for effective ammonia excretion. Animals that are ureotelic expel urea as their main nitrogenous waste. In addition to requiring less water for excretion, urea is less hazardous than ammonia. Both frogs and people are ureotelic. Urea is passed through the skin by frogs and the urine of humans. Uricotelic animals excrete uric acid as their primary nitrogenous waste. Uric acid is relatively non-toxic and can be excreted with minimal water loss. Birds, reptiles (including lizards), and insects like cockroaches are uricotelic. Pigeons are also uricotelic birds.

12. Correct option: (B) Podocytes: Create minute spaces (slit pores) for the filtration of blood into the Bowman's capsule

Podocytes are specialized cells found in the nephron's Bowman's capsule. Pedicels, which are special foot-like extensions, are attached to them and wrap around the glomerulus capillaries. Filtration slits are the little apertures seen on these pedicels. The filtration barrier that enables tiny molecules and ions from the blood to be filtered into the Bowman's capsule is made up of these filtration slits, glomerular endothelial cells, and the basement membrane. The kidneys' first stage in producing urine is the first filtration of blood.

The other options are incorrect. The artery that supplies the glomerulus with blood is called the afferent arteriole. Blood is removed from the glomerulus through the efferent arteriole, which is an artery. The Henle's loop, a U-shaped component of the nephron, aids in the concentration of urine. A portion of the water and solutes from the filtrate are reabsorb able by the distal convoluted tubule of the nephron.

13. Correct option: (B) When someone drinks lot of water, ADH release is suppressed

The pituitary gland stores antidiuretic hormone (ADH), also known as vasopressin, which is a hormone generated by the brain. In reaction to elevated blood osmolality, ADH is released. The concentration of solutes in a solution is measured by osmolality. The hypothalamus receives a signal to release ADH when blood osmolality rises. ADH then goes to the kidneys where it increases the water permeability of the collecting ducts. As a result, the kidneys are able to reabsorb more water from the urine, increasing its concentration.

Option A is inaccurate since the blood's osmolality rises when the body loses a lot of water through evaporation. The kidneys reabsorb more water and create more concentrated urine as a result of this increased ADH release. Because exposure to low temperatures does not induce the production of ADH, Option C is inaccurate. Because an increase in glomerular blood flow does not trigger the production of angiotensin II, Option D is untrue.

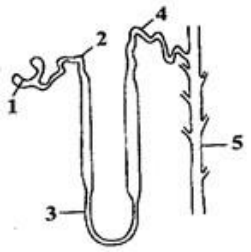
14. Correct option: (A) Uric acid can be excreted in almost solid form

The fact that uric acid may be expelled in a virtually solid state and is comparatively insoluble in water makes it advantageous to expel nitrogenous wastes in this form. This helps creatures with limited access to water because it enables them to preserve water by excreting waste that is concentrated. The main nitrogenous waste product emitted by

animals like birds, reptiles, insects, and certain other species that have trouble conserving water is uric acid.

Option B is inaccurate since little energy is needed to create uric acid. Option C is inaccurate because uric acid is not the first byproduct of the metabolic breakdown of acids. Because uric acid is not eliminated through the lungs, Option D is untrue. It is eliminated by way of the kidneys and bladder.

15. Correct option: (C) 2 and 3



Both the descending loop of Henle and the proximal convoluted tubule (PCT) include cells with many microvilli and mitochondria.

The PCT is the nephron's initial structure. The majority of the water, glucose, and amino acids in the filtrate are reabsorbed by it. The many microvilli seen in PCT cells improve the surface area for absorption. They also have a large number of mitochondria, which provide the process of absorption with energy.

The section of the nephron that descends into the kidney's medulla is known as the descending loop of Henle. It is in charge of making the pee more concentrated. Additionally lined with microvilli, the cells of the descending loop of Henle assist to improve the surface area for water absorption.

There are fewer microvilli and mitochondria in the Bowman's capsule, the distal convoluted tubule (DCT), and the collecting duct. This is due to the fact that they are not as effective in absorbing water and solutes.

16. Correct option: (D) a, b and c

a) The expulsion of urine stored in the urinary bladder is called micturition

This statement is true. The process of releasing urine from the bladder is known as micturition. It entails a synchronized set of actions, such as the internal urethral sphincter relaxing and the bladder muscles contracting to force urine out the body.

b) Urine is collected in the nephrons and flows into the ureters

The kidneys generate urine. After that, it is gathered in the renal pelvis and passes via the ureters. The muscular tubes known as ureters transport urine from the kidneys to the bladder.

c) Micturition can also be initiated voluntarily by contracting the abdominal muscle which applies pressure over the urinary bladder, activating the stretch receptors

The abdominal muscles can be consciously contracted to start urinating. The bladder experiences pressure as a result of the contraction of the abdominal muscles. The bladder wall's stretch receptors are activated by this pressure, which then sets off the micturition reflex.

17. Correct option: (C) Only i

The juxtaglomerular apparatus (JGA) is activated when the glomerular filtration rate (GFR) drops. The enzyme renin is released as a result of this activation. Angiotensin I is created through a sequence of events involving angiotensinogen that are triggered by renin. The angiotensin-converting enzyme (ACE) subsequently converts angiotensin I into angiotensin II. Aldosterone is secreted as a result of the powerful vasoconstrictor angiotensin II stimulating the adrenal cortex. In the kidneys, aldosterone encourages the reabsorption of water and sodium ions.

Antidiuretic hormone (ADH) promotes water reabsorption from the tubule's last sections, avoiding diuresis. Angiotensinogen in the blood is converted by renin to angiotensin I, and then to angiotensin II. Due to its potent vasoconstrictor properties, angiotensin II raises GFR and glomerular blood pressure. Vasodilation (dilation of blood vessels) brought on by ANF can lower blood pressure.

18. Correct option: (C) Both statements are correct

Statement 1 is correct. Osmoregulation is the process of maintaining the quantity of water and dissolved solutes in balance in the body. This is important for a number of reasons, including maintaining blood pressure, cell volume, and electrolyte balance.

Statement 2 is correct. The excretion of water and dissolved solutes is achieved through organs, such as the skin and the kidneys. The skin helps to regulate water loss through sweating, and the kidneys help to regulate water and electrolyte balance by filtering blood and producing urine.

The major organs involved in osmoregulation are the skin and kidneys, however they don't function independently. Water and electrolyte balance is also maintained by other organs such the liver, lungs, and intestines.

19. Correct option: (B) Only 1 and 3

The capillary network of the glomerulus' capillary network is divided into several branches, while the rubber tubing at B is not. As the glomerulus is a cluster of capillaries that filters blood into the nephron, this is a significant defect in the concept. To reflect this, the rubber tubing should be divided into several branches.

To symbolize the tangled tubules of the nephron, the stem of the thistle funnel (E) is not coiled. The nephron's convoluted tubules are in charge of reabsorbing water and other solutes from the filtrate. To depict this, curl the stem of the thistle funnel.

The other statements in the question are not accurate. The rubber tubing at B should be continuous, but it should contain small pores to allow the filtration of blood into the nephron.

20. Correct option: (D) PCT, DCT and Malpighian corpuscle

In the cortical part of the kidney are the Proximal Convoluted Tubule (PCT), Distal Convoluted Tubule (DCT), and Malpighian corpuscle (which includes the glomerulus and Bowman's capsule).

In the cortical area of the kidney, there is a Malpighian corpuscle. It consists of a Bowman's capsule, which encloses the glomerulus, and a glomerulus, which is a ball of capillaries.

The PCT, which is likewise found in the kidney's cortical area, receives the filtrate thereafter. The PCT reabsorbs the majority of the electrolytes, amino acids, and glucose in the filtrate.

The DCT, which is likewise situated in the kidney's cortical area, receives the filtrate after that. Some water and electrolytes are reabsorbed by the DCT from the filtrate.

21. Correct Matching is: a-T; b-R; c-U; d-Q; e-S; f-P

Answer:

a) Glomerulus - T) Ultrafiltration
b) Loop of Henle - R) Concentrated urine
c) Proximal convoluted Tubule - U) Reabsorption of glucose, amino acids
d) Distal convoluted tubule - Q) Regulate K, Na and pH+
e) Collecting duct - S) Absorption of water
f) Vasopressin - P) Urine formation

a) Glomerulus - T) Ultrafiltration

The glomerulus, a network of tiny blood arteries (capillaries) in the kidney, is crucial to the ultrafiltration process. The process of making urine begins with ultrafiltration, in which blood is filtered under pressure to separate bigger molecules—such as proteins and blood cells—from smaller ones, such as waste particles, ions, and water. The glomerulus functions as a filter that lets smaller molecules through while keeping bigger ones in the circulation.

b) Loop of Henle - R) Concentrated urine

The kidney tubules' Loop of Henle is a part that establishes a gradient of concentration in the tissue around it. Urine must be concentrated using this concentration gradient. The Loop of Henle aids in the reabsorption of water and ions when urine passes through the tubules, which helps to concentrate urine by eliminating extra water from the filtrate.

c)Proximal convoluted Tubule - U) Reabsorption of glucose, amino acids

An essential component of the nephron is the Proximal Convoluted Tubule (PCT), which is where crucial chemicals like glucose and amino acids are reabsorb able. To avoid losing important nutrients in the urine, these compounds are selectively reabsorbed back into the circulation from the tubular fluid.

d) Distal convoluted tubule - Q) Regulate K, Na and pH+

The Distal Convoluted Tubule (DCT) is in charge of controlling the body's ion balance, including the ratio of potassium (K+) to sodium (Na+). The DCT controls the acid-base balance and adjusts the amounts of these ions, both of which are essential for proper physiological operation.

e) Collecting duct - S) Absorption of water

The Collecting Duct, the nephron's last component, is essential for reabsorbing water. The amount of water reabsorbed back into the circulation by the collecting duct can be altered in response to the body's level of hydration and hormonal cues such antidiuretic hormone (vasopressin). Both the water balance and urine concentration are regulated by this mechanism.

f) Vasopressin - P) Urine formation

Antidiuretic hormone (ADH), also known as vasopressin, is a hormone produced by the hypothalamus and released by the pituitary. By adjusting the collecting ducts' permeability in reaction to the body's level of hydration, it manages the water balance. Vasopressin reduces urine production and aids in the formation of concentrated urine when it is released by increasing the reabsorption of water from the filtrate.

5. Life Process in Plants

Solutions

PHOTOSYNTHESIS

EXERCISE

1. Correct option: (C) Ananbolic, endergonic, oxidation – reduction process

Anabolic, endergonic, oxidation-reduction is the basis of photosynthesis. In this process, glucose and other organic molecules are used to transform light energy into chemical energy. In the same way that glucose is created during photosynthesis from carbon dioxide and water, anabolic refers to the process of creating complex molecules from simpler ones. Endergonic denotes that the process needs an energy input, which in this case is given by sunshine. Redox processes, often referred to as oxidation-reduction reactions, are essential to photosynthesis because they entail the transfer of electrons from water to carbon dioxide, which produces glucose and releases oxygen.

Other options are incorrect. Photosynthesis is an endergonic process, not an exergonic one, hence option A is erroneous. Photosynthesis is an anabolic process, not a catabolic one, hence option B is wrong. Due to the fact that photosynthesis is an oxidation-reduction process rather than a non-oxidation-reduction process, Option D is wrong.

2. Correct option: (B) 400 – 700 nm

PAR (Photosynthetically Active Radiation) has a wavelength range of 400–700 nm. The area of the electromagnetic spectrum known as PAR is essential for photosynthesis in plants. In order to power the process of photosynthesis, where light energy is transformed into chemical energy to fuel the production of glucose and other organic molecules required for plant growth and development, these wavelengths are absorbed by the pigments in plant cells, chiefly chlorophyll.

Option A is incorrect because it is the wavelength of ultraviolet light.

Option C is incorrect because it is the wavelength of infrared light.

Option D is incorrect because it is the wavelength of X-rays.

3. Correct option: (B) Humidity

The other factors, light intensity, temperature, and carbon dioxide concentration, affect the rate of photosynthesis.

The quantity of light that is available to the plant is known as light intensity. The rate of photosynthesis increases with increasing light intensity.

What determines how hot or cold anything is its temperature. Up until a certain point, the rate of photosynthesis increases with temperature. A temperature that is too high might harm the plant and hinder photosynthesis.

The level of carbon dioxide in the air is known as its concentration. The rate of photosynthesis increases with increasing carbon dioxide levels.

Although plant transpiration and water intake can be influenced by humidity, photosynthesis is not directly impacted.

4. Correct option: (B) Remove starch from leaves.

The process through which plants use sunlight to convert carbon dioxide and water into glucose and oxygen is known as photosynthesis. Plants create the kind of carbohydrate known as starch during photosynthesis. Photosynthesis cannot take place while a plant is in the dark, and the starch that is stored in the leaves will be depleted. In order to estimate the amount of starch generated by photosynthesis precisely, it is important that the leaves be free of starch before the experiment is carried out.

Option A is incorrect because it would stop photosynthesis if the leaves' chlorophyll were removed. Option C is incorrect because, although it happens considerably more slowly in the dark, photosynthesis may still take place there. Because it is impossible to guarantee that no photosynthesis took place, option D is incorrect. The plant may still do some photosynthesis even if it is kept in the dark.

5. Correct option: (A) C.B. Van Neil

Dutch-American scientist C.B. Van Neil is most recognized for his contributions to the photosynthesis equation. He put out the equation in 1931.

$CO_2 + 2H_2O \xrightarrow{\text{Light}} CH_2O + H_2O + O_2$

According to this equation, one molecule of carbohydrate (CH_2O), one molecule of oxygen (O_2), and one molecule of water are created for every molecule of carbon dioxide (CO_2) and two molecules of water (H_2O) that are consumed.

The 17th-century description of photosynthesis attributed to von Helmont is considered to be the first. The discovery that plants create oxygen during photosynthesis is attributed to Jan Ingenhousz. They also used wrong equations for photosynthesis, though. The function of chloroplasts in photosynthesis was first identified by Engelman. He did not, however, offer a photosynthesis equation.

6. Correct option: (C) Starch

Starch may be detected using iodine tests. This is due to the fact that when iodine reacts with starch molecules, a blue-black complex is created.

This is possible because the iodine molecules can accomodate the starch molecules' helical configuration.

An iodine solution is applied to a sample, such as a plant leaf or a piece of food, that may contain starch as part of the test. A trustworthy visual signal for the presence of starch in the tested sample is the color change from red to blue-black when starch is present.

Glucose, light, and carbon dioxide are not able to form a complex with iodine. Therefore, iodine cannot be used to test for the presence of these substances.

7. Correct option: (C) Joseph priestly

The statement "Plants restore to the air whatever breathing animals and burning candles remove" was hypothesized by Joseph Priestley. He carried out a series of experiments in 1772 in which he enclosed a mouse in a container with a burning candle. The candle rapidly burned out once the mouse passed away. The mouse survived and the candle kept burning when he added a mint plant to the same container. This prompted Priestley to postulate that plants replace the air pollution that animals and burning candles remove by breathing.

The rule of conservation of mass is thought to have been discovered by French scientist Lavoisier. The first description of photosynthesis is attributed to a Flemish physician and chemist named von Helmont in the 17th century. His assumption regarding the procedure, nevertheless, was incorrect. Ingenhousz showed that plants produce oxygen during photosynthesis, which is the process by which they convert sunlight, water, and carbon dioxide into food.

8. Correct option: (B) Jan Ingenhousz

The late 1770s saw a series of experiments by Dutch physician and physicist Ingenhousz to investigate the mechanism of photosynthesis. In one experiment, he exposed a hydrilla sprig to sunlight in a sealed water container. He noticed bubbles forming around the plant and recognized the bubbles as being oxygen. He also demonstrated how the plant's ability to produce oxygen was impaired when it was left in the dark.

The investigations of Ingenhousz provide convincing proof that plants create oxygen during photosynthesis. His discoveries contributed to our growing understanding of this critical procedure, which is necessary for life on Earth.

The part chloroplasts play in photosynthesis was identified by Engelman. In the 17th century, von Helmont gave the earliest account of photosynthesis. The cell hypothesis has benefited from Van Sachs' ideas. On photosynthesis, he didn't perform any tests.

9. Correct option: (C) Engelman

The experimental material for T.W. Engelmann's groundbreaking experiment was reticulated filamentous green algae called Cladophora suspended with aerobic bacteria. He lighted the bacterial-algae solution by using a prism to break up light into its individual colors, allowing the detection of oxygen-evolving sites. When exposed to red and blue light wavelengths, these aerobic bacteria gathered in greater numbers. As a result, this study created the first action spectrum of photosynthesis, which shows graphically how the rate of photosynthetic activity and the wavelength of light relate to one another.

Experiments by Jan Ingenhousz shown that plants emit oxygen during photosynthesis. The Calvin cycle, which describes how plants utilize carbon dioxide to make glucose, was discovered by Melvin Calvin. In 1817, Pelletier, a French chemist, and Joseph Caventou successfully isolated chlorophyll.

10. Correct option: (A) Pelletier and caventou

In 1817, French chemists Pelletier and Caventou successfully extracted chlorophyll from plants. The Greek words chloros (green) and phylon (leaf) were used to give the pigment its name, chlorophyll.

Chlorophyll was painstakingly collected from leaves by Pelletier and Caventou. In order to dissolve the chlorophyll, they first crushed the leaves to make a pulp, which they then combined with alcohol. To get rid of any solid components, the resultant solution was filtered. They produced a concentrated chlorophyll extract by removing the alcohol from the filtered solution.

Experiments by Jan Ingenhousz have shown that plants emit oxygen during photosynthesis. The research of Engelman demonstrated the involvement of two photosystems in photosynthesis. Botanists Von Helmont and Van Sachs both failed to isolate chlorophyll. Chemical investigations by Lavoisier and Joseph Priestley demonstrated that plants take up carbon dioxide and release oxygen during photosynthesis. They failed to separate chlorophyll, though.

11. Correct option: (A) Daniel I. Arnon

By effectively removing chloroplasts from plant cells in 1954, Daniel I. Arnon made an important contribution to the field of plant biology. His procedure required gently mechanically disrupting the plant cells, extracting the chloroplasts from the cell debris using differential centrifugation, and then reassembling the plant cells. A greater comprehension of the structure, functionality, and significance of chloroplasts in photosynthesis resulted from this ground-breaking accomplishment, which enabled the study of chloroplasts in isolation.

Von Heil, Von Sachs, and Darwin did not remove chloroplasts from plant cells.

12. Correct option: (B) Photolysis

The crucial process during photosynthesis in which water molecules are divided or broken down into their component constituents, namely oxygen, protons (H+ ions), and electrons, is referred to as photolysis. The chlorophyll molecules' absorption of light energy triggers this process in the thylakoid membranes of the chloroplasts. The photosynthetic electron transport chain relies on high-energy electrons from water

molecules to produce ATP and reduced NADPH, which are both necessary for the synthesis of glucose and other energy-dense organic molecules. In plants and other photosynthetic organisms, photolysis is a basic mechanism that helps convert light energy into chemical energy.

Emulsification is the process of dissolving big fat droplets into smaller ones. Electrolysis is the process of breaking down a material by running an electric current through it. By adding water, a molecule is broken down by hydrolysis.

13. Correct option: (B) ATP, NADPH

The assimilatory capabilities that result from photosynthesis's final light reaction. Chlorophyll and other pigments absorb light energy during the light reaction, creating high-energy molecules as a result. Through a process known as photophosphorylation, energy from light is utilized to add a phosphate group to the energy-dense molecule adenosine diphosphate (ADP), resulting in the production of ATP. As electrons are transported from water molecules to NADP+ ions during the light reaction, NADPH, a molecule that transports high-energy electrons, is also produced.

O2 is also produced during the light reaction, but it is not an assimilatory power. O2 is released into the atmosphere as a waste product of photosynthesis.

14. Correct option: (A) Less than even one thousand of a second

In a chloroplast, photosynthesis takes place through two distinct processes called the light reaction and the dark reaction. The stroma is where the dark reaction occurs, whereas the thylakoid membranes are where the light response occurs. The dark reaction is far slower than the light reaction, and there is barely a thousandth of a second between them.

The light reaction generates ATP and NADPH using light energy. The dark process uses these molecules after that to turn them into glucose. The dark reaction doesn't rely on light energy directly; hence it moves more slowly.

15. Correct option: (C) Blue and red

The primary pigment involved in photosynthesis in plants, chlorophyll, absorbs blue and red light more efficiently than other hues. This is so because the visible spectrum's blue and red portions are where chlorophyll's absorption spectrum peaks. The graph of a pigment's absorption spectrum demonstrates how much light of a certain wavelength is absorbed by the pigment.

Chlorophyll reflects green light, reducing its absorption by the pigment. Because of this, plants seem green. The other choices are false. Chlorophyll absorbs blue and red light more efficiently than violet and green light. Chlorophyll can absorb red and green light, but not as much as it can absorb blue and red light.

16. Correct option: (B) C6H12O6

The second stage of photosynthesis is known as the dark reaction or Calvin cycle. Carbon dioxide (CO2) is fixed and transformed into glucose and other carbs during this phase. The first product of the Calvin cycle is phosphoglycerate (PGA), which is subsequently transformed into glucose, a six-carbon sugar molecule, through a sequence of enzyme-mediated processes. As a key source of energy for the plant and the necessary fuel for growth, metabolism, and many cellular functions, glucose is an essential byproduct of the dark reaction.

The other choices are false. The Calvin cycle's intermediate product is PGA. The Calvin cycle's first component is CO2. The starting material for the first phase of the Calvin cycle is ribulose biphosphate.

17. Correct Matching is: a- R, b-Q, c-T, d-P, e-S

a) Light reaction - R) Grana

The granum, a stack of thylakoid membranes, is where light response occurs. Chlorophyll and other photosynthetic pigments that absorb light energy are found in the thylakoid membranes. In order to create oxygen gas, light energy is utilized to divide water molecules.

b) Dark reaction - Q) Stroma

The fluid surrounding the thylakoids, the stroma, is the site of the dark reaction. The dark reaction converts carbon dioxide into glucose by using the ATP and NADPH generated in the light reaction as energy.

c) Photolysis - T) Release of oxygen

The splitting of water molecules by light energy is known as photolysis. This procedure causes the emission of oxygen gas during the light response.

d) Photon - P) Smallest energy of light

The smallest type of light energy is called a photon. Chlorophyll and other photosynthetic pigments absorb photons.

e) Stomata - S) Exchange of gases

Small pores called stomata in the leaves allow gases to move back and forth. When a plant needs to absorb carbon dioxide and release oxygen, stomata open.

RESPIRATION IN PLANTS

EXERCISE

1. Correct option: (B) Glycolysis

In the cytoplasm of cells, glucose is broken down into two molecules of pyruvate during a metabolic process known as glycolysis. A tiny quantity of ATP and NADH are produced during this process; these molecules can then be utilized to produce more energy. Without the use of oxygen, pyruvate is transformed into a number of byproducts in anaerobic circumstances, such as during fermentation. Unaffected by the presence of oxygen, cells may produce energy through glycolysis, a key metabolic route.

The other options are incorrect. PPP, TCA cycle, and ETS are all part of aerobic respiration, but they do not occur in anaerobic respiration.

2. Correct option: (C) Fermentation

Cells use glycolysis to create energy during fermentation, which is followed by non-oxygenated processes. For organisms that cannot rely on aerobic respiration because of a lack of oxygen or other limiting circumstances, this mechanism is especially crucial. Microorganisms like yeast use fermentation to make energy and regenerate NAD+ from NADH, which is necessary for maintaining glycolysis.

By heating food to a high temperature for a brief period of time, the pasteurisation procedure renders microorganisms inert. Microorganisms break down organic substances to create decay. The breakdown of organic materials by bacteria and fungus is called putrefaction.

3. Correct option: (B) H2O

H2O, or water, is not a product of fermentation. Fermentation is a metabolic process that occurs in the absence of oxygen, converting sugars into other substances such as alcohol, CO2, and ATP. During fermentation, yeast or bacteria break down sugar molecules into alcohol and CO2, releasing energy in the form of ATP. Water, on the other hand, is not directly produced during this process. Instead, it is a byproduct of other metabolic reactions or is present as a solvent in the fermentation environment. Therefore, H2O is not considered a product of fermentation."

4. Correct option: (C) Latin

It comes from the Latin word "respirare," which translates to "to breathe." The biological process by which living things exchange gases—inhaling oxygen and exhaling carbon dioxide—is accurately described by this word. The processes of inhaling and exhalation are all included in the concept of respiration, as are the cellular metabolic activities that use oxygen to create energy-rich molecules like adenosine triphosphate (ATP) and release carbon dioxide as waste.

5. Correct option: (B) Fermentation

The process of fermentation, particularly yeast fermentation, includes the anaerobic respiration of sugars found in the dough to produce alcohol and carbon dioxide (CO_2). Small bubbles are formed in the dough as a result of the trapped carbon dioxide gas produced during this process.

These bubbles grow as the carbon dioxide is produced, which causes the dough to rise and swell. The characteristically light and airy texture of bread is a result of this occurrence. Additionally, the alcohol created during fermentation dissipates as the bread is baking, leaving just the delicious flavor and scent of freshly baked bread.

The transfer of molecules from a high-concentration area to a low-concentration area is known as diffusion. Distillation is the procedure of boiling material and condensing the vapours to separate the different components. A liquid turning into a gas is called evaporation.

6. Correct option: (A) Lavoisier

Experiments by Antoine Lavoisier demonstrated that plants emit oxygen during photosynthesis. In addition, he said that "respiration is a combustion process" in a compilation from 1783. Here, oxygen not only joins with carbon but also with hydrogen in a very sluggish process.

The man who is credited with discovering oxygen is Joseph Priestley. Engelman demonstrated the involvement of two photosystems in photosynthesis. Through his research, Jan Ingenhousz demonstrated how plants use photosynthesis to take up carbon dioxide and expel oxygen. They didn't do any respiration-related tests.

7. Correct option: (D) All the above

Gas exchange in plants takes place through stomata, the surface of roots, and lenticels on stems.

A) Stomata: Stomata are tiny pores that are typically located on the surfaces of leaves and allow gases to enter and leave the plant. They control how much CO_2 and O_2, which are generated as byproducts of photosynthesis, are exchanged with the environment.

B) Surface of roots: Although to a lesser extent, roots may also exchange gases. The ability of the roots to absorb oxygen from the soil around them is crucial for the root cells' metabolic functions.

C) Lenticles on stem: Small pores or apertures called lenticles can be seen on the stems of several plants, especially woody plants like trees. They permit the entry of O_2 and the release of CO_2 generated during cellular respiration.

8. Correct option: (A) Photosynthesis

Photosynthesis takes place in plants when there is sunlight present during the day. Plants use light energy, typically from the sun, during the process of photosynthesis to change carbon dioxide and water into glucose and oxygen. This process generates the oxygen we breathe and supplies energy for the development and growth of the plant.

Due to the need for light energy to power the processes involved, photosynthesis normally happens more often during the daytime when sunlight is available. In

contrast, respiration—the continual process of releasing energy by dissolving glucose—takes place constantly both throughout the day and the night. However, during the day, photosynthesis may have a greater effect on the plant's total energy balance than respiration.

TRANSPORT IN PLANTS

LEVEL-I

1. Correct option: (B) Solvent

A semi-permeable membrane is one that lets solvent molecules flow through but not solute molecules. Solvent molecules are the molecules that dissolve the solute molecules, whereas solute molecules are the compounds that are dissolved in a solution.

In osmosis, water molecules pass across a semi-permeable membrane from a location of high-water concentration to a region of low water concentration. This is due to the constant movement of water molecules from areas of high concentration to areas of low concentration.

Osmosis uses a semi-permeable membrane that lets water molecules through but prevents solute molecules from doing the same. This explains why the solute molecule concentration is constant on both sides of the membrane.

2. Correct option: (C) Hypertonic

A plant cell's cytoplasm shrinks as a result of water loss, a process known as plasmolysis. This happens when a cell is positioned in a solution with a greater solute concentration than the cell. The cytoplasm shrinks and the cell membrane separates from the cell wall as a result of the solutes' osmotic water removal from the cell.

A solution that contains more solutes than the cell does is said to be hypertonic. Therefore, a hypertonic salt solution is required to start cell plasmolysis.

In an isotonic solution, plasmolysis would not happen since there would be no net wate r transport into or out of the cell.

Osmosis caused the cytoplasm to enlarge and the cell membrane to break when a hypot onic solution of water entered the cell. None of the above is also incorrect, as plasmolysis can only occur when the cell is placed in a hypertonic solution.

3. Correct option: (B) Increase

Increases in wall pressure, commonly referred to as turgor pressure, follow increases in turgidity. Turgor pressure is the force that water within a plant cell applies to its cell wall. The cell becomes turgid when water enters it through osmosis, forcing the cell membrane to push up against the cell wall. The stiffness and structural integrity of plant cells are preserved by this increased pressure.

Decrease is inaccurate because as the turgidity of the cell grows, the wall pressure rises. Because the wall pressure does not change along with the cell's turgidity, the word "fluctuate" is erroneous. Not changing is incorrect since the cell's turgidity causes the wall pressure to rise.

4. Correct option: (A) Endosmosis

Endosmosis causes water to enter a cell when it is put in a solution that is less concentrated than the cell's sap. Endosmosis is a form of osmosis when water molecules pass through a selectively permeable membrane from an area of lower solute concentration (in this example, the solution) to an area of higher solute concentration (inside the cell). To balance the solute concentration on each side of the membrane, this movement takes place. Water will migrate into the cell and essentially cause it to swell and expand since the solution outside the cell has a lower concentration of solutes than the cell's interior.

Diffusion is the movement of molecules from a region of high concentration to a region of low concentration. Imbibition is the absorption of water by a solid. Plasmolysis is the shrinking of a cell due to the loss of water.

5. Correct option: (D) Chloroplast

Having chloroplasts sets guard cells apart from epidermal cells. Photosynthesis is carried out by chloroplasts, specialized organelles that are present in plant cells. The guard cells, a pair of specialized cells that are present in the epidermis of plant leaves and stems, control the opening and shutting of stomata, which are tiny openings that permit gas exchange and water management. Because they can photosynthesize and generate energy-rich molecules like ATP that are required for the active transport of ions and water, chloroplasts are essential for guard cells' ability to control stomatal aperture. This influence ultimately affects the turgor pressure and shape of the guard cells, which controls stomatal opening and closing.

The other options are incorrect. Mitochondria are found in all cells, including guard cells and epidermal cells. Vacuoles are also found in all cells, but they are larger in guard cells than in epidermal cells. Cell walls are found in all plant cells.

6. Correct option: (D) Guard cells

In the epidermis of plant leaves and stems, there are specialized cells called guard cells. Stomata, tiny pores that enable the exchange of gases and water vapour between the plant and its surroundings, are controlled in their opening and shutting. The turgor pressure inside the guard cells regulates the morphology of the guard cells and the opening of the stomata. The guard cells enlarge and become turgid as they absorb water by osmosis, which opens the stomatal hole. On the other hand, water loss causes guard cells to contract and the stomatal hole to close. It is essential for plants to be able to regulate their water loss and gas exchange in order to survive and adapt to changing environmental circumstances.

The other options are incorrect. Epidermal cells do not control the opening and closing of stomata. Palisade cells and spongy parenchyma cells are not involved in the opening and closing of stomata.

7. Correct option: (D) Imbibition

Imbibition is the process through which solids like colloids absorb water and swell up as a result. The capillary action and adhesion forces between the water molecules and the surface of the solid substance are to blame for this phenomenon. Imbibition is frequently noticed in porous materials including seeds, wood, and gel-like things. The expansion of wood after being submerged in water, the swelling of seeds during germination, and the uptake of water by substances like cotton or paper are all caused by it. Water molecules are pulled into the voids of the solid matrix during this process, causing the substance to swell and expand.

The other options are incorrect. Plasmolysis is the shrinking of a cell due to the loss of water. Diffusion is the movement of molecules from a region of high concentration to a region of low concentration. Osmosis is the movement of water from a region of low solute concentration to a region of high solute concentration through a semipermeable membrane.

8. Correct option: (D) Ascent of sap

Water moves upward in plants, typically through xylem vessels, and this is referred to as the ascent of sap. The processes that propel this process include capillarity, cohesion, and transpiration. Water molecules may travel through the xylem vessels' tiny tubes thanks to capillary action, adhesion, and cohesion, while cohesion makes sure that water molecules stick together and enables the upward movement of a

continuous column of water. Water is drawn up from the roots as a result of transpiration, the loss of water vapour from the stomata of leaves. Together, these activities allow plants to move water from their roots to various sections of the plant, supporting critical functions including photosynthesis and preserving the plant's turgidity and structure.

The other options are incorrect. Translocation is the movement of food from the leaves to the other parts of the plant. Suction is the force that pulls water up from the roots. Flight is not a biological process in context of plants.

9. Correct option: (B) The assertion and reason are true and the reason is the correct explanation of the assertion.

The assertion is that the movement of materials inside the phloem is bi-directional, both upwards and downwards. This is true. Food and other organic materials are transported throughout the plant through the vascular tissue known as phloem. A pressure gradient, which may be produced by a multitude of processes including photosynthesis, respiration, and growth, is what propels the flow of materials in the phloem. Depending on the demands of the plant, the phloem can transfer materials both upwards and downwards.

The reason given is that the movement of molecules inside the xylem is unidirectional, always upwards. This is also true. A vascular tissue known as the xylem carries water and minerals from the roots to the leaves. Transpiration, or the evaporation of water from the leaves, propels the transport of water and minerals in the xylem. Water is drawn up from the roots by the transpirational pull, which induces a negative pressure in the xylem.

So, while both the assertion and the reason are true, the reason is not the correct explanation of the assertion.

10. Correct option: (B) Capillary water

Water that is accessible for root absorption by soil is referred to as capillary water. Due to capillary action, which allows water to rise in small spaces, it is kept in the soil pores in opposition to the pull of gravity. The cohesive and sticky characteristics of water molecules help to assist this capillary action.

The other options are incorrect. Gravitational water, on the other hand, is the excess water that drains downward due to gravity after the soil becomes saturated. Hygroscopic water is the water that strongly adheres to soil particles and is not available for plant uptake. Combined water is generally found in minerals and is not relevant to soil water availability for plants.

Therefore, capillary water is the primary source of moisture that plants can access for their growth and survival.

11. Correct option: (A) Relative humidity

The most significant element in the control of transpiration is relative humidity. Plants exhale water vapour into the atmosphere through a process known as transpiration. The term "relative humidity" describes the quantity of moisture in the air as compared to the maximum amount that it might contain at a specific temperature. The gradient between the moist air inside the leaf and the surrounding air is lessened by high relative humidity since the air is already saturated with moisture. Transpiration is driven by this gradient, therefore when relative humidity levels are high, transpiration rates drop to limit excessive water loss. In contrast, when relative humidity is low, the gradient is steep, which causes increased transpiration rates as the plant tries to keep its water balance.

The other factors, temperature, light, and wind, also affect the rate of transpiration. However, their effects are much smaller than the effect of relative humidity.

12. Correct option: (B) Diffusion

Diffusion is the transfer of free atoms from a higher concentration to a lower concentration. Particles, like atoms or molecules, travel haphazardly down their concentration gradient, from places of greater concentration to areas of lower concentration, as part of the fundamental process known as diffusion. This movement results from the random movement of particles and eventually equalizes concentrations within a particular region.

The other options are incorrect. Osmosis is the movement of water from a region of low solute concentration to a region of high solute concentration through a semipermeable membrane. Endosmosis is the movement of water into a cell through a semipermeable membrane. Exosmosis is the movement of water out of a cell through a semipermeable membrane.

13. Correct option: (D) Transpiration

Plant wilting happens as a result of excessive transpiration. Plants release water vapour from their leaves into the atmosphere through a process called transpiration. A water deficit develops inside the tissues of the plant when transpiration rates are higher than the plant's capacity to absorb water from the soil. This causes the cell turgor pressure to drop, which weakens the plant and causes the leaves to droop or wilt. This visible stress indicator occurs when the physiological processes of the plant that regulate water balance are unable to keep up with the pace of water loss. While respiration, photosynthesis, and absorption are vital plant activities, an imbalance between transpiration and water uptake is the main contributor to wilting.

14. Correct option: (D) Closing and opening of stomata

Stomata are controlled by guard cells to close and open. To exchange gases (including oxygen and carbon dioxide) and water vapour with the environment, plants have tiny holes called stomata that are located on the surface of their leaves and stems. Each stomatal hole has a guard cell, which is a specialized cell. The size of the stomatal opening is controlled by guard cells, which alter their form. To facilitate gas exchange and water loss through transpiration, guard cells' stomata open and they become turgid when they osmotically acquire water. The guard cells' shrinkage and closing of the stomatal hole, on the other hand, reduce water loss by causing water loss from them.

While photosynthesis, light intensity, and leaf color are important aspects of plant physiology, guard cells' primary role is in regulating stomatal aperture for optimal water and gas exchange.

15. Correct option: (C) Osmosis

Osmosis is the process through which water diffuses across a membrane that is only partially permeable. Osmosis is a particular form of passive transport in which water molecules travel across a semi-permeable membrane from an area of lower solute concentration to an area of greater solute concentration. The goal of this movement is to balance the solute content on each side of the membrane. Osmosis is crucial for many biological processes in living things, including maintaining the water balance in cells.

While diffusion is a broader term referring to the movement of particles (not limited to water) from areas of higher concentration to lower concentration, imbibition is the absorption of water by solids causing them to swell, and translocation refers to the transport of nutrients in plants, which is different from the process of osmosis.

16. Correct option: (A) Exosmosis

Exosmosis is the cause of plasmolysis. When water is lost from plant cells as a result of an external environment that is too hypertonic, a condition known as plasmolysis takes

place. The protoplast (cell contents) of the cell contracts away from the cell wall in this case as a result of water leaving the cell through osmosis. Plasmolysis is the name for this. When a cell is put in a hypertonic solution, the process known as exosmosis particularly refers to the outward flow of water from the cell that causes this shrinking effect.
Endosmosis, on the other hand, refers to the inward movement of water into a cell when it is placed in a hypotonic solution. Osmosis is the overall process of water movement across a semi-permeable membrane, while adsorption is the adherence of molecules to the surface of a substance, which is different from the processes involved in plasmolysis.

LEVEL – II

1. Correct option: (D) (a), (b) and (d) are correct
The proton gradient is created by the splitting of water molecules in the thylakoid membrane. The protons are then pumped into the lumen of the thylakoid, creating a high concentration of protons in the lumen. The ATP synthase enzyme is located on the stroma side of the membrane. The protons flow down their concentration gradient through the ATP synthase enzyme, which creates ATP. In the context of ATP formation in the chloroplast during photosynthesis, (a) protons accumulate in the thylakoid lumen as a result of the electron transport chain and water-splitting reactions, (b) water molecules are indeed split on the inner side of the thylakoid membrane to release protons and oxygen, and (d) NADP reductase, an enzyme involved in electron transfer, is located on the stroma side of the thylakoid membrane. These processes collectively contribute to the establishment of a proton gradient that drives ATP synthesis through ATP synthase.

2. Correct option: (A) Chloroplast, mitochondria, peroxisome
When a cell is photo respiring, its organelles should be arranged in the following order: A) chloroplast, mitochondria, and peroxisome. When the enzyme RuBisCO accidentally binds oxygen instead of carbon dioxide during the Calvin cycle of photosynthesis, photorespiration—a metabolic process—occurs in plants. As a result of this process, a molecule is created that must be processed by metabolism. The first stage of photorespiration occurs in the chloroplasts, where oxygen is absorbed and carbon dioxide is expelled. The metabolic byproducts are then transported to the mitochondria, where further processes take place. Finally, the peroxisome is involved in the detoxification of some photorespiration byproducts.
These organelles' sequential engagement enables plants to control the unwanted effects of photorespiration and preserve metabolic equilibrium.

3. Correct option: (A) b and d
In photosynthesis, statement b is correct because cyclic photophosphorylation involves only Photosystem I (PS I), as Photosystem II (PS II) is not involved in this process. Statement d is also correct because stroma lamellae lack both PS II and the enzyme NADP reductase. Cyclic photophosphorylation, as mentioned in statement c, does not result in the synthesis of NADPH2; instead, it generates ATP. Statement a is incorrect because the Z scheme of the light reactions involves both Photosystem I and Photosystem II, not just PS I. Therefore, options b and d are the only ones that include correct statements.

4. Correct option: (B) Excitement of chlorophyll molecules due to absorption of light
The first step for the initiation of photosynthesis is the excitement of chlorophyll molecules due to the absorption of light, as indicated by option B. The activation of chlorophyll molecules within chloroplasts as a result of light energy absorption causes the generation of high-energy electron states. This first action sets off a series of

processes that eventually power the light-dependent reactions of photosynthesis, which include the flow of electrons, production of ATP, and the breaking apart of water molecules (photolysis) to liberate oxygen.

Options A, C, and D are not the initial steps for photosynthesis; photolysis of water, ATP formation, and glucose formation occur as subsequent events within the broader process of photosynthesis, following the initial absorption of light energy by chlorophyll molecules.

5. Correct option: (B) A – ii, B – iii, C – iv, D – i

Column A	**Column B**
A. Grana of chloroplast	ii. Light reaction
B. Stroma of chloroplast	iii. Dark reaction
C. Cytoplasm	iv. Glycolysis
D. Mitochondrial matrix	i. Krebs cycle

A. Grana of chloroplast - ii. Light reaction

The initial phase of photosynthesis is the light response. The process occurs in the grana of the chloroplast, where solar energy is converted to ATP and NADPH.

B. Stroma of chloroplast - iii. Dark reaction

The second phase of photosynthesis is known as the dark reaction. The light reaction generates ATP and NADPH, which are used to synthesize glucose in the stroma of the chloroplast where carbon dioxide is fixed.

C. Cytoplasm - iv. Glycolysis

The initial stage of cellular respiration is glycolysis. The breakdown of glucose into pyruvic acid and the release of energy in the form of ATP occur in the cytoplasm of the cell.

D. Mitochondrial matrixi. i. Krebs cycle

The second stage of cellular respiration is the Krebs cycle. Pyruvic acid is further broken down and energy is released in the form of ATP in the mitochondrial matrix.

6. Correct option: (D) 1 – Photosystem II, 2 – e- acceptor, 3 – Photosystem I, 4 – NADPH+ H+

This sequence represents the pathway of light reactions in photosynthesis. In this process, Photosystem II (1) captures light energy and initiates the flow of electrons, which are passed to an electron acceptor (2). The excited electrons are then transferred through an electron transport chain. Meanwhile, Photosystem I (3) also capture light energy and further boosts the energy of the electrons, which are then transferred to NADP+ to form NADPH, along with the addition of protons (H+) from the surrounding environment. This NADPH serves as a carrier of high-energy electrons for the later stages of photosynthesis. This sequence is a crucial part of the light-dependent reactions, where light energy is harnessed to generate ATP and NADPH, essential for the production of sugars in the Calvin cycle.

7. Correct option: (A) Respiration rate is faster than animals in roots, stems and leaves

The rate of respiration in plants is often lower than that in mammals. The pace at which an organism utilizes oxygen and releases carbon dioxide is known as its respiration rate. Temperature, light intensity, and water accessibility are only a few of the variables that affect a plant's respiration rate.

The pace at which plants breathe also depends on the availability of water. Water is necessary for plants to breathe. Plants breathe less quickly when there is a lack of water.

Options B, C, and D are correct statements: plants utilize the oxygen released during photosynthesis for respiration, the loose packing of parenchyma cells facilitates respiration, and there is indeed limited transport of gases between different plant parts.

8. Correct option: (D) Only d

The statement d) ETS (Electron Transport System) gives 11ATP for each Krebs cycle is unrelated to fermentation. This remark alludes to the non-directly related aerobic respiration process, which produces energy through the electron transport system. In fact, statements a, b, and c are all related to fermentation: a) fermentation is responsible for a portion of the breakdown of glucose; b) only 2 ATP is produced as a net yield from the conversion of glucose to pyruvic acid; and c) NADH is oxidized more slowly in fermentation than it is during aerobic respiration.

9. Correct option: (B) The assertion and reason are true, and the reason is the correct explanation of the assertion.

Assertion: Glycolysis occurs in cytoplasm.

This is true. Glycolysis is the first step in both aerobic and anaerobic respiration. It is a series of chemical reactions that break down glucose into pyruvate. The enzymes for glycolysis are found in the cytoplasm of the cell.

Reason: Enzymes for glycolysis are found in cytoplasm. It is common in aerobic/anaerobic respiration.

This is also true. The enzymes for glycolysis are found in the cytoplasm of the cell. Glycolysis is a common process in both aerobic and anaerobic respiration.

The reason is the correct explanation of the assertion because the enzymes for glycolysis are found in the cytoplasm, which is where glycolysis occurs.

10. Correct option: (C) The rate of respiration is reduced

To reduce the rate of respiration, green-colored apples are frequently kept at low temperatures, as in a refrigerator. All living cells, including those in fruits, go through respiration, a metabolic process in which carbohydrates are broken down to produce energy. The metabolic processes of apple cells, including respiration, are slowed down by lowering the temperature. This aids in preserving the apples, preventing over-ripening or premature spoilage, and preserving their quality and freshness for a longer time. Apples should be kept in the refrigerator to minimize the rate of respiration, even if lower temperatures and less sun exposure may also slow down photosynthesis.

Low temperatures have little effect on photosynthesis's pace. Low temperatures can somewhat restrict photosynthesis, but not entirely. Low temperatures have little effect on the rate of photosynthesis, but they do slow down the rate of respiration.

11. Correct option: (B) Only c

Chemical energy is broken down during respiration to release energy: Respiration is a process that breaks down food molecules to release energy. This energy is used by the cell to carry out its activities.

A and D are not totally true. Statement an is untrue since respiration involves both catabolic and anabolic processes, not simply catabolic ones. Statement d is also untrue since fungi are heterotrophic animals that cannot manufacture their own food through sunlight; instead, they receive energy by decomposing organic substances through procedures like respiration. Because food oxidation releases energy rather than reduction, statement b is not part of the correct answer.

12. Correct option: (D) Ethyl alcohol, CO_2, or lactic acid

Anaerobic respiration occurs in the absence of oxygen and is a less efficient process compared to aerobic respiration. In different organisms, anaerobic respiration can lead to the production of various byproducts depending on the type of metabolism and the

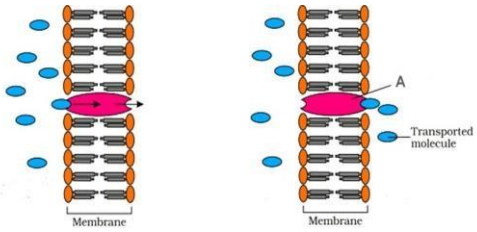
specific conditions. Depending on the organism, anaerobic respiration's byproducts change. In yeast, anaerobic respiration results in the production of carbon dioxide and ethanol. Lactic acid is produced in muscle cells by anaerobic respiration.

13. Correct option: (A) Cu, Mn, Fe

These elements are important cofactors in several enzymes and proteins involved in both photosynthetic and mitochondrial electron transport, which explains why they have a substantial influence on both of these processes. Three minerals—copper (Cu), manganese (Mn), and iron (Fe)—are crucial building blocks of several enzymes and proteins involved in both photosynthesis and mitochondrial electron transport. For example, iron is a vital component of cytochromes involved in the electron transport chain of both photosynthesis and cellular respiration, while manganese is necessary for the water-splitting complex in photosystem II during photosynthesis. In these processes, copper is also necessary for certain enzyme activity connected to electron transport.

Therefore, deficiencies in these micronutrients can greatly hinder the efficiency of both photosynthetic and mitochondrial electron transport, leading to significant impacts on plant growth and vital functions.

14. Correct option: (B) Facilitated diffusion; A – Transport protein



Molecules penetrate a cell membrane by a process known as "facilitated diffusion" that is aided by transport proteins. Some compounds can only travel across the membrane with the help of the transport proteins, which are specialized for those molecules.

Transport protein is the entity designated as structure A. For the reason that it is a transmembrane protein, it crosses the cell membrane. For the chemical it is carrying, the transport protein has a particular binding site. A molecule can travel across a membrane when it attaches to a transport protein, which alters shape to allow the molecule to do so.

15. Correct option: (B) Only b

The statement that is not true is b. Maximum during the day and minimum during night. Actually, when there is less transpiration at night, root pressure is produced. Negative pressure is produced in the xylem vessels by the water that is lost from the leaves throughout the day when transpiration is high. The root pressure is offset by this negative pressure, which has the potential to possibly force water to flow down the xylem channels.

Other statements are true. Root pressure is a positive hydrostatic pressure that is generated in the roots. The magnitude of root pressure is typically 1-2 bars. Root pressure is developed due to the metabolic activity of the roots. The roots absorb water and minerals from the soil, and this process uses energy. The energy released by this process creates the positive hydrostatic pressure that is root pressure.

16. Correct option: (B) Only C is incorrect

The answer to the question is B) Only C is incorrect. This is so that it is not true that cobalt chloride paper turns blue as it absorbs water as stated in option c. In fact, when cobalt chloride paper absorbs water, it turns pink. Cobalt chloride paper has a characteristic that is used to gauge how quickly plants transpire. According to the

cohesion tension transpiration pull model, transpiration acts as the primary force pulling water upward through the plant. The majority of the water that reaches the leaves is lost through transpiration, with only a little portion being utilized for plant growth and photosynthesis.

17. Correct option: (B) Guard cells are always surrounded by subsidiary cells

For stomatal apparatus, it is not true that guard cells are always encircled by subsidiary cells. While it is true that subsidiary cells are frequently located in the vicinity of guard cells, there are some circumstances in which this is not the case. Some plants, especially monocots, may only have a pair of guard cells and no subsidiary cells in their stomatal system. Different plant species exhibit this variety in subsidiary cell arrangement, which is not a common feature of the stomatal system.

The other statements about the stomatal apparatus are all true. Guard cells invariably possess chloroplasts and mitochondria, which they need to carry out photosynthesis. Stomata are involved in gaseous exchange, allowing the plant to take in carbon dioxide and release oxygen. The inner wall of guard cells is thick, which helps to create the pressure gradient that opens and closes the stomata.

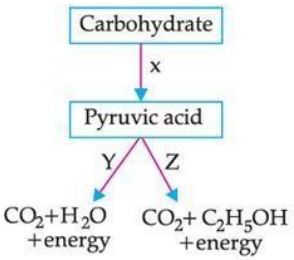
18. Correct option: (B) Active transport

Active transport is a technique that uses energy to move molecules across a cell membrane against their concentration gradient. In this case, the group of students noticed that when the cells were exposed to a poison that prevents energy production, the transfer of molecules slowed down. This implies that since active transport needs energy to take place, it is likely how molecules are normally transported within cells. Simple diffusion, enhanced diffusion, and osmosis do not need energy and are unaffected by the suppression of energy generation. Therefore, active transport is the most likely mechanism for molecule transportation under normal circumstances.

19. Correct option: (B) Only A

Minerals are absorbed from the soil by passive transport, and they cannot follow active transport. This claim is false because, depending on the particular mineral and its concentration in the soil, minerals are absorbed from the soil by both passive and active transport pathways. When minerals need to be moved up against a concentration gradient but are present in lower concentrations in the soil, active transport is used. On the other hand, passive transport happens when minerals are abundant in the soil and can go into the roots of the plant while adhering to the natural concentration gradient. The other statements (b, c, and d) are correct.

20. Correct option: (C) X – Glycolysis, Y – Aerobic Z- Anaerobic



X – Glycolysis. The initial stage of cellular respiration, known as glycolysis, can take place both under aerobic and anaerobic circumstances.

Y- Aerobic. The process of metabolizing glucose in the presence of oxygen is known as aerobic respiration. It is the most effective means to make ATP, the cell's energy currency. Aerobic respiration produces carbon dioxide and water as its byproducts.

Z is anaerobic. In the absence of oxygen, glucose is broken down by anaerobic respiration. Ethanol is created during anaerobic respiration.

6. Control and Coordination

Solutions

LEVEL-I

1. Correct option: (C) 24

Every mammal, including man, has 24 cranial nerves. Cranial nerves are a group of nerves that leave the brain immediately and are in charge of carrying sensory and motor information to and from the area of the head and neck. The cranial nerves in mammals, including humans, come in pairs of 12. Each pair performs a separate function, such as regulating taste and smell sensations, transferring visual information, directing facial expressions, and aiding eye and tongue movement. The total number of cranial nerves in a mammal, including man, is 24, and they all serve a significant role in preserving the proper operation of numerous sensory and motor functions in mammals.

2. Correct option: (D) Hypothalamus

The midbrain and the cerebral hemispheres are separated by a region of the brain called the diencephalon. The thalamus and hypothalamus are two of its many components. Particularly, the diencephalon and the hypothalamus share a strong relationship. It is a small but significant area that regulates numerous biological processes and preserves equilibrium. The hypothalamus regulates the pituitary glands hormone secretion, which in turn affects the actions of the body's other endocrine glands. In addition, it aids in the regulation of emotions, appetite, thirst, and sleep-wake cycles. Therefore, in terms of its role and significance in preserving general physiological balance, the diencephalon is intimately related to the hypothalamus.

Instead of the diencephalon, the corpora quadrigemina is a component of the midbrain. Instead of the diencephalon, the basal ganglia are a collection of nuclei found in the cerebrum. The pons is a part of the hindbrain, which is located below the diencephalon.

3. Correct option: (B) Nodes of Ranvier

The Nodes of Ranvier are the gaps in the myelin sheath. The myelin sheath, which surrounds and insulates the axons of nerve cells, is a protective coating. It facilitates a faster and more effective passage of nerve impulses. The myelin coating does not, however, extend the full length of the axon. The myelin sheath experiences irregular breaks known as Nodes of Ranvier. These nodes are essential for the transmission of nerve impulses. They enable electrical signals to move swiftly from one node to the next, hastening the movement of the nerve impulse along the axon. This saltatory conduction provides effective communication between nerve cells and aids in energy conservation.

The gap between one neuron's axon and another neuron's dendrite is known as the synaptic cleft. The enlarged end of the axon that releases neurotransmitters into the synaptic cleft is known as the synaptic knob. Schwann cells are the cells that produce the myelin sheath.

4. Correct option: (D) a – s, b – t, c – p, d – q

	List – I	**List – II**
a)	Cerebrum	s. Seat of intelligence
b)	Cerebellum	t. Maintain body posture
c)	Hypothalamus	p. Control pituitary
d)	Midbrain	q. Control vision of hearing

a) Cerebrum - s. Seat of intelligence

The cerebrum, which makes up the majority of the brain, is in charge of higher-order mental processes including reasoning, thinking, and language. It is in charge of regulating free will movements as well.

b)Cerebellum - t. Maintain body posture

The cerebellum, which is situated near the back of the brain, controls balance and coordination. It contributes to motor learning as well.

c)Hypothalamus - p. Control pituitary

Body temperature, appetite, thirst, and sleep are just a few of the crucial processes that the hypothalamus, a tiny structure near the base of the brain, regulates. Additionally, it contributes to the functioning of the endocrine system by secreting hormones that regulate the release of hormones from other glands.

d)Midbrain - q. Control vision of hearing

The cerebrum, pons, and medulla oblongata are all connected by a little portion of the brain called the midbrain. It comprises a number of significant structures, including the inferior and superior colliculi, which regulate hearing and eye movements, respectively.

5. Correct option: (D) Cerebellum

The cerebellum, an essential component of the mammalian brain, regulates muscular coordination, equilibrium, and posture. It is situated immediately above the brainstem at the back of the brain. The cerebellum receives information from various sensory systems, such as the inner ear, to help maintain balance and coordination. It also receives input from the motor cortex and spinal cord, allowing it to fine-tune and coordinate muscle movements. Damage to the cerebellum can result in difficulties with balance, coordination, and posture, leading to problems with movement and motor skills. Overall, the cerebellum is essential for ensuring smooth and coordinated movements in mammals.

The cerebellum, an essential component of the mammalian brain, regulates posture, equilibrium, and muscular coordination. Just above the brainstem, it is situated at the rear of the brain. For balance and coordination, the cerebellum gets data from a variety of sensory systems, including the inner ear. It also gets information from the spinal cord and motor cortex, which enables it to coordinate and fine-tune muscle actions. Problems with movement and motor abilities can come from damage to the cerebellum, which can also cause balance, coordination, and posture issues. Overall, mammals' ability to move smoothly and in unison is dependent on the cerebellum.

6. Correct option: (A) Stimulus → Sensory →Motor →Response

Reflex action is a quick, unconscious reaction to a stimulus. The reflex arc, a particular process it follows, has four key steps: stimulus, sensory, motor, and reaction. First, specific sensory receptors in the body, such the skin or sense organs, detect a stimulus. Through sensory neurons, these receptors subsequently transmit messages to the brain or spinal cord. Motor neurons in the brain or spinal cord process the information and immediately transmit a motor command. Finally, effectors like muscles or glands carry out the response, resulting in a prompt and automated response to the stimuli. This reflex arc allows the body to respond swiftly to potential dangers or threats, without the need for conscious thought or decision-making.

7. Correct option: (D) All the above

A critical component of the brain that is essential for many physiological processes is the hypothalamus. It affects sleep regulation by modulating the sleep-wake cycle. The hypothalamus also has parts that control appetite and thirst, which aid in regulating food intake and preserving fluid balance. By regulating the body's thermoregulatory

systems, it also has a big impact on regulating body temperature. In order to keep the body's equilibrium, the hypothalamus is in charge of coordinating and integrating these vital processes.

8. Correct option: (D) Dura mater – arachnoid – pia mater

Dura mater, arachnoid, and pia mater are the proper order of the meninges of the brain from the outside to the interior. The Pia mater is the innermost layer, followed by the arachnoid, the Dura mater, and then the arachnoid. The brain is shielded by the dura mater, a robust and fibrous membrane. It creates many divisions within the cranial cavity and is joined to the inner surface of the skull. Under the Dura mater is the arachnoid, a thin and web-like membrane. It contains the cerebrospinal fluid and acts as a cushion for the brain. The Pia mater is a thin and delicate membrane that adheres closely to the surface of the brain, providing it with nourishment and protection.

9. Correct option: (C) Midbrain, pons, medulla

The midbrain, pons, and medulla are the three primary parts of the brain stem. The midbrain, which is situated near the top of the brain stem, is in charge of sending sensory data to various areas of the brain. Additionally, it is essential for regulating eye movement and coordinating motor activities. The pons, which is below the midbrain, serves as a link between various regions of the brain. It facilitates the control of breathing, sleep, and alertness. Last but not least, the medulla, which is found at the base of the brain stem, regulates critical processes including breathing, blood pressure, and heart rate. These three parts together make up the brain stem, which is crucial for sustaining fundamental physical functions and communicating information between the brain and the rest of the body.

10. Correct option: (A) Balancing

The main function of the cerebellum is balancing. At the back of the brainstem is a part of the brain called the cerebellum. It is essential for planning and controlling voluntary motions, such as preserving balance and posture. To ensure fluid and synchronized motions, it integrates information from multiple sensory systems, including the inner ear. Additionally, the cerebellum collaborates with other brain areas to polish motor abilities and modify motions in response to signals from the body. The cerebellum is crucial for preserving stability and balance throughout the body.

11. Correct option: (B) Olfactory lobes

The answer to the question "Centre for the sense of smell is" is (B) Olfactory lobes. The signals associated to the sense of smell are processed and interpreted by the olfactory lobes, also referred to as the olfactory bulbs. These lobes, which are found near the front of the brain, collect data from the nasal cavity's olfactory receptors and send it to other parts of the brain for additional processing and perception. The olfactory lobes play a crucial role in our ability to detect and distinguish different odors, contributing to our sense of taste, memory, and emotional responses.

The center for the sense of smell is not among the other choices. The cerebellum is responsible for coordination and balance. The midbrain is involved in vision and hearing. The cerebrum is the largest part of the brain and is responsible for many higher-order functions, such as thinking, reasoning, and language

12. Correct option: (A) Medulla oblongata

The brainstem's medulla oblongata is in charge of regulating vital processes like respiration, heartbeat, and peristalsis. It serves as the body's control center, regulating these operations by receiving signals from various body areas and sending out the proper signals in response. It is essential for preserving homeostasis and making sure the body is operating properly. Although the cerebrum and cerebellum are significant

brain structures as well, they are not directly responsible for these particular processes.

The cerebrum, which makes up the majority of the brain and is in charge of numerous higher-order processes like reasoning, thinking, and language. The cerebellum, which is situated near the back of the brain, controls balance and coordination. They do not control respiration, heartbeat, or peristalsis

13. Correct option: (A) Corpora quadrigemina

Four little structures collectively known as the corpora quadrigemina are found in the midbrain. They are in charge of processing both auditory and visual information. The corpora quadrigemina's superior colliculi, which are involved in processing visual information and controlling eye movement, have an important function. The corpora quadrigemina's inferior colliculi, which are also involved in auditory processing, transfer auditory data to higher brain areas. Overall, the midbrain, in particular the corpora quadrigemina, is essential for processing and integrating sensory information.

Between the midbrain and the cerebrum is a part of the brain called the diencephalon. The greatest portion of the brain is called the cerebrum. The pons is a region of the brain that is located between the midbrain and the medulla oblongata

14. Correct option: (C) Hypothalamus

The hypothalamus is where hunger and thirst are controlled. A tiny area of the brain called the hypothalamus is essential for controlling several body processes, including appetite and thirst. It has specialized cells called neurons that can recognize alterations in the bodily environment, such low blood sugar or dehydration. When these neurons fire, they send messages to other regions of the brain and body to start the proper reaction, like boosting appetite or making someone want to drink more water. In order to make sure that the body's demands for food and water are met, the hypothalamus serves as the control center for hunger and thirst.

The largest portion of the brain, the cerebrum, is in charge of higher-order mental processes including thinking, reasoning, and language. Behind the cerebrum, in the cerebellum, which controls balance, coordination, and motor skills. The medulla, which is found near the base of the brain, controls automatic processes including breathing, heart rate, and blood pressure.

15. Correct option: (B) Medulla – temperature regulation

Breathing, heart rate, and blood pressure are controlled by the medulla, which is a structure in the brainstem. It doesn't directly affect how the body regulates temperature. The hypothalamus, which is located in the forebrain, plays a major role in temperature regulation. As the body's thermostat, the hypothalamus monitors temperature fluctuations and sets in motion reactions to keep homeostasis. As a result, (B) is the appropriate response due to the incompatibility between the medulla and temperature regulation.

All of the other claims are true. Memory, balance, and smell are all functions of the cerebrum, cerebellum, and olfactory lobes, respectively.

16. Correct option: (D) Vasopressin

Vasopressin, also referred to as antidiuretic hormone, is hypo secreted in diabetes insipidus (ADH). The posterior pituitary gland is where vasopressin is kept after it is created in the hypothalamus. By regulating the reabsorption of water in the kidneys, it controls the body's water balance. Vasopressin is a hormone that the kidneys need to adequately reabsorb water, therefore when there is a deficit it causes excessive urination and thirst, which are the main signs of diabetes insipidus. Numerous things, including harm to the hypothalamus or pituitary gland, interactions with particular

drugs, or inherited conditions, can contribute to this illness. To restore the body's normal vasopressin levels, treatment usually entails hormone replacement therapy.
The thymus gland produces a hormone called thymosin. It contributes to the immune system's growth. The posterior pituitary gland secretes the hormone oxytocin, which is made by the hypothalamus and released by it. A hormone called insulin is created by the pancreas. It has a role in controlling blood sugar levels.

17. Correct option: (A) Hormone of the adrenal gland

The adrenal gland produces the hormone epinephrine. It also goes by the name adrenaline. The body's reaction to stress and emergencies is greatly influenced by this hormone. Epinephrine speeds up heartbeat, tightens blood vessels, and widens airways when it enters the bloodstream, preparing the body for a "fight or flight" reaction. It also aids in controlling metabolism and blood sugar levels. Epinephrine is a vital hormone that aids the body in adjusting and reacting to a variety of stresses and difficulties.
The mesoderm's nephrostomal portion does not produce hormones. It is a section of the embryo where the kidneys are formed. Renal corpuscles are groups of glomeruli in mammalian kidneys. Nephridia's frontal lobe is also not a hormone. Nephridia are excretory organs found in some invertebrates.

18. Correct option: (A) Increases water absorption

The pituitary gland secretes ADH, also known as antidiuretic hormone, which is a hormone made by the hypothalamus. Its primary purpose is to improve renal water absorption. ADH levels increase in a dehydrated body, which prompts the kidneys to reabsorb more water from the urine into the bloodstream. This promotes water conservation and minimizes excessive fluid loss. Conversely, when the body is sufficiently hydrated, ADH levels fall, allowing for a greater amount of water to be passed through the urine. In general, ADH is essential for preserving the body's water balance and avoiding dehydration.

19. Correct option: (D) Kidney

Over the kidneys are the adrenal glands. These tiny glands, which have a triangle shape, are found above each kidney. They are a vital component of the endocrine system and produce and release hormones like cortisol, adrenaline, and aldosterone. The kidneys and adrenals can communicate and work together effectively because of their close closeness. The adrenal glands respond to brain impulses by releasing hormones that control a number of body processes, including as metabolism, blood pressure, and stress response. Therefore, D) Kidney is the right response.

20. Correct option: (B) Androgens are produced by the ovary

Males' testes are the primary source of a set of male sex hormones known as androgens, including testosterone. While the ovaries do make a minor quantity of androgens, the testes are where most of the hormone is made. Specifically, estrogen and progesterone, which are female sex hormones, are produced by the ovaries. Therefore, it is untrue to say that the ovary produces androgens.
All of the other claims are true. The pituitary secretes oxytocin, the thyroid secretes thyroxine, and the pancreas secretes glucagon.

21. Correct option: (A) Pineal body

The pineal gland, which is found in the brain, is the main gland responsible for producing the hormone melatonin. The sleep-wake cycle is one of many biological rhythms that the pineal gland controls. Melatonin is a hormone that the body produces in reaction to darkness to tell the body that it is time to rest. Additionally, it affects physiological systems like immunological performance and antioxidant activity. Melatonin is not predominantly secreted by the skin, despite the fact that the skin does

contribute to the production of several hormones. Similar to the hypothalamus, the pituitary gland secretes a number of hormones, although melatonin is not one of them. Therefore, (A) Pineal body is the appropriate response.

22. Correct option: (D) Melatonin

Melatonin is secreted by the pineal gland, a tiny endocrine gland that is situated in the brain. The circadian rhythm, generally known as the sleep-wake cycle, is regulated by the hormone melatonin. It is created in reaction to darkness and aids in the promotion of sleep by reducing body temperature and causing tiredness. The night is when melatonin levels are highest, and they drop during the day. This hormone is frequently used as a supplement to treat sleep disorders including insomnia because it is essential for maintaining a healthy sleep pattern.

The thyroid gland produces the hormone thyroxine. The testes in men and the ovaries in women both generate a group of hormones known as androgens. Prolactin is a hormone that is produced by the pituitary gland.

23. Correct option: (A) Glucagon

α-cells of the pancreas secrete the hormone called glucagon. Glucagon plays a crucial role in regulating blood sugar levels in the body. When blood sugar levels are low, glucagon is released into the bloodstream and signals the liver to break down stored glycogen into glucose, which is then released into the bloodstream to increase blood sugar levels. This hormone works in opposition to insulin, which is secreted by β-cells of the pancreas and helps lower blood sugar levels by promoting the uptake and storage of glucose by cells. Together, glucagon and insulin maintain a delicate balance of blood sugar levels in the body to ensure proper functioning and energy production.

The other options are incorrect. Insulin is secreted by beta cells in the pancreas. Androgens and testosterone are hormones produced by the testes in males.

24. Correct option: (D) Prolactin

The hormone known as prolactin stimulates females to make milk. In reaction to a newborn sucking, it is created by the pituitary gland. Prolactin is released when a baby latches onto the mother's breast and starts to suckle. The mammary glands in the breasts are then stimulated by this hormone to create and release milk. As long as the baby is nursing, prolactin levels stay high, maintaining a consistent supply of milk for the child. Prolactin is a crucial hormone for breastfeeding women because it is necessary for the production and secretion of milk.

The uterus contracts more forcefully during childbirth and breastfeeding thanks to oxytocin. The maintenance of pregnancy is accomplished by progesterone. LH, also known as luteinizing hormone, is what causes females to ovulate.

25. Correct option: (A) Childbirth

The hypothalamus produces the hormone oxytocin, which the pituitary gland then secretes. Its main purpose is to facilitate childbirth. Oxytocin induces uterine muscular contractions during birth, assisting with the progression of labor and baby delivery. Oxytocin also helps to release breast milk by stimulating the muscles around the milk ducts, which makes it possible for milk to be expelled. While oxytocin is most commonly linked to delivery, it also plays other roles in the body, including fostering social connections and bonding, controlling mood, and affecting sexual behavior. But without a question, its engagement in birthing plays the most significant function.

26. Correct option: (B) Adrenal

The adrenal gland is referred to as the body's emergency gland since it is essential to the body's reaction to stress and emergencies. The adrenal gland, which is situated above the kidneys, creates the stress-related chemicals cortisol and adrenaline that are

then released into the bloodstream. In order to get the body ready for a fight-or-flight response, these hormones help to raise energy levels, blood pressure, and heart rate. The adrenal gland also controls the body's immune system, electrolyte balance, and metabolism. Overall, the adrenal gland is crucial for preserving the body's general health and emergency reaction.

A male reproductive organ that makes testosterone and sperm is the testis. A gland in the chest called the thymus plays a role in the maturation of the immune system. Several hormones are produced by the pituitary gland, a gland at the base of the brain that regulates the activity of other glands.

27. Correct option: (C) β- cells

Insulin is a hormone that is secreted by the β-cells of the pancreas. These cells are responsible for producing and releasing insulin into the bloodstream. Insulin plays a crucial role in regulating blood sugar levels by allowing cells in the body to take in glucose from the bloodstream and use it as a source of energy. Without insulin, glucose cannot enter the cells, leading to high blood sugar levels and potentially causing diabetes. Therefore, the correct answer is (C) β-cells.

Alpha cells of the pancreas secrete glucagon, which raises blood sugar levels. S cells of the pancreas secrete somatostatin, which inhibits the secretion of glucagon and insulin.

28. Correct option: (C) a – iv, b – iii, c – ii, d – i

a. Testis - (iv) Testosterone

Sperm and testosterone are produced by the male gonads known as the testes. Men's secondary sex characteristics, such as facial hair and a deep voice, are brought on by the hormone testosterone.

b. Ovaries - (iii) Estrogen

Female gonads called ovaries produce estrogen and eggs. The hormone estrogen is in charge of the growth of female secondary sex organs including the breasts and hips.

c. Thymus - (ii) Atrophies in adult

A gland in the chest called the thymus plays a role in the maturation of the immune system. Adulthood causes the thymus to atrophy.

d. Melanin - (i) Pigmentation

The pigment melanin is what gives skin its color. Melanocytes are the type of cells that create it.

29. Correct option: (C) Differentiation of T lymphocytes

T lymphocyte differentiation is brought about by thymosin. White blood cells called T lymphocytes, sometimes referred to as T cells, are an essential component of the immune response. The thymus gland, which is found in the chest area, produces the hormone known as thymosin. It aids in T lymphocyte maturation and development, making them into functional immune cells with the ability to recognize and eradicate external invaders like bacteria or viruses. Thymosin is necessary for the differentiation of T cells; otherwise, the immune system would be compromised, making people more vulnerable to illnesses and infections.

30. Correct option: (A) Androgens

Androgens are a class of hormones that are produced by Leydig cells, which are only found in the testes of males. In the hormone class known as androgens, testosterone is one of the key players. It is essential for the growth and upkeep of male reproductive organs and secondary sexual traits. In addition to the development of sperm production and other aspects of the male reproductive system, such as voice deepening, facial and

body hair growth, and muscular mass, these hormones also have a role in the growth and maturation of the male genitalia. So, androgens are the proper response (A).
The ovaries in females are the organs that create estrogens. In the adrenal cortex, aldosterone is made. The anterior pituitary gland makes gonad-stimulating hormones. So, the only option left is androgens, which are produced by Leydig cells.

31. Correct option: (C) Melatonin

Humans' sleep-wake cycles are regulated by the hormone melatonin. The pineal gland, a tiny gland in the brain, produces it. The body receives a signal to go to sleep when melatonin levels rise in the evening. This hormone aids in synchronizing the circadian rhythm, the body's internal clock, with the regular day-night cycle. Melatonin encourages alertness and wakefulness when levels are low during the day. However, melatonin levels rise in the evening, encouraging sluggishness and aiding in sleep. Melatonin is therefore essential for sustaining a regular cycle of sleep and wakefulness.
A hormone called epinephrine is released in the presence of stress or danger. It aids in raising heart rate, blood pressure, and blood sugar levels, putting us in a fight or flight position. A hormone called gastrin promotes the formation of stomach acid. A hormone called insulin aids in controlling blood sugar levels.

32. Correct option: (C) a - ii, b - iv, c - iii, d - i

a. Pineal - (ii) Melatonin
Melatonin is produced by the pineal gland in the brain, which also aids in controlling the body's sleep-wake cycle.
b. Thyroid - (iv) Tetraiodothyronine
In the body, tetraiodothyronine (T4) is converted to triiodothyronine (T3) by the thyroid gland, which is situated in the neck. The hormones T3 and T4 aid in controlling the body's metabolism.
c. Ovary - (iii) Estrogen
The ovaries, which are found in the female pelvis, create estrogen, a hormone that aids in controlling the menstrual cycle and the development of female sexuality.
d. Adrenal medulla - (i) Epinephrine
Above the adrenal glands, which are situated above the kidneys, lies the adrenal medulla. Epinephrine is a hormone that the adrenal medulla produces to aid the body's reaction to stress.

33. Correct option: (B) Thyroxine

The hormone thyroxine, sometimes referred to as T4, is important for maintaining basal metabolic rate (BMR). BMR is the term used to describe how much energy is used by the body while it is at rest to maintain essential physiological processes like breathing, circulation, and cell creation. The thyroid gland produces the hormone thyroxine, which boosts the body's ability to produce heat and consume oxygen, therefore regulating metabolism. It promotes the digestion of proteins, lipids, and carbs, enabling the body to turn these foods into energy. So, it is essential to maintain a healthy BMR and general metabolic function that thyroxine levels are adequate.

34. Correct option: (D) Thymus gland

Under the breastbone is where you'll find the thymus gland. It is a little organ with a significant impact on the immune system. The thymus gland is in charge of the development and production of T-cells, a subset of white blood cells that aid in the defense against infections and sickness. The thymus gland gradually shrinks and becomes less active as we age. However, it is extremely active during childhood and adolescence and necessary for the building of a robust immune system. Because of

this, the thymus gland's location beneath the breastbone makes it convenient for its immune system tasks.

The adrenal glands are located on top of the kidneys. Calcitonin is a hormone produced by the thyroid gland. The thyroid gland is located in the neck.

35. Correct option: (B) Acromegaly

Acromegaly, a rare illness, is brought on by excessive growth hormone production by the pituitary gland (GH). A little gland called the pituitary is found near the brain's base. It creates a variety of hormones, such as GH, which aids in controlling growth and development. Enlarged hands and feet, facial abnormalities like a projecting jaw and an enlarged nose, joint pain, and thickening skin are all typical signs of acromegaly. Acromegaly can cause major health issues like cardiovascular disease, diabetes, and respiratory issues if it is not treated.

When growth hormone is oversecreted at the time of skeletal growth, it causes gigantism, which results in excessive bone growth and organ expansion. Due to a lack of growth hormone, a person with dwarfism has much shorter height than the average individual. Midget is a term that is no longer used to describe people with dwarfism.

36. Correct option: (B) Simple goitre

Simple goitre is a disorder marked by thyroid gland hypertrophy brought on by insufficient dietary iodine. Iodine is necessary for the thyroid gland to create thyroid hormones, which are crucial for controlling growth and metabolism. The thyroid gland enlarges when there is an iodine deficit in an effort to make up for the deficiency. Goitre is the medical term for this growth. Iodine shortage frequently leads to hypothyroidism, or an underactive thyroid, which is characterised by symptoms like fatigue, weight gain, and cold intolerance. It is crucial to remember that while iodine shortage is the root cause of simple goitre, other disorders including Graves' disease can also induce an enlarged thyroid.

The disorder known as gigantism is brought on by the overproduction of growth hormone throughout childhood, which causes abnormal and excessive bone and tissue growth. Exophthalmic goitre, or Graves' disease, is a condition that causes bulging eyes and hyperthyroidism (an overactive thyroid gland) (exophthalmos). Congenital hypothyroidism, another name for cretinism, is a condition in which the thyroid gland is not fully developed from birth.

37. Correct option: (A) Decreased level of glucose in the blood

A condition known as insulin shock occurs when the blood's glucose level falls. When the body produces too much insulin, the glucose is quickly absorbed by the cells and not enough remains in the bloodstream, which leads to this condition. Dizziness, confusion, sweating, and in severe situations, loss of consciousness, are just a few of the symptoms that can result from this. To quickly treat insulin shock and return blood sugar levels to normal, consume foods or beverages high in glucose.

Increased blood glucose levels are not the best course of action. Low blood sugar levels, not high blood sugar levels, are related to insulin shock. In the setting of insulin shock, glucose loss through urine excretion is incorrect. A sign of diabetes, more specifically diabetes mellitus, in which the body is unable to control blood sugar levels, is the loss of glucose through the urine.

38. Correct option: (C) Cortisol

Cortisol and adrenaline are the hormones that are released in reaction to stress and dangerous emergency conditions. The fight-or-flight response, often known as the body's natural stress response mechanism, includes several chemicals. Adrenaline, which raises heart rate, blood pressure, and energy levels under stressful or dangerous

situations, is released by the adrenal glands and primes the body for quick action. Stress causes the body to release cortisol, which helps control blood sugar levels, reduce inflammation, and improve the body's response to stress. When these hormones are together, the body is better able to handle and react to emergency conditions.

Androgens are hormones that are made by both the ovaries and the testes in both men and women. They are in charge of the growth of male secondary sex traits like muscular mass and facial hair. The thymus gland produces a hormone called thymosin. It aids in immune system regulation.

39. Correct option: (A) Isthmus

The thyroid gland is a neck-based, butterfly-shaped gland. Two lobes, one on either side of the trachea, are what make up the isthmus, a slender flap of connective tissue that connects the two lobes. The thyroid glands isthmus serves as a structural support and aids in holding the two lobes together. In order to ensure adequate hormone production and secretion, it also enables communication and coordination between the lobes. The thyroid lobes would be divided and unable to work together properly without the isthmus.

The thyroid gland's functional units are called follicles. Iodine and tyrosine are combined to create thyroid hormones in them. Diuresis is an increase in urine production. The thyroid gland does not contain it. Thyroid-stimulating hormone is abbreviated as TCT. The pituitary gland releases a hormone that prompts the thyroid to make thyroid hormones.

LEVEL-II

1. Correct option: (A) Early embryos

Only early embryos contain unipolar neurons. The neural system is still growing and diffusing in the early stages of embryonic development. One process protrudes from the cell body of unipolar neurons, a kind of sensory neuron. The axon and dendrites are eventually produced by the same mechanism. These unipolar neurons will undergo additional specialization and differentiation as the embryo develops to create the intricate neural networks observed in the mature nervous system. As a result, unipolar neurons are fragile and are only seen in early embryos during the beginning of the nervous system's development. As the embryo develops, the unipolar neurons differentiate into other types of neurons, such as bipolar neurons and multipolar neurons.

Bipolar neurons make form the retina of the eye and the olfactory epithelium. Multipolar neurons, which comprise a cell body with one axon and two or more dendrites, make up the spinal cord.

2. Correct option: (D)Malathion

Acetylcholinesterase is inhibited by malathion. Acetylcholine, a neurotransmitter used in signal transmission between nerve cells, is broken down by the enzyme acetylcholinesterase. Malathion blocks the breakdown of acetylcholine by blocking this enzyme, which causes the neurotransmitter to build up in the synaptic cleft. Because of the continual stimulation of the postsynaptic cell caused by this extended cholinergic presence, the nervous system is overstimulated. The consequences of this overstimulation might range from muscular spasms to paralysis to, eventually, death. As a result, malathion's capacity to block acetylcholinesterase makes it a potent pesticide since it prevents bugs' nerve systems from operating normally, which causes them to die.

Aldrin, BHC, and endosulfan are also pesticides, but they are not acetylcholinesterase inhibitors.

3. Correct option: (B)Voluntary motor impulses

Loss of voluntary motor impulses would come from the death of the spinal cord's anterior horn cells. The motor neurons, sometimes referred to as the anterior horn cells, are in charge of carrying messages from the brain to the muscles so that we may consciously regulate our actions. The connection between the brain and the muscles is hampered when these cells are injured or eliminated, which results in a lack of voluntary motor control. This may cause paralysis, muscular wasting, and the inability to carry out voluntary motions like walking, reaching, or grabbing.

Integrating impulses are utilized to synchronize muscular movements. Commissural impulses are impulses that go back and forth down the spinal cord between the left and right sides. The brain receives information from the body via sensory impulses.

4. Correct option: (C) Cerebellum

A component of the brain called the cerebellum is essential for maintaining proper posture and bodily balance. It gathers data from a variety of sensory systems, including the inner ear, the eyes, and the muscles, and uses this data to coordinate and control movement. To maintain stability and guarantee synchronized, smooth movements, the cerebellum continually modifies muscle tone and activity. Balance problems, unstable walking, and trouble maintaining good posture can all be brought on by damage to or malfunction of the cerebellum. In order to maintain a steady and balanced bodily posture, the cerebellum is essential.

Although the pons and medulla oblongata are significant brain structures, they are not the main generators of balance and posture. The cerebrum, which makes up the majority of the brain, is in charge of higher-order processes including reasoning, organizing, and language. Balance and posture are not its primary functions.

5. Correct option: (B) under secretion of thyroxin

Thyroxin, a hormone produced by the thyroid gland, is under secreted in cretinism, a disorder that results from this. Thyroxin is essential for controlling the body's development and metabolism. Particularly in children, a thyroxin shortage can result in different developmental defects and limited growth. Lack of this hormone impairs the body's ability to operate normally, particularly the brain, which causes mental retardation, postponed physical maturation, and impairments in speech and hearing. Furthermore, those who have cretinism may have physical traits such a small stature, enlarged facial features, and a protruding tongue. Early detection and thyroxin replacement medication can help treat the symptoms of cretinism and stop future issues from developing.

Excess secretion of growth hormone can lead to a condition called acromegaly. Acromegaly is characterized by enlargement of the bones and soft tissues. Excess secretion of adrenaline can lead to a condition called pheochromocytoma. Pheochromocytoma is characterized by high blood pressure and other symptoms.

6. Correct option: (C) Amacrine cell

One of the cell types present in the retina are amacrine cells. They are interneurons that are very important for the retina's visual processing and information integration. Bipolar and ganglion cells provide input to amacrine cells, which modulates the transfer of visual signals between these cells. They are in charge of lateral inhibition, which raises contrast and brings edges of visual objects into sharper focus. Amacrine cells also play a role in controlling the retina's sensitivity to various amounts of light and help us perceive motion and directionality. Generally speaking, amacrine cells are

necessary for the retina's normal operation and the processing of visual information prior to transmission to the brain.

In the retina, neuroglial cells are present. The retina's neurons are supported and fed by these cells, which are the most prevalent cell type there. They do not, however, participate in the processing of visual data. Not the retina, but the peripheral nervous system contains Schwann cells. The neurons in the peripheral nervous system's axons are protected by them. Not the retina, but the cerebellum has Purkinje cells. They assist with motor control.

7. Correct option: (A)somatic N.S

The somatic nervous system regulates the skeletal muscles. This system is controlled consciously and is in charge of voluntary motions. Our ability to do numerous motions, like walking, running, and lifting items, is made possible by the somatic nervous system, which is made up of motor neurons that send signals from the brain and spinal cord to the skeletal muscles. Acetylcholine in particular, a neurotransmitter that promotes muscular contraction, is released to achieve this regulation. As a result, the somatic nervous system is essential for directing and coordinating the actions of our skeletal muscles.

The autonomic nervous system controls involuntary actions, such as breathing and digestion. Glial cells are supportive cells in the nervous system. They do not control movement.

8. Correct option: (A) Hypochondria

A psychosomatic condition, hypochondria. Psychosomatic disorders are diseases or physical symptoms that are brought on or made worse by psychological elements like stress, worry, or emotional anguish. The hallmark of hypochondria, sometimes referred to as sickness anxiety disorder, is extreme worry and dread about having a major medical condition, but having little to no medical evidence to back up these fears. Physical symptoms experienced by hypochondriacs are frequently exaggerated or misread, which results in ongoing concern and obsession with their health. These symptoms can seriously impair a person's daily functioning and general well-being. As a result of the effect of psychological variables on the physical symptoms, hypochondria is classified as a psychosomatic condition.

Other options are incorrect. Anxiety, sadness, and obsessive-compulsive behaviors and thoughts are frequently present in neuroses. A significant mental illness that affects a person's thoughts, feelings, and behaviors is schizophrenia. Hallucinations, delusions, and disordered thinking are common in people with schizophrenia. Numerous causes, including mental illnesses, substance misuse, and physical ailments, can lead to psychosis.

9. Correct option: (C) rapid cell divisions in tendriller cells that are away from support

Rapid cell divisions in the tendrilled cells that are furthest from the support cause tendrils to form on pea plants. Plants can climb and hold themselves thanks to unique structures called tendrils. The cells in the tendrils quickly divide, lengthening and growing as a result. The tendrils may stretch out and locate a structure to attach to thanks to these divisions that especially take place in the tendrilled cells that are furthest from the support. With the help of this growth mechanism, the pea plant may efficiently climb and increase its exposure to sunlight, which is necessary for photosynthesis and the general growth and development of the plant.

The other options are incorrect. Option (A) is incorrect because the effect of gravity is not involved in the growth of tendrils. Option (B) is incorrect because chemicals are not

involved in the growth of tendrils. Option (D) is incorrect because rapid cell division in tendriller cells in contact with the support would cause the tendril to move away from the support.

10. Correct option: (A) Neurotransmitter

A synapse is a tiny space or gap between two adjacent neurons where nerve impulses are transmitted. When an electrical signal, known as an action potential, reaches the end of a neuron, it triggers the release of chemical messengers called neurotransmitters. These neurotransmitters then diffuse across the synapse and bind to specific receptors on the neighboring neuron, allowing the transmission of the signal to continue. The synapse acts as a junction or connection point between neurons, enabling the flow of information throughout the nervous system. It plays a crucial role in processes such as learning, memory, and coordination of bodily functions. Neurotransmitters are chemicals that are released by the axon of the first neuron and received by the dendrites of the second neuron. Dendrites are the branches of a neuron that receive signals from other neurons. Axons are the long fibers that transmit signals from one neuron to another.

11. Correct option: (C) Ear: Adequate of sylvius

The other options in the question are all related to specific structures within the body part mentioned. Nervous tissue: Node of Ranvier is associated with the nervous system; it is a gap in the myelin sheath that surrounds an axon. Bone: Volkman's refers to a type of blood vessel found in bones, it is a canal that runs through the bone and contains blood vessels and Eye: Canal of Schlemm is a structure in the eye, it is a canal that drains fluid from the eye. However, "Ear: Adequate of Sylvius" does not correspond to any known anatomical structure or term. The adequate of Sylvius is a structure found in the brain, not the ear. It is a small canal that connects the third and fourth ventricles.

12. Correct option: (B) PNS

The Peripheral Nervous System, or PNS, is the Peripheral Nervous System. All of the body's nerves that are not found in the brain or spinal cord make up the peripheral nervous system. It consists of the spinal nerves, which emerge from the spinal cord to innervate the rest of the body, as well as the cranial nerves, which link the brain to the head and neck. The PNS is in charge of delivering motor instructions from the brain to the muscles and organs as well as sensory data from the body to the brain. It is essential for the coordination and regulation of body processes like movement, sensation, and organ control. Overall, the PNS acts as a communication network that connects the central nervous system (brain and spinal cord) to the rest of the body.

13. Correct option: (B) 31

The answer is 31. A long, cylindric bundle of nerves, the spinal cord runs from the brain's base to the lower back. It is in charge of sending messages from the brain to the rest of the body. The 31 pairs of nerves that make up the spinal cord each originate from a distinct section of the cord. In order to transmit sensory data from the body to the brain and motor impulses from the brain to the muscles, these nerves branch out and innervate various parts of the body. There are 31 pairs of nerves that emerge from the spinal cord as a result.

14. Correct option: (D) Abscisic acid

Abscisic acid is the substance that triggers the fall of mature leaves and fruits from plants. This plant hormone plays a crucial role in regulating various physiological processes, including leaf and fruit abscission. When the concentration of abscisic acid increases in a plant, it signals the cells at the base of the leaf or fruit to break down the

cell walls, leading to detachment. This process is important for plants to conserve energy and resources during unfavorable conditions, such as drought or winter. Abscisic acid also helps in maintaining water balance, seed dormancy, and stress responses in plants."

Auxin, cytokinin, and gibberellin are other plant hormones, but they do not trigger the fall of mature leaves and fruits from plants.

15. Correct option: (A) Iron is essential for the synthesis of thyroxin

The statement that iron is essential for the synthesis of thyroxin is not true. Thyroxin, also known as thyroxine, is a hormone produced by the thyroid gland. It is synthesized from the amino acid tyrosine and iodine, not iron. Iron plays a crucial role in the body, particularly in the production of red blood cells and oxygen transportation, but it is not directly involved in the synthesis of thyroxin. The thyroid gland requires iodine to produce thyroxin, which is then released into the bloodstream to regulate metabolism, growth, and development. Therefore, the statement that iron is essential for the synthesis of thyroxin is incorrect.

The other statements are all true. Thyroxine regulates carbohydrates, protein, and fat metabolism in the body. It is also called thyroid hormone.

16. Correct option: (D) Photonasty

Photonasty is the term used to describe the response of dandelion flowers to light. This phenomenon refers to the opening and closing of the petals in response to the presence or absence of bright light. During the daytime when there is ample light, dandelion flowers open their petals to maximize exposure to sunlight for photosynthesis and pollination. However, at night or in darkness, the flowers close their petals to protect their reproductive structures and conserve energy. This adaptive behavior allows dandelion flowers to optimize their chances of successful reproduction and survival in varying light conditions.

Phototropism is a directional movement of plant parts in response to light. Chemotropism is a directional movement of plant parts in response to chemicals. Thigmonasty is a non-directional movement of plant parts in response to touch.

17. Correct option: (A) Scala vestibuli and Scala media

Scala vestibuli and scala media in the inner ear are divided by Reisner's membrane. The basilar membrane, which is found in the cochlea, is parallel to this membrane, which is a thin structure. Two of the three chambers of the cochlea that are filled with fluid are the scala media and scala vestibuli. The intermediate chamber is called the scala medium, while the top chamber is called the scala vestibuli. The fluids in each of these two compartments are kept apart by Reisner's membrane, which serves as a barrier between them. The capacity of the inner ear to operate properly and send sound impulses to the brain depends on this separation.

The other choices are false. Option (B) is erroneous because the eustachian tube divides the throat from the tympanic cavity. Option (C) is erroneous since sound waves can reach the inner ear through holes in the cochlea called the fenestra rotundas and fenestra ovalis. The basilar membrane divides the scala media and scala tympani, making option (D) erroneous.

18. Correct option: (C) Testosterone from testes and estrogen from ovary

The secretions of testosterone and estrogen from the ovary play a major role in the significant changes in physical characteristics that accompany puberty. As the body gets ready for reproductive maturity throughout puberty, considerable physical changes occur. The formation of secondary sexual traits including facial hair, a deeper voice, and muscular mass is caused by testosterone, which is generated by the testes in

males. The development of secondary sexual features including the enlargement of the breasts, broadening of the hips, and the beginning of menstruation are all influenced by the hormone estrogen, which is generated by the ovaries in females. In moulding people's physical development throughout puberty and separating them into their distinct genders, these hormones are essential.

19. Correct option: (D) Insulin

A lack of the hormone insulin causes diabetes in patients. Blood sugar levels are controlled by insulin, which is produced by the pancreas. In people with diabetes, either the pancreas does not generate enough insulin or the body does not utilize the insulin it does make properly. This results in elevated blood glucose levels, which, if addressed, can have a negative impact on a number of bodily functions. By replacing or enhancing the body's natural insulin production, insulin treatment is frequently used to control diabetes.

Thyroxine is a hormone produced by the thyroid gland that regulates metabolism. Testosterone is a hormone produced by the testes in men and the ovaries in women that is responsible for sexual development and reproduction. Estrogen is a hormone produced by the ovaries in women that is responsible for sexual development and reproduction.

20. Correct option: (B) Differentiate the smell of a perfume from that of agarbatti

When someone has a cold, they cannot tell the difference between the fragrance of perfume and agarbatti. This is due to the fact that a cold frequently results in nasal congestion, which blocks the nasal passages and impairs one's ability to smell. It is difficult for the nose's olfactory receptors to correctly detect and differentiate between many odours. As a result, the person has much less capacity to sense and distinguish between particular odours, such as the distinction between a perfume and agarbatti. A temporary loss or modification in one's sense of smell results from the obstruction and inflammation in the nasal passages making it difficult for the person to distinguish minute differences in aroma.

The other options are incorrect. Option (A) is incorrect because taste is detected by taste buds in the tongue. Option (C) is incorrect because vision is not affected by a cold. Option (D) is incorrect because touch is not affected by a cold.

21. Correct option: (A) CNS, PNS, the brain and spinal cord; nerves around the body

The central nervous system (CNS) and the peripheral nervous system (PNS) are two different parts of the nervous system. The brain and spinal cord, which are parts of the CNS, are in charge of processing and delivering information. The PNS, on the other hand, is made up of nerves that run the length of the body and connect the CNS to numerous muscles, organs, and sensory receptors. The control center, or CNS, is responsible for receiving information from the PNS and responding appropriately. In contrast, the PNS acts as a network of communication, transmitting sensory data to the CNS and transferring motor instructions from the CNS to the rest of the body. These two sections of the nervous system act in concert to control body processes, plan motions, and permit sensory experience.

Therefore, the nervous system is divided into the Central nervous system (CNS) and the peripheral nervous system (PNS); the former consists of the brain and spinal cord and the latter consists of nerves around the body.

22. Correct option: (C) Dendrite -> cell body -> axon -> nerve ending

Neural transmission is the movement of information inside a neuron. Dendrites, which are branch-like extensions of the neuron that receive impulses from neighboring

neurons, are where it all begins. These signals are subsequently sent to the soma, the cell's body, where they are digested and integrated. The generation of an electrical impulse known as an action potential depends on whether the signals are powerful enough to cross the threshold. After then, the action potential moves up the axon, a long, tubular structure that protrudes from the cell body. Myelin, a fatty material that coats the axon, aids in accelerating the transmission of the action potential.

The nerve ending, also referred to as the axon terminal, at the end of the axon releases chemical messengers known as neurotransmitters into the synapse, which is the minuscule space between the nerve ending and the dendrites of the following neuron. The signal is then sent to the following neuron in the circuit by these neurotransmitters binding to receptors on the dendrites of the neighboring neuron.

23. Correct option: (A) Bright light → receptors in eyes → sensory neuron –> spinal cord → motor neurons → eyelid closes

The option (A) correctly shows the order of events when a bright light is focused on our eyes. When a bright light enters our eyes, it first stimulates the receptors in our eyes, specifically the rods and cones. These receptors then send signals to the sensory neurons, which are responsible for transmitting information from the sensory organs to the central nervous system. The sensory neurons carry the signals to the spinal cord, where they are processed and interpreted. Once the spinal cord receives the signals, it activates the motor neurons, which are responsible for transmitting signals from the central nervous system to the muscles. In this case, the motor neurons signal the muscles responsible for controlling our eyelids, causing them to close as a protective reflex to shield the eyes from the bright light. Therefore, the correct order of events is bright light → receptors in eyes → sensory neuron spinal cord → motor neurons → eyelid closes.

24. Correct option: (D) A part of midbrain

The forebrain, not the midbrain, contains the thalamus. It is situated near the cerebrum at the apex of the brainstem. A crucial center for the coordination of sensory and motor signaling is the thalamus. It gathers sensory data from all across the body and transmits it to the cerebral regions that need it. The cerebrum also communicates motor commands to the rest of the body.

The midbrain is a small part of the brain that is located between the thalamus and the pons. It is involved in a variety of functions, including vision, hearing, and motor control.

So, the thalamus is not a part of the midbrain. It is a part of the forebrain and is located between the midbrain and the cerebrum.

25. Correct option: (C) Pons

A complex network of brain regions called the limbic system or limbic lobe is responsible for emotions, memory, and behavior. It contains a number of important regions, including the hippocampus, amygdala, and inner cerebral hemispheres. The pons, however, is not regarded as a component of the limbic system or lobe. The brainstem's pons is a structure that plays a role in signal transmission between various regions of the brain and spinal cord. The limbic system, which controls emotions and memory, is not directly affected by this structure, despite the fact that it is a crucial component of overall brain function. Therefore, the correct answer is C) PONS.

The other options are all parts of the limbic lobe. The amygdala is a small almond-shaped structure that is involved in emotion. The hippocampus is a seahorse-shaped structure that is involved in memory.

26. Correct option: (A) Spinal cord

A reflex arc is a neural pathway that controls a reflex. It is a simple pathway that does not involve the brain. The reflex arc consists of the following parts:

- Receptor: This is the part of the body that detects the stimulus.
- Sensory neuron: This neuron carries the signal from the receptor to the spinal cord.
- Interneuron: This neuron is optional and may be present in some reflex arcs. It helps to integrate the signal from the sensory neuron with other signals.
- Motor neuron: This neuron carries the signal from the spinal cord to the effector. The effector is the muscle or gland that produces the response.

The spinal cord is the part of the nervous system that carries signals between the brain and the rest of the body. It is the site of the reflex arc. So, the missing term in the reflex arc is the spinal cord

27. Correct option: (B) Ganglion cells - Bipolar cells - Photoreceptor cells

From inside to outside, the retina's cells should be arranged in the following order: B) Ganglion cells - Bipolar cells - Photoreceptor cells. Bipolar cells provide impulses to ganglion cells, which are the retina's deepest layer of cells. Photoreceptor cells send signals to bipolar cells, which in turn receive them. The retina's outermost layer of cells, the rods and cones, are photoreceptor cells that are in charge of detecting light and turning it into electrical impulses. Ganglion cells, bipolar cells, and ultimately photoreceptor cells are the proper sequence of cells in the retina.

28. Correct option: (D) Perikaryon and dendrites

The "perikaryon" (sometimes referred to as the "cell body" or "soma") and "dendrites" are the two components that set a nerve cell (or "neuron") apart from other types of cells. Dendrites are the branching extensions that receive messages from other neurons or sense receptors, whereas the perikaryon is where the nucleus and other cellular organelles are found. The distinct mix of features that distinguishes neurons from other cell types is a crucial characteristic of neurons.

Flagellum and medullary sheath are not found in all nerve cells. Flagellum is a whip-like structure that helps cells move, and medullary sheath is a fatty covering that insulates axons. Vacuoles are sacs that store nutrients and waste products, and they are found in all cells.

29. Correct option: (A) 1

A neuron has a 1:1 ratio of intracellular potassium to extracellular sodium. This indicates that the sodium ions outside the cell are balanced by the potassium ions present inside the neuron. Ion channels and pumps operating in the cell membrane help to maintain this ratio. To maintain a balanced concentration, these channels and pumps actively move potassium ions into the cell and sodium ions out of the cell. As it permits the creation and transmission of electrical impulses known as action potentials, this equilibrium is essential for the neuron to operate properly. Various neurological diseases and reduced neuronal function may result from any change in this concentration ratio.

30. Correct option: (C) 2,3

Electric synapses are faster: True. When compared to chemical synapses, electric synapses—also known as gap junctions—allow for direct electrical connection between cells, resulting in quicker information transmission.

It is more difficult to regulate an electric synapse w.r.t. a chemical synapse: True. In terms of regulation, chemical synapses are more complicated and adaptable. Greater control over signal transmission is made possible by the ability to fine-tune and modify

the release and reception of neurotransmitters at chemical synapses. Electric synapses lack this degree of control and regulation since they are more direct.

Acetylcholine is released at the electric synapse: False. Acetylcholine is not released at the electric synapse. Electric synapses do not use neurotransmitters to transmit signals. Instead, they use gap junctions, which are channels that directly connect the cytoplasm of two cells.

31. Correct option: (A)1 and 2

Humans sweat (1) and dilate their skin arteries (2) to lower their body temperature when it slightly rises. When sweat evaporates, it removes heat from the body and aids in cooling it down. Sweating helps to cool the body by evaporating. Moreover, widening the skin arteries boosts blood flow close to the skin's surface, promoting heat dissipation. Together, these processes control body temperature and keep it within a relatively small range.

Constricting the skin arteries and tightening the skin's muscles are two strategies utilized to retain heat rather than dissipate it. As a result, they are not engaged in reversing a little increase in body temperature.

32. Correct option: (C)2-1-4-3

This pattern captures the physiological process of stimulus-induced changes in the permeability of a neuron's cell membrane. Due to the opening of voltage-gated sodium channels in response to a stimulus, there is first an increase in permeability to sodium ions (2). The resting membrane potential (RMP) is depolarized by this inflow of sodium ions, which depolarizes the membrane potential (3). The restoration of RMP (4) then occurs as a result of the diffusion of potassium ions (1) outside the cell, which is made possible by the opening of voltage-gated potassium channels. For neurons to operate properly in conveying electrical impulses, this sequence of events—in which sodium ions enter the cell first, followed by potassium ions exiting and ultimately restoring the RMP—must occur.

7. How do organisms reproduce?

Solutions

LEVEL I

1. Correct option: (B) Viviparous

The animals which give birth to their young ones are called viviparous animals or mammals. Animals like humans, cows, dogs, cats, lions, etc., give birth to their young ones.

2. Correct option: (C) Leopard

A Leopard is an example of a Viviparous animal. Viviparous animals are those that reproduce by giving birth to live young. This is the reproductive method of all placental mammals, some fish, reptiles and invertebrates.

3. Correct option: (D) Cat

Viviparous animals, who reproduce by giving birth to young ones, have epidermal hair on their skin and external ears. It's seen in dogs, cats, lions, monkeys, panda, platypus.

4. Correct option: (A) Hydra

Hydra, Algae and Fungi, all of them can reproduce by asexual mode of reproduction, Hydra reproduces by budding, Algae by simple division, fragmentation or sometimes by spores, they even have a sexual mode of reproduction, Fungi too have both asexual and sexual modes of reproduction.

5. Correct option: (B) Amoeba

Amoeba Reproduction-Binary Fission in Amoeba

In this method, two similar individuals are produced from a single parent cell. An amoeba that is about to undergo division grows larger, and eventually, its nucleus extends and divides into two. The division of cytoplasm follows the division of the nucleus.

6. Correct option: (D) Stamen

Most flowers have four main parts: sepals, petals, stamens, and carpels. The stamens are the male part whereas the carpels are the female part of the flower. Most flowers are hermaphrodite where they contain both male and female parts. Others may contain one of the two parts and may be male or female.

7. Correct option: (B) Testes

The male reproductive system contains the external genitals (the penis, testes and the scrotum) and internal parts, including the prostate gland, vas deferens and urethra. A man's fertility and sexual traits depend on the normal functioning of the male reproductive system, as well as hormones released from the brain.

8. Correct option: (C) Uterus

Inside the uterus, the blastocyst implants in the wall of the uterus, where it develops into an embryo attached to a placenta and surrounded by fluid filled membranes.

9. Correct option: (A) Frog

In most species of frogs, fertilization is external. The male frog grabs the female's back and fertilizes the eggs as the female frog releases them.

10. Correct option: (B) Tadpole

Most frogs and other amphibians hatch as fish-like larvae called tadpoles, or "pollywogs." Tadpoles exist to eat and grow. They eventually transform into froglets, a process called metamorphosis.

11. Correct option: (A) Male flower

Pollen grains are produced by male flowers.

Through the process of meiosis, pollen is formed in the pollen sacs (microsporangia) inside the anthers of a flower. The anther, while it is developing from anther primordium (undifferentiated cells), forms two clusters of cells.

12. Correct option: (C) 270-280 days
Pregnancy lasts for about 280 days or 40 weeks. A preterm or premature baby is delivered before 37 weeks of your pregnancy.
13. Correct option: (D) Camel
Camel is the only viviparous animal amongst the others, rest all are oviparous.
14. Correct option: (D) Crow
Crow is oviparous in the following option as it lay eggs.
15. Correct option: (D) Reproduction
Reproduction ensures the continuity of various species on the Earth. In the absence of reproduction, the species will not be able to exist for a long time and may soon get extinct.
16. Correct option: (C) Metamorphosis
The transformation of larva into adult through drastic changes is called metamorphosis.
17. Correct option: (A) Asexual
Asexual reproduction is a type of reproduction that does not involve the fusion of gametes or change in the number of chromosomes.
18. Correct option: (B) Bisexual animals
The organisms in which both the male and female reproductive system are present in the same individual are called bisexual organisms or hermaphrodites. For example, earthworm, leech, starfish, tapeworm etc.
19. Correct option: (B) Budding
Yeast usually asexually reproduces by a method called budding. A small knob or bud forms on the parent cell, grows, and finally separates to become a new yeast cell. This new yeast cell is genetically identical to the parent cell.
20. Correct option: (A) vasa efferentia → Epididymis → vas deferens → urethra
The male reproductive system contains the external genitals (the penis, testes and the scrotum) and internal parts, including the prostate gland, vas deferens and urethra. A man's fertility and sexual traits depend on the normal functioning of the male reproductive system, as well as hormones released from the brain.
21. Correct option: (C) On either side of vagina in human
The Bartholin's glands (or greater vestibular glands) are important organs of the female reproductive system. The glands are located in the vulvar vestibule, at either side of the external orifice of the vagina.
22. Correct option: (B) Epididymis
Epididymis is a long and a coiled tube that receives sperms from the vasa efferentia. It leads to the vas deferens. It is responsible for storing and nourishing the sperms until they are moved further to the vas deferens. Thereafter, the secretions of the prostate gland and the seminal vesicles will provide nourishment to the sperms.
23. Correct option: (A) Testes with epididymis
The vasa efferentia connects the rete testis to the epididymis and then to the vas deferens. The sperm produced in the seminiferous tubules are transported through accessory ducts.

LEVEL- II

1. Correct option: (D) Sexual reproduction
Changes in the structure of DNA leads to variations. It is because of sexual reproduction.
2. Correct option: (A) Has either carpel or stamen
A flower that possesses either stamens or carpels but not both are called unisexual flowers. Examples include coconut flowers, papaya, watermelon etc.

3. Correct option: (A) Contraceptive devices

It's sometimes referred to as a nonhormonal IUD option. The ParaGard device is a T-shaped plastic frame that's inserted into the uterus. Copper wire coiled around the device produces an inflammatory reaction that is toxic to sperm and eggs (ova), preventing pregnancy.

4. Correct option: (C) Oviduct

Fertilization usually takes place in a fallopian tube that links an ovary to the uterus. If the fertilized egg successfully travels down the fallopian tube and implants in the uterus, an embryo starts growing.

5. Correct option: (A) Copper-T

There are mainly two types of IUCD: copper T and hormonal.

6. Correct option: (A) Hepatitis

Hepatitis is a viral infection of liver which is caused by different hepatitis virus, it is generally food or blood borne infection. Rest all are exclusively sexually transmitted.

7. Correct option: (C) Oviduct

Fertilization usually takes place in a fallopian tube that links an ovary to the uterus. If the fertilized egg successfully travels down the fallopian tube and implants in the uterus, an embryo starts growing.

8. Correct option: (B) Kala azar

Kala-azar is a slow advancing and naturally occurring disease that is caused by a protozoan parasite of the Leishmania genus. Kala-azar is not categorized as a sexually transmitted disease (STD

9. Correct option: (D) To make them sticky

A seminal vesicle is a part of the anatomy of the male reproductive system. The two seminal vesicles are glands that produce the fluids that will turn into semen. The role of seminal vesicles is to provide nourishment to the sperm instead of making the semen sticky.

10. Correct option: (B) Contraception

An Intrauterine Contraceptive Device (IUCD) is a small, plastic, copper or hormone containing device, which is placed inside the uterus to prevent pregnancy. It comes in different types, sizes and shapes to suite different women. Depending upon your choice of contraception period, it can be inserted for 3-5-10 years

11. Correct option: (A) 2:1

The human sperm and egg have 23 chromosomes each. So, when they fuse together, the zygote formed has 23+23=46 chromosomes. So, the required ratio is 46:23, i.e., 2:1.

12. Correct option: (B) Halving of chromosomes during gamete formation

New offsprings are formed by the fusion of male and female gametes. Gametes are formed by meiotic division. Meiosis is a reduction division where the number of chromosomes gets halved. Gametes formed are haploid in number(n). When both gametes are fused, chromosome numbers get restored to diploid (2n).

13. Correct option: (B) When she is in her mother's womb

In any one human generation, the egg's development starts before the female that carries it is even born; 8 to 20 weeks after the fetus has started to grow, cells that are to become mature ova have been multiplying, and by the time that the female is born, all of the egg cells that the ovaries will release during the active reproductive years of the female are already present in the ovaries.

14. Correct option: (A) Testis → vas deferens → urethra

The sperms are formed in the testes and delivered through the vas deferens which unites with a tube coming from the urinary bladder. The urethra thus forms a common passage for both the sperms and urine.

15. Correct option: (B) Cracking of voice

During adolescence, several changes occur in the human body, one change associated with sexual maturation in boys is the cracking of voice

16. Correct option: (D) Chromosomes get halved during gamete formation

The gametes have half the number of chromosomes. After their fusion, the zygote thus formed has a full set of chromosomes. Therefore, the numbers of chromosomes in parents and offsprings of a particular species remain constant

17. Correct option: (A) spores divide and grow into new individual

The spores develop into Rhizopus by dividing and growing into a new individual. Explanation: Rhizopus is a fungus species. They reproduce asexually by spore production.

18. Correct option: (B) Yeast fungus

Yeast fungus does not reproduce by spore formation method.

19. Correct option: (B) Hydra

Hydra can reproduce by the method of budding similar to yeast, and also by regeneration as in the case of Planaria.

20. Correct option: (D) testis → vas deferens → urethra → penis

The path of sperm during ejaculation is as follows: the sperm travels from the testes to the epididymis, then to the vas deferens, then to the prostate, and ultimately to the urethra.

21. Correct option: (D) 28 days

The average length of a menstrual cycle is 28 days. However, a cycle can range in length from 21 days to about 35 days and still be normal.

22. Correct option: (D) Sellaginella- Sucker

Selaginella shows vegetative as well as sexual reproduction. Vegetative reproduction occurs through tubers, buds and fragmentation. Sexual reproduction occurs through spore formation.

23. Correct option: (C) Both similarities and variations with parents

Characters that are transmitted from parents to offspring shows both similarity and variation. They show similarity because half of the chromosome comes from father and half from the mother. They show variation because of genetic recombination, mutation and natural selection.

24. Correct option: (A) Vas deferens

The vasa efferentia connects the rete testis to the epididymis and then to the vas deferens. The sperm produced in the seminiferous tubules are transported through accessory ducts. Vas deferens is a part of male genitals.

25. Correct option: (A) linear growth phase

The growth of the plant is plotted against time and if it shows a linear curve then the growth is called arithmetic growth. In this growth, the rate of growth is constant. It is found in the root or shoots elongating at a constant rate.

26. Correct option: (C) Urethra

Urethra, in a male urinary and reproductory system, is a tube that passes through the penis. It carries urine from the urinary bladder to the outside of the body. It also helps in the transfer of sperms to the vagina. Thus, the urethra is the common pathway for both urine and semen.

27. Correct option: (C) Such plants produce seeds and fruits much earlier than other methods

Vegetative propagation is an asexual method of plant reproduction that occurs in its leaves, roots and stem. This can occur through fragmentation and regeneration of specific

vegetative parts of plants. Such plants produce seeds and fruits the same as any other methods.

28. Correct option: (C) The nucleus divides repeatedly inside the cell to form new daughter cells.

In Plasmodium blood-stage schizogony, the nuclear membrane divides to separate daughter genomes and the cell does not divide until several cycles of mitosis have produced a multinuclear cell.

29. Correct option: (D) iii and iv

Development of the placenta and secretion of estrogen is not the function of the testes at puberty.

30. Correct option: (C) i and iii

The factors responsible for the rapid spreading of bread mould on slices of bread are the presence of a large number of spores in the air and the presence of moisture and nutrients in bread that act as food for the fungi.

8. Heredity and Evolution

Solutions

LEVEL-I

1. Correct option: (B) Responsible for character

The alleles are responsible for character. Mendel studied peas and deduced that one is responsible for structure, the other for colour, and so on. All these are characters that are shown by alleles.

2. Correct option: (D) Both A & B

Natural selection is the process through which populations of living organisms adapt and change. Individuals in a population are naturally variable, meaning that they are all different in some ways. This variation means that some individuals have traits better suited to the environment than others.

3. Correct option: (B) Fossil evidence

A palaeontologist is a scientist who studies the history of life on earth. They study the history of life on earth with the help of fossil records as fossils are the evidence of past life on the planet and they can include animal bodies or their imprints.

4. Correct option: (C) Mendel

Gregor Johann Mendel is known as the father of genetics.

5. Correct option: (D) 22

In humans, each cell normally contains 23 pairs of chromosomes, for a total of 46. Twenty-two of these pairs, called autosomes, look the same in both males and females.

6. Correct option: (A) Lamarck

Inheritance of acquired characters is the once widely accepted idea that physiological modifications acquired by an organism can be inherited by the offspring. Lamarck presented the idea of the inheritance of acquired characters as self-evident.

7. Correct option: (D) All of the above

Vestigial organs can be considered as evidence of evolution because vestigial structures are often homologous to structures that are functioning normally in other species.

Homologous organs are organs which are similar in structure and developmental origin but perform different functions. For example, a human arm and a bat's wing. These homologous organs point towards a common ancestor from which these organisms evolved. Hence it is considered as evidence of evolution.

Organs which have different origin and dissimilar fundamental structure but have similar functions are called analogous organs. This means that analogous organs do not show common ancestor though they show convergent evolution. Hence analogous organs are considered as evidence of evolution.

Hence, homologous organs, analogous organs and vestigial organs is a set of evidence of evolution.

So, the correct answer is 'All of the above'.

8. Correct option: (B) Darwin

Charles Darwin was a British naturalist who proposed the theory of biological evolution by natural selection. Darwin defined evolution as "descent with modification," the idea that species change over time, give rise to new species, and share a common ancestor.

9. Correct option: (C) Pea plants

Mendel carried out his key experiments using the garden pea, Pisum sativum, as a model system. Pea plants make a convenient system for studies of inheritance, and they are

still studied by some geneticists today. Useful features of peas include their rapid life cycle and the production of lots and lots of seeds.

10. Correct option: (A) Male

The human being with XY pair of chromosomes is called a male. The sex of the human offspring is determined by the set of chromosomes inherited from father.

11. Correct option: (D) Part of a chromosome that transmits a trait

A gene is the basic physical and functional unit of heredity. Genes are made up of DNA. Some genes act as instructions to make molecules called proteins. Gene is a part of a chromosome that transmits a trait.

12. Correct option: (C) Phenotype

Visible characteristics of an organism are called phenotype.

13. Correct option: (A) One

The number of pair(s) of sex chromosomes in the zygote of humans is one.

14. Correct option: (A) Copies of the same chromosome

The two versions of a trait are brought in by the male and female gametes situated on copies of the same chromosome. During fertilization one of the chromosomes comes from the father through the male gamete and the other chromosome comes from the mother through the female gamete.

15. Correct option: (B) Mendel

Gregor Johann Mendel is known as the father of genetics.

16. Correct option: (A) Homologous organs

The organs that perform different functions but have the same basic structure are known as homologous organs. For example, forelimbs of humans and lizards are homologous.

17. Correct option: (B) Analogous organs

The organs which perform a similar function, but have different basic structures are called analogous organs. For example, wings of insects and birds are analogous.

18. Correct option: (A) Analogous organs

Wings of an insect and forelimbs of a bird are analogous organs, as they have different structures but perform the same function of flying.

19. Correct option: (C) S Miller

In 1953, Stanley L. Miller and Harold C. Urey, working at the University of Chicago, conducted an experiment which would change the approach of scientific investigation into the origin of life. Miller took molecules which were believed to represent the major components of the early Earth's atmosphere and put them into a closed system

20. Correct option: (D) Convergent evolution

The bodies of whales, seals, and sharks are streamlined and smooth. Whales, seals, and sharks have a single tail fin which is used for propulsion. These similar methods in whales, seals, and sharks for locomotion are examples of convergent evolution.

21. Correct option: (A) Possible origin of life by the combination of chemicals under suitable environmental conditions

During the course of evolution, the reducing environment of earth favoured the accumulation and reaction of elements to form bioorganic compounds which then formed the building blocks of life.

22. Correct option: (C) Australia

Every surviving marsupial is endemic to Australasia and the Americas. An unmistakable trademark normal to these species is that the majority of the youthful are conveyed in a pouch.

23. Correct option: (B) Homology

Homologous organs: The organs of different species of common descent which look different and perform different functions, but have the similar basic structure, similar topographic relationship, and similar embryonic origin.

24. Correct option: (A) i – b, ii – a, iii – d, iv – c

(i) Darwin - b) Origin of species

Charles Darwin which is considered to be the foundation of evolutionary biology is the author of 'The Origin of Species'. Darwin's book introduced the scientific theory that populations evolve over the course of generations through a process of natural selection.

ii) Lamarck - a) Use and disuse theory

The theory of use and disuse was proposed by Lamarck. He suggested that changes in traits or characters occur in organisms depending on whether an organ is used or disused continuously

iii) Hugo de Vries - d) Mutation theory

The mutation Theory of Evolution was proposed by Hugo de Vries.

iv) A. S. Oparin - c) Origin of life

Oparin believed that the life developed from microscopic spontaneously formed spherical lipid molecules held together by electrostatic forces, probably the earliest form of cells. These molecules most likely functioned as enzymes, essential for the biochemical metabolic reactions necessary for life's evolution.

25. Correct option: (B) Biogenetic law → Recapitulation theory

Biogenetic law, also called Recapitulation Theory, postulation, by Ernst Haeckel in 1866, that ontogeny recapitulates phylogeny—i.e., the development of the animal embryo and young traces the evolutionary development of the species.

26. Correct option: (A) Mesozoic

Amid the Mesozoic Era, life broadened quickly, and monster reptiles, dinosaurs, and different tremendous brutes meandered the Earth. The period, which ranges from around 252 million years back to around 66 million years prior, was otherwise called the time of reptiles or the time of dinosaurs.

27. Correct option: (B) Mutations are random and directionless

Mutations are random changes taking place at the level nucleotide sequence in DNA. Mutation is directionless means that nobody knows if it might result in something beneficial for the organism or harmful.

28. Correct option: (B) Analogous

The potato underground stem has evolved, allowing it to store food in the form of starch. Sweet potato roots, on the other hand, have been modified for the same purpose. The edible parts are the modified potato stem and the modified sweet potato root.

29. Correct option: (B)Oparin and Haldane

According to chemical theory of origin of life proposed by Oparin and Haldane, there formed the evolution of chemical substances that preceded the actual origin of life.

30. Correct option: (A) Charles Darwin

Darwin applied the phrase “survival of fittest” to refer to the unequal ability of individuals to survive and reproduce based on suitability of their inherited characters to prevailing environmental conditions. Lamarck viewed the inheritance of acquired characters as the basis of evolution.

31. Correct option: (D) All the above

The classical example of adaptive radiation during formation of new species is marsupials of Australia, Darwin's finches and giant Tortoise.

32. Correct option: (C) Darwin

English naturalist Charles Darwin developed the idea of natural selection after a five-year voyage to study plants, animals, and fossils in South America and on islands in the Pacific. In 1859, he brought the idea of natural selection to the attention of the world in his best-selling book, On the Origin of Species.

33. Correct option: (A) Both 'A' & 'R' are true, and 'R' is the correct explanation of 'A'

Assertion is correct, as natural selection is the outcome of difference in survival and reproduction among individuals that show variation in one or more traits. It is a key mechanism of evolution, the change in heritable traits of a population over time.

Reason is also correct and correct explanation of assertion as adaptive forms of a given trait tend to become more common, less adaptive ones become less common or disappear. The unequal ability of individuals to survive and reproduce will lead to a gradual change in population with favourable characteristics accumulating over the generation, which leads to the formation of new species.

34. Correct option: (C) Theory of mutations

The theory of mutation given by Hugo de Vries suggested that evolution is fuelled by the occurrence of the discontinuous process called mutation, that arises in the genetic material suddenly and are transmitted to the next generation.

35. Correct option: (A) Homologous organs

Tendrils of Cucurbita and thorns of Bougainvillea are homologous organs because they are similar in origin as they are modified branches and axial in position. Their different function is modified for protection and in Cucurbita, tendrils are for climbing.

36. Correct option: (D) All the above

Synthetic theory is considered as Modern theory and the main contributor of this theory was G.L.Stebbins. They considered 5 factors, responsible for organic evolution. They are as follows 1. Mutation 2. Genetic recombination 3. Chromosomal aberration 4. Reproductive isolation 5. Natural Selection

37. Correct option: (C) Adaptive radiation

Adaptive radiation is a developmental example described by a fast increment in the number of sorts of firmly related species.

38. Correct option: (C) Theory of natural selection

The theory of evolution by natural selection, first formulated in Darwin's book "On the Origin of Species" in 1859. It is the process by which organisms change over time as a result of changes in heritable physical or behavioural traits.

39. Correct option: (D) Darwin

On the Origin of Species', published on 24th November 1859, is a work of scientific literature by Charles Darwin which is the foundation of evolutionary biology. Its full title was 'On the Origin of Species by Means of Natural Selection or the Preservation of Favoured Races in the Struggle for Life'. Darwin's book introduced the scientific theory that natural selection and presented a body of evidence gathered from the Beagle expedition, stating that the diversity of life descended from a common ancestor through evolution.

40. Correct option: (D) Stanley Miller

In 1953 American chemists Harold C. Urey and Stanley Miller tested the Oparin-Haldane theory and successfully produced organic molecules from some of the inorganic components thought to have been present on prebiotic Earth. This became known as the Miller-Urey experiment.

LEVEL II

1. Correct option: (A) Cabbage

Wild cabbage is the common ancestor of broccoli, kohlrabi and kale.

2. Correct option: (B) Africa

According to the paleontological evidence, the fossils of the earliest man, Homo naledi was found in the regions of South Africa. The human species has genetic roots in Africa.

3. Correct option: (D) All

Excavating, time-dating and studying fossils, as well as determining DNA sequences – have been used for studying human evolution.

4. Correct option: (B) Big nose

Big nose is an inherited trait which can be passed from parents to offspring.

5. Correct option: (D) fossils

The remains (or impressions) of dead animals or plants that lived in the remote past are known as fossils.

6. Correct option: (B) dinosaur and fish (Knightian)

Fish (Knightian) and dinosaurs are now extinct and are studied from their fossils.

7. Correct option: (A) weight of a person

Height of a person depends on genes, that's why Our height depends on height of our parents. Genes also control many other things like colour of eye, body colour can also be inherited. Weight is acquired during life due to factors including lifestyle.

8. Correct option: (B) Size of body

Body size as it can change during the lifetime of a person (an individual may become fat or thin during his or her lifetime) but it cannot be passed to the next generation i.e., it is not inheritable.

9. Correct option: (B) self-pollination

Two pink coloured flowers on crossing resulted in 1 red, 2 pink, and 1 white flower progeny. The nature of the cross will be self-pollination.

10. Correct option: (A) 50%

Since the ratio of X chromosome and those containing Y chromosome is 50:50 in a male gamete, the statistical probability of male or a female offspring also becomes 50:50. In human beings, the statistical probability of getting either a male or female child is 50: 50.

11. Correct option: (B) girl

Father can donate either X or Y to the ovum with X chromosome, if zygote is of XX each from father and mother respectively, then the child must be a girl with XX chromosome.

12. Correct option: (D) Natural selection can alter frequency of an inherited trait.

The fourth statement, natural selection can alter the frequencies of an inherited character, is true with respect to theories of evolution and is not a direct conclusion of Mendelian experiments.

13. Correct option: (D) All

All of them can be called as characteristics.

14. Correct option: (A) Gall bladder

Because the gallbladder can be removed from humans without much effect and is not present in many animals, it is sometimes mistakenly classified as a vestigial organ—a body part that has lost all or most of its original function through evolution.

15. Correct option: (B) 9: 3: 3: 1

Mendel's cross on pea plants produced an observable ratio of 9: 3: 3: 1 as a result of the independent inheritance of two distinct traits, seed shape, and colour.

16. Correct option: (A) Chromosomes get halved during gamete formation

The number of chromosomes in parents and offsprings of a particular species remains constant due to the halving of chromosomes during gamete formation.

17. Correct option: (D) Equal chance of acquiring blood group A or blood group O.

The given information is not enough to determine the traits- blood group A or O is dominant.

In ABO blood grouping type, A is always dominant whereas, the blood type O is always recessive.

Here the father's blood group is A hence, it can either be homozygous (AA) or heterozygous (AO).

The mother's blood group is O and it is only possible if it is heterozygous (OO).

For the daughter to have blood group O the father must be heterozygous (AO) and mother must be (OO). but there are equal chances for acquiring blood group A and O.

18. Correct option: (B) Yeast fungus

Yeast fungus does not reproduce by spore formation method.

19. Correct option: (C) Genetic drift does not involve competition between members of a species.

Genetics drift is a sudden and random process, it may cause loss of some significant and vital genes in the process of evolution whereas natural selection is a non-random process due to which there is always the survival of the fittest and useful genes are added up in the process of evolution.

20. Correct option: (C) TtWW

A Mendelian experiment consisted of breeding tall pea plants bearing violet flowers with short pea plants bearing white flowers. The progeny all bore violet flowers, but almost half of them were short. This suggests that the genetic make-up of the tall parent can be depicted as TtWW.

21. Correct option: (D) Overproduction of offspring

22. Correct option: (A) A Chinese boy

We have more in common with a Chinese school-boy because both belongs to the same species of Homo sapiens, and thus, share a common ancestor.

23. Correct option: (D) birds have evolved from reptiles

Here, evolution means that birds have evolved from reptiles both mammals and birds evolved from reptiles, though not the same ones - mammals evolved from the mammal-like reptiles, whilst birds evolved from dinosaurs.

24. Correct option: (B) Round and yellow

If a round, green seeded pea plant (RRyy) is crossed with wrinkled, yellow seeded pea plant, (rrYY) the seeds produced in F1generation are round and yellow.

Parents: RRyy X rrYY

Gametes: Ry rY

F1: RrYy (Round yellow)

25. Correct option: (A) two individuals of a species

Species is the lowest level of the taxonomic classification hierarchy and the members of the species share the most common characteristics with themselves. Thus, any two individuals of a species share the maximum numbers of related or common characters.

26. Correct option: (C) All variations in a species have equal chance of survival

To survive, nature selects the most adaptable variant. This is Darwin's theory of Natural selection.

27. Correct option: (B) accumulation of variations over several generations

According to the evolutionary theory, the formation of a new species is generally due to accumulation of variations over several generations.

28. Correct option: (D) the extinction of organism has occurred thousands of years ago

If the fossil of an organism is found in the deeper layers of the Earth, then we can predict that the extinction of organisms occurred thousands of years ago.

29. Correct option: (A) Only 1

1. Weight of a person

The correct answer is weight of a person. Height of a person depends on genes, that's why Our height depends on height of our parents.

30. Correct option: (C) (iii) and (iv)

These unpaired chromosomes are: Y-chromosome and. X-chromosome.

9. Our Environment – The Concern Solutions

LEVEL - I

1. Correct option: (C) The producer

The producer, which is typically a plant or autotrophic organism, is where a food chain always begins. Producers are able to produce their own food and turn sunlight into energy through a process known as photosynthesis. As the main source of energy for all other organisms in the ecosystem, they are at the bottom of the food chain. These producers are eaten by herbivores (animals that eat plants), and the herbivores may then be eaten by carnivores (animals that eat other animals). The foundation of a food chain is this orderly transfer of energy from one organism to another, with producers always marking the beginning of the energy cycle through an ecosystem.

2. Correct option: (B) Food

Due to their ability to produce their own food through the process of photosynthesis, plants do not compete for resources. Plants produce glucose and oxygen through the process of photosynthesis, which involves sunlight, carbon dioxide, and water. Their source of food and energy is this glucose. Animals must find and eat outside sources of food, but plants do not. Instead, they can produce their own nutrients by utilizing the energy from the sun.

However, there is competition between plants for other resources like water, sunlight, and space. Water is necessary for photosynthesis, so plants with greater access to water will flourish over those without it. Because there isn't much room, plants that can grow taller or wider will be able to absorb more sunlight. Because photosynthesis depends on sunlight, plants that can receive more of it will be able to produce more food.

3. Correct option: (C) Promote eco-friendly agricultural practices

If all pesticides were banned, farmers would no longer be able to control pests and diseases in their crops with pesticides. In order to do this, they would need to find alternative pest control strategies, like crop rotation, biological control, and integrated pest management. Although these techniques are less effective and frequently require more work than pesticides, they are less harmful to the environment and people's health.

Controlling the use of pesticides, which is Option A, is not the same as outright banning them. By imposing restrictions on where, when, and how much pesticides can be used, it is possible to manage their use. Nevertheless, this does not make pesticide-related risks obsolete.

Preventing pesticide use in option B is also not feasible. In addition to agriculture, many other industries also use pesticides. It would be very challenging to completely stop using them.

Option D, shutting down biochemical plants, has nothing to do with outlawing pesticides. Pesticides are among the many products produced by biochemical factories. A ban on pesticides would not necessarily result from closing down these factories.

4. Correct option: (B) Energy trapping is high at the top of the pyramid

The correct answer is B) Energy trapping is high at the top of the pyramid. Ecologist Charles Elton is renowned for his studies of ecological pyramids, particularly the pyramid of energy. His ecological pyramid theory states that as you move up the food chain in an ecosystem, the amount of energy available to each trophic level decreases.

The producers, usually plants, capture solar energy and use photosynthesis to transform it into chemical energy. Because not all of the energy from one trophic level is transferred to the next as you move up to herbivores and then carnivores, there is a progressive loss of energy. This means that relatively little energy is still available when compared to the lower levels by the time you reach the top of the pyramid, which is why it's important to understand how energy flows through ecosystems. Carnivores are typically at the top of the food chain, but Elton's theory does not center on them, so Option A is not entirely accurate. Option C is also incorrect because the top of the pyramid, which belongs to consumers instead of producers, is incorrect.

5. Correct option: (B) Grass, Goat and Human

Each organism acts as a link in the flow of nutrients and energy in this chain, which represents a food chain. The primary producer of solar energy that transforms it into plant matter is grass. The primary consumer of grass, the goat uses it as a source of energy and nutrients. Last but not least, the human is a secondary consumer who ingests the goat, thus obtaining energy and nutrients from the grass indirectly. The basic structure of a food chain, which depicts the flow of energy through an ecosystem, is the linear progression of energy transfer from one organism to another.

6. Correct option: (A) Photosynthetic

Primary producers, or photosynthetic organisms, typically occupy the first trophic level in a food chain. Through the process of photosynthesis, these organisms, primarily plants and specific kinds of algae and bacteria, obtain energy from the sun by converting light, carbon dioxide, and water into organic material (food) and releasing oxygen. They serve as the base of the food chain and supply the nutrients and energy needed by organisms at higher trophic levels, like herbivores. Since they are the main sources of energy entering the food chain, these photosynthetic organisms are essential to the ecosystem's ability to transfer energy.

7. Correct option: (D) They are made of non - biodegradable materials

The correct answer is D) They are made of non-biodegradable materials. Due to their typical construction of materials like polyethylene or polypropylene, which are difficult to decompose in the environment, disposable plastic plates are not environmentally friendly. Non-biodegradable plastics persist in the environment for hundreds of years, causing pollution and harming wildlife, in contrast to biodegradable materials that can naturally decompose and return to the Earth without causing long-term harm. When they are not properly discarded or recycled, they can build up in ecosystems, clog waterways, and pose a serious threat to the environment and wildlife. To lessen the environmental impact caused by their persistence in the environment, the use of disposable plastic plates that are not biodegradable is discouraged.

8. Correct option: (A) Carnivores

Carnivores typically make up the third trophic level in a food chain. The energy and nutrients for these organisms come from the organisms at lower trophic levels. They are primary consumers that feed on herbivores or other carnivores. The fact that carnivores consume animal flesh and other byproducts defines them. They are essential for keeping herbivore populations under control and ecosystems in balance. The flow of energy and nutrients through the food chain as they are transferred from one organism to another is represented by the progression from producers (typically plants) at the first trophic level to herbivores at the second trophic level and then to carnivores at the third trophic level.

9. Correct option: (C) Biomagnification

In the food chain, a process known as biomagnification takes place where the concentration of some non-biodegradable pollutants or chemicals, such as pesticides or heavy metals, rises with each higher trophic level. This occurs because, despite being exposed to these substances in their environment, organisms at lower trophic levels are unable to effectively metabolize or eliminate them. The pollutants present in their prey are accumulated by organisms at the next trophic level when they consume multiple members of the lower level. As you move up the food chain, this process keeps going, which causes a higher concentration of the non-biodegradable substance in the top predators. Biomagnification can have detrimental effects on wildlife and humans who consume contaminated organisms, as the concentration of these pollutants can reach toxic levels and pose health and environmental risks.

10. Correct option: (C) Primary consumer to secondary consumer

The correct answer is C) Primary consumer to secondary consumer. Since the grasshopper consumes plants, the scenario's primary producers in the ecosystem, it is thought of as the primary consumer. Given that it is feeding on a primary consumer when the frog eats the grasshopper, it becomes the secondary consumer. This illustrates how energy moves up the food chain, first going to the grasshopper (the primary consumer) and then to the frog from where it was originally captured by the producers, the plants, through photosynthesis (secondary consumer). This progression is an illustration of how energy is typically transferred along a food chain, where each trophic level corresponds to a different level of consumption within the ecosystem.

11. Correct option: (C) Both living organisms and non-living objects

The correct answer is C) Both living organisms and non-living objects. An ecosystem is the collection of all interactions and connections among living things in a particular habitat or region, including plants, animals, microorganisms, and more. However, it's equally important to take into account the non-living elements, such as soil, water, air, rocks, and climate, which give life its physical and chemical necessities and affect how the ecosystem works. Ecosystems depend on both biotic (living) and abiotic (non-living) elements. For the majority of terrestrial ecosystems, sunlight is a vital source of energy. Therefore, the dynamic interaction between living things and their physical environment is included in the definition of an ecosystem.

12. Correct option: (D) The population of tigers decreases and the population of grass increases

The correct answer is D) The population of tigers decreases, and the population of grass increases. The primary producer in this food chain is grass, the primary consumer is a deer, and the top predator or tertiary consumer is a tiger. The tiger would lose its main source of prey if deer were to disappear from the food chain. Because of a lack of food, the number of tigers would probably decline as a result. On the other hand, since deer primarily consume plants, including grass, their absence would not have a direct effect on the population of grass. The population of grass might increase if there are fewer deer grazing on it because there would be less pressure from herbivores. With a decrease in the top predator and an increase in other species, this change in population dynamics would have a cascading effect on the ecosystem.

13. Correct option: (B) Producers

The correct answer is B) Producers. Producing organisms are those that create carbohydrates using radiant energy, usually from sunlight. Plants, algae, and specific bacteria make up the majority of these organisms. Through the process of photosynthesis, which involves converting carbon dioxide and water into glucose and

other carbohydrates, they play a crucial ecological role by absorbing energy from the sun. Having the capacity to produce their own food makes them the base of the food chain because they supply the nutrients and energy needed to sustain all other organisms in the ecosystem. Producers are essential to ecosystems because they provide food for herbivores, which in turn provide food for carnivores and other higher trophic level organisms in the food chain.

14. Correct option: (B) Food web

The correct answer is B) Food web. A food web is an ecosystem's intricate network of various connected food chains. It demonstrates the interactions and trophic level-spanning feeding relationships between various species. A food web offers a more accurate and thorough view of how energy and nutrients flow through an ecosystem than a single linear food chain, which only depicts a straightforward predator-prey relationship. It takes into account the fact that numerous species in an ecosystem can belong to various food chains and have access to a variety of food sources. Because changes in one species can have an impact on the entire web, this complexity better captures the complexities of actual ecological relationships and aids in our understanding of the dynamic balance within ecosystems.

15. Correct option: (B) Producers prepare their own food from readymade organic compounds

The incorrect statement is B) "Producers prepare their own food from readymade organic compounds." Producers like green plants and blue-green algae create their own food through the process of photosynthesis rather than preparing it from readymade organic compounds. Utilizing the energy from sunlight, photosynthesis is the process of converting inorganic substances like carbon dioxide and water into organic substances, especially carbohydrates like glucose. Producing food this way enables farmers to use raw materials and radiant energy rather than pre-existing organic compounds. Therefore, statement B is untrue because it misrepresents the method by which producers obtain their food.

16. Correct option: (C) ii and iii

The correct answer is C) ii and iii. The linear transfer of nutrients and energy from one organism to another within an ecosystem is represented by food chains. The food chain in option ii, "Plankton, Man, Fish, Grasshopper," is not simple because it also includes humans, who can eat a wide range of foods in addition to plankton, fish, and grasshoppers. The transfer of energy from one trophic level to another is therefore not obvious or linear. It is also not a food chain in option iii, "Wolf, grass snake, tiger," as there is no transfer of energy from producers to consumers. Without taking into account the intermediate trophic levels that are ordinarily present in a food chain, it directly connects a wolf to a tiger. Options I and iv both depict legitimate food chains because they demonstrate

17. Correct option: (C) T_1

The correct answer is C) T1. Each trophic level in an ecological pyramid, such as the one in the question, corresponds to a different level of the food chain. For example, T1 represents the primary producers (typically plants or autotrophic organisms), T2 the primary consumers (herbivores), T3 the secondary consumers (carnivores or omnivores that eat herbivores), and T4 the tertiary consumers (carnivores that eat other carnivores or omnivores). The energy available decreases as you move up the trophic levels due to energy loss from metabolic processes and heat production, as shown by the pyramid. Therefore, T1, the primary producers, have the maximum energy available because they directly harness energy from the sun through photosynthesis, while

energy diminishes as it is transferred to higher trophic levels, making T1 the trophic level with the most energy.

18. Correct option: (C) Aluminum foil

The correct answer is C) Aluminum foil. Due to the fact that aluminium foil is a metallic substance, it cannot be biodegraded. The term "biodegradability" describes a material's capacity to gradually and naturally decompose into harmless substances with the aid of microorganisms and natural environmental processes. Aluminum foil is not biodegradable, whereas cotton, animal bones, and wood are organic materials that can break down and reintegrate into the environment because they are made of carbon-based compounds. If not properly disposed of or recycled, it can persist in the environment for a very long time and is difficult for natural ecosystems to break down. If this leads to pollution.

19. Correct option: (C) Jute carry bag

Jute is a naturally occurring fiber that is derived from the stems of the jute plant and is thought to be biodegradable. When exposed to the environment, it can decompose over time and transform into organic matter without leaving behind any harmful residues. While synthetic materials like nylon and polyester are not biodegradable, options A, B, and D (Nylon socks, Plastic school bag, and Polyester clothing) are. These synthetic materials can persist in the environment for long periods of time and, if improperly disposed of or recycled, can cause pollution because they do not decompose easily in natural ecosystems. The jute carry bag is therefore biodegradable and not non-biodegradable.

20. Correct option: (C) Nylon cloth carry bags

The correct answer is C) Nylon cloth carry bags. Because of its long-lasting durability and lack of biodegradability, nylon is a synthetic material. When not properly managed, its durability, which can be advantageous in terms of reuse, turns into a serious environmental issue. When improperly disposed of or left as litter, nylon cloth carry bags can linger in the environment for many years and add to the plastic pollution. They may degrade into smaller microplastic particles over time, further polluting ecosystems and endangering wildlife. Options A (Paper carry bags), B (Cotton cloth carry bags), and D (Jute carry bags), on the other hand, are made of natural or biodegradable materials and, as a result, are typically more environmentally friendly options as they decompose more quickly and have less of a negative impact on the environment.

21. Correct option: (C) Algae

The correct answer is C) Algae. In terms of ecology, algae are not regarded as consumers. They are chief producers instead. Algae are photosynthesis-capable organisms that use water, carbon dioxide, and sunlight to create organic compounds that are high in energy. By giving energy and nutrients to herbivores, who are the main consumers and suppliers of food for them, they serve as the foundation of aquatic food chains. The other options (Giraffe, Antelope, and Alligator) are consumers, in contrast. Alligators are carnivores, whereas giraffes and antelopes are herbivores. They are all consumers in their respective ecosystems because they all get their energy from eating other living things.

22. Correct option: (B) Zooplankton

The correct answer is B) Zooplankton. Zooplankton are consumers, not producers. Zooplankton are minuscule aquatic organisms that float in bodies of water. Their main sources of food are detritus and phytoplankton, which are microscopic organisms that are capable of photosynthesis. They are members of the heterotrophic group of

organisms, which means that they get their energy and nutrients from eating other organisms or organic matter. Options A (Grass), C (Phytoplankton), and D (Paddy) on the other hand, are all producers. Paddy refers to rice plants, which are also producers because they produce their own food through photosynthesis. Grass is a terrestrial plant that uses photosynthesis to produce its own food. Phytoplankton are aquatic photosynthetic microorganisms.

23. Correct option: (B) Saprotrophs

Saprotrophs, also referred to as decomposers or detritivores, are essential to an ecosystem's ability to recycle materials. Through the process of decomposition, these organisms transform dead organic matter, such as fallen leaves, dead animals, and plant debris, into less complex substances. By doing this, they return nutrients to the environment where plants and other primary producers can absorb them. Because it ensures that vital elements like carbon, nitrogen, and phosphorus are continuously cycled through the ecosystem, supporting the growth of new organisms and maintaining ecological balance, nutrient recycling is crucial for the sustainability of ecosystems. Autotrophs (A), omnivores (C), and carnivores (D) have important roles in ecosystems but are not primarily responsible for material recycling through decomposition.

24. Correct option: (C) Frog

The frog is the secondary consumer in the given food chain, which also includes a snake, grass, insect, and a snake. Organisms known as secondary consumers eat primary consumers, in this case the insect. Since it provides nutrients and energy at the base of the food chain, grass serves as the primary producer. The main consumer is an insect that eats grass as an herbivore. Because it preys on the frog, which is the secondary consumer, the snake is the third consumer. Since the frog feeds on primary consumers (insects) and is then preyed upon by the snake, which is higher in the trophic hierarchy, it occupies the position of a secondary consumer in this food chain.

25. Correct option: (C) Excessive killing of large carnivores

The Sahara Desert was not formed due to excessive killing of large carnivores or any uncontrolled activity related to them. The Sahara Desert's formation is primarily attributed to long-term changes in Earth's climate patterns and geological processes. Sahara desertification is thought to have resulted from changes in Earth's orbital parameters, variations in ocean currents, and changes in solar radiation. Due to a combination of these factors, the area gradually dried out, changing it from a more hospitable, vegetated environment into the dry desert that we see today. The formation of the Sahara Desert, which is primarily the result of natural climatic and geological forces, is not the result of human activities such as deforestation (option A), overhunting of large herbivores (option B), and the use of pesticides (option D).

26. Correct option: (D) Sunlight

The correct answer is D) Sunlight. Sunlight provides the initial source of energy that flows through a food chain and fuels all life on Earth. Plants, algae, and some bacteria use the process of photosynthesis to absorb solar energy and transform it into chemical energy in the form of glucose and other carbohydrates. The primary consumers (herbivores) receive this stored energy when they consume the plants, which is how these primary producers create their own food using sunlight. As herbivores are consumed by carnivores or omnivores at higher trophic levels, the energy continues to move up the food chain. In the end, sunlight serves as the primary source of energy for all living things in an ecosystem, whether they use it directly or indirectly to power ecological processes and maintain life.

27. Correct option: (C) CFC

The greatest contributor to the depletion of the ozone layer from refrigeration and air conditioning systems is Chlorofluorocarbons (CFCs). CFCs were frequently used in aerosol sprays and as refrigerants, but when they were released into the atmosphere, the ozone layer suffered severe damage. When CFCs are released, they eventually ascend to the stratosphere where ultraviolet (UV) radiation breaks them down. Ozone (O3) molecules are destroyed as a result of the reaction between chlorine atoms, which are released by this breakdown, and ozone molecules. Increased exposure to the sun's harmful UV radiation, which can lead to health issues like skin cancer and damage ecosystems, is one of the negative effects of the ozone layer's depletion. International agreements like the Montreal Protocol have been established to gradually phase out the use of CFCs and other ozone-depleting substances as a result of these negative effects.

28. Correct option: (C) 5000kJ

In a typical food chain, about 10% of the energy from one trophic level is transferred to the next. If the fourth trophic level has 5kJ, then the third trophic level would have had 10 times that amount, or 50kJ. Similarly, the second trophic level would have had 500kJ, and the first trophic level (the producer level) would have had 5000kJ.

So, the correct answer is C) 5000kJ.

In the given food chain, the energy transfer is as follows:

Producers (1st trophic level) = 5000kJ

Primary Consumers (2nd trophic level) = 10% of 5000kJ = 500kJ

Secondary Consumers (3rd trophic level) = 10% of 500kJ = 50kJ

Tertiary Consumers (4th trophic level) = 10% of 50kJ = 5kJ.

29. Correct option: (A) Pb>Cr>Fe>Ni> Cd

The correct answer is A) Pb > Cr > Fe > Ni > Cd. This pattern shows how the reservoir's EBWR (East Branch White River) reservoir's metal concentration rises over time. Lead (Pb) is the element with the highest concentration, followed by chromium (Cr), iron (Fe), nickel (Ni), and finally cadmium (Cd), which has the lowest concentration. Given that different metals can have varying degrees of toxicity and effects on aquatic ecosystems and human health, this information is crucial for understanding the water quality and potential environmental effects of these metals in the reservoir.

30. Correct option: (A)

Cd > Pb >Ni > Fe >Cv

The correct answer is A) Cd > Pb > Ni > Fe > Cv. The bioaccumulation of heavy metals in fish tissues is represented by this sequence. Fish, for example, accumulate substances like heavy metals from their environment over time through a process known as bioaccumulation. In this instance, lead (Pb), nickel (Ni), iron (Fe), and copper have the highest levels of cadmium (Cd) in fish tissues (Cv). Because some of these metals can be toxic and their accumulation in the food chain can present risks, this order is significant because it shows the relative levels of these metals in the fish and may have implications for ecological and human health concerns.

LEVEL – II

1. Correct option: (D) All of these

The answer is D) All of these. All the components mentioned in the options are included in the broad concept of the environment. It includes the abiotic factors of the environment, which are the physical elements like land, air, and water (Option A). Additionally, it incorporates elements that are important abiotic elements influencing ecosystems, such as light, temperature, and rainfall (Option B). In addition, the

environment is made up of biotic components like plants, animals, and microbes (Option C), which stand in for the interactions between living things found in ecosystems. The environment therefore includes all of these elements, making option D the best one because it takes into account the full range of elements that make up our surroundings as well as the ecological systems in which they exist.

2. Correct option: (C) Energy level

The correct answer is C) Energy level. A trophic level in an ecosystem refers to a particular place in the food chain or food web and is connected to the movement of energy from one group of organisms to another. A different stage in the flow of energy through the ecosystem is represented by each trophic level. In the first trophic level are producers like plants that use photosynthesis to absorb solar energy. Due to their consumption of producers, herbivores live in the second trophic level. The following trophic levels are occupied by carnivores or omnivores that consume herbivores. The idea of trophic levels helps us comprehend how energy flows and is distributed within an ecosystem. With each trophic level, there is a transfer of energy from one set of organisms to another.

3. Correct option: (A) Using CFLs at homes and offices

The environment-friendly practice is A) Using CFLs (Compact Fluorescent Lamps) at homes and offices. Because they are energy-efficient and leave a significantly smaller carbon footprint than conventional incandescent bulbs, CFLs are regarded as being environmentally friendly. Because CFLs use less electricity to produce the same amount of light, they contribute less to the greenhouse gas emissions linked to the production of electricity. Options B and C, in contrast, call for energy-intensive actions like using a car for short trips or running the air conditioner in every room, both of which increase energy consumption and have a negative impact on the environment. Option D, burning plastics, is extremely harmful to the environment because it emits hazardous chemicals, causes pollution, and accelerates climate change. As a result, using CFLs is a sensible and sustainable option for lowering energy consumption and fostering sustainability.

4. Correct option: (A) Skin Cancer

The answer is A) Skin Cancer. Sunlight's UV radiation has the potential to harm skin cells' DNA. Skin cancer risk can be raised by prolonged and excessive exposure to UV radiation, especially UVB and UVA rays. The most prevalent form of cancer in the world is skin cancer, which can take many different forms including basal cell carcinoma, squamous cell carcinoma, and melanoma. Skin cancer can develop as a result of DNA mutations brought on by UV radiation that can cause unchecked cell growth in skin cells. In order to lower the risk of skin cancer, it is crucial to protect the skin from excessive sun exposure by taking precautions like donning sunscreen, wearing protective clothing, and finding shade.

5. Correct option: (B) DDT

The answer is B) DDT. Dichlorodiphenyltrichloroethane, or DDT, is a synthetic pesticide that was widely used to control insects and pests in the middle of the 20th century. The persistence of DDT in the environment is one of its worrying traits. DDT can build up in the fatty tissues of organisms after being sprayed on crops and become more concentrated as it moves up the food chain. Biomagnification is the name given to this process. As a result, when apex predators like humans eat creatures lower on the food chain, they may end up with high levels of DDT in their bodies. Since DDT has been discovered to have negative effects on both human health and the environment, many nations have banned or severely restricted its use.

6. Correct option: (D) 5000KJ

The energy at the producer level will be 5000KJ. This is due to the fact that energy decreases as it rises through a food chain's trophic levels. Only about 10% of the energy at each trophic level comes from the level beneath it. The energy at the first trophic level (the producer level) would be 5000KJ if the energy at the fourth trophic level is 5KJ, 50KJ at the third trophic level, 500KJ at the second trophic level, and 5000KJ at the fourth trophic level.

7. Correct option: (A) Abiotic components

The answer is A) Abiotic components. The non-living, abiotic elements of an ecosystem that affect its dynamics and characteristics are referred to as edaphic factors. The properties of the soil, including its texture, composition, pH, moisture content, temperature, and mineral content, are among the factors that primarily affect the soil and substrate within an ecosystem. Because they directly affect the availability of vital nutrients, water retention, and physical support for plants and other organisms in the ecosystem, edaphic factors are significant. These abiotic factors have a significant impact on the kinds of plant species that can flourish in a specific location and, consequently, have an impact on the ecosystem's overall structure and function.

8. Correct option: (B) convert organic material to inorganic forms

The answer is B) Convert organic material to inorganic forms. By dissolving organic materials like dead plants, animals, and other organic matter, decomposers like bacteria and fungi play a critical role in the ecosystem. They achieve this through the process of decomposition, in which complex organic compounds are enzymatically broken down into more basic, inorganic forms like carbon dioxide, water, and minerals. For nutrients to be recycled back into the ecosystem and made available for other organisms to use, organic material must be transformed into inorganic forms. It is crucial to the sustainability of ecosystems as a whole and the cycling of nutrients.

9. Correct option: (B) Needle of the syringe

The correct answer is B) Needle of the syringe. A technique for getting rid of waste called incineration involves burning stuff at really high temperatures, usually in a controlled setting like an incinerator. Only the syringe's needle can be safely burned when choosing from the options offered. Although paper can be burned, recycling it is frequently more environmentally friendly. Contrarily, medicine bottles are typically made of plastic, which when burned can emit dangerous fumes and is typically better suited for recycling or proper disposal in accordance with local laws. Syringe needles can be safely destroyed in a medical waste incinerator, lowering the danger of contamination or injury from used medical sharps.

10. Correct option: (C) (A) and (B)

The answer is C) (A) and (B). Both the statements (A) and (B) are true in a food web. As you move up the food chain or web, there is a loss of energy at each trophic level due to metabolic processes and heat production, which results in less energy being transferred to higher levels. Energy moves progressively through different trophic levels as organisms consume one another, with predators gaining energy from their prey, as stated in (B). The interconnectedness and energy transfer within ecosystems are demonstrated by the flow of energy through various levels, which is a fundamental characteristic of food webs.

11. Correct option: (D) This will reduce the release of CFCs that reacts with ozone molecules

The correct answer is D) This will reduce the release of CFCs that react with ozone molecules. The production of CFC-free refrigerators is required globally because doing

so helps stop ozone depletion. Synthetic CFCs can migrate into the stratosphere after being released into the atmosphere, where they are destroyed by ultraviolet (UV) light. Chlorine atoms are released during this breakdown, and when they come into contact with ozone (O3) molecules, they cause the ozone molecules to be destroyed. Because ozone depletion in the stratosphere causes the ozone layer to thin, which is problematic because the ozone layer is essential for shielding Earth from harmful ultraviolet (UV) radiation from the sun. By mandating CFC-free refrigerators and reducing the release of CFCs into the atmosphere, the potential for ozone depletion is significantly reduced, thereby protecting the ozone layer and mitigating the harmful effects of increased UV radiation on the environment and human health.

12. Correct option: (D) Frog: Lizard

The correct answer is D) Frog: Lizard. The frog and the lizard in this pair are both secondary consumers at the same trophic level. While lizards also primarily eat insects and small animals like rodents, frogs primarily eat insects and other small invertebrates. As consumers of comparable prey items, frogs and lizards share the same trophic level and occupy a similar ecological niche in their respective ecosystems. The other options, however, involve a predator-prey relationship between creatures from various trophic levels, such as a tiger preying on a rabbit, a vulture preying on a rat, or a hawk hunting a grasshopper, indicating various positions in the food chain.

13. Correct option: (A) Decrease in energy at higher trophic level

The correct answer is A) Decrease in energy at higher trophic levels. As you move up the trophic hierarchy, the amount of energy available decreases, which places a cap on the number of trophic levels in a food chain or food web. Consuming organisms from the lower trophic level provides energy to organisms at each subsequent trophic level. However, metabolic functions, heat production, and other inefficiencies result in energy loss at every transfer. As a result, organisms at higher trophic levels have less energy available to them. Many trophic levels in a food chain cannot exist due to this energy restriction. The majority of ecosystems have three to four trophic levels, with top predators occupying the highest level because there isn't enough energy to support any more levels below that.

14. Correct option: (D) Both (B) and (C)

The correct answer is D) Both (B) and (C). Sulfur and nitrogen oxides, in particular, are two pollutants that are released into the atmosphere and are the main culprits behind acid rain. These pollutants are frequently released by vehicles, industrial processes, and the burning of fossil fuels like coal and oil. These oxides can react with atmospheric moisture when released into the atmosphere to create sulfuric acid (H2SO4) and nitric acid (HNO3). These acids can then travel great distances by wind and atmospheric conditions before being deposited as acid rain on the surface of the Earth. Acid rain is a problem for the environment because it can harm forests, infrastructure, and aquatic ecosystems.

15. Correct option: (C) Treatment

The correct answer is C) Treatment. Treatment is typically the most expensive step in the handling of solid waste. The treatment of solid waste involves a number of procedures designed to minimize the amount and negative effects of waste materials on the environment. Composting, incineration, landfilling, and other waste processing techniques fall under this category. Treatment procedures frequently call for specialized tools, cutting-edge technology, and skilled labor, all of which raise costs. Furthermore, it can be expensive to treat waste to conform to environmental regulations and safety standards. While treatment is typically the most expensive option due to the complexity

and resource-intensive nature of waste management processes involved in transforming or disposing of solid waste in an environmentally responsible manner, options A, B, and D for collection, storage, and separation all incur costs as well.

16. Correct option: (B) Crop field

The correct answer is B) Crop field. An artificial or manmade ecosystem is a field of crops. In a crop field, humans cultivate and manage crops for agricultural purposes, and human activities like planting, harvesting, irrigation, and the use of fertilizers and pesticides greatly influence and control the ecosystem. A crop field represents a purposeful alteration of the environment to support the growth of particular plant species for food production, in contrast to natural ecosystems like ponds, lakes, and forests (options A, C, and D), which develop and evolve without direct human intervention. Although crop fields are an integral part of agriculture and contribute significantly to the world's food supply, they differ from natural ecosystems because of the extensive human management and manipulation they undergo.

17. Correct option: (D) Prevention-Reuse-Recycle-Disposal

The correct order of the waste management hierarchy, from most to least favored in terms of sustainability and environmental impact, is D) Prevention-Reuse-Recycle-Disposal.

Prevention: The preferred and most environmentally friendly action, it focuses on lowering waste generation at its source. It entails actions like cutting back on excessive consumption, selecting goods with minimal packaging, and using resources more effectively to reduce waste production.

Reuse: Reusing items is the next best course of action after prevention. It entails putting goods or resources to use for their original purpose once more or reusing them for a different purpose to increase their lifespan and decrease the need for fresh resources.

Recycle: The next step is recycling, which involves gathering and processing materials to produce new goods. By doing this, raw material extraction has a minimal negative impact on the environment, consumes less energy, and conserves resources.

Disposal: The least preferred option is disposal, which includes incineration or sending waste to landfills because it frequently results in the waste of valuable resources and may have negative environmental effects. Only when prevention, reuse, and recycling are not practical should it be taken into account.

The waste management hierarchy promotes responsible resource management, waste reduction, and environmental protection by prioritizing practices that minimize waste generation and promote resource conservation.

18. Correct option: (C) Chemical energy

The correct answer is C) Chemical energy. Energy moves up and down the trophic levels of the food chain in an ecosystem. Primary producers, like plants, first take in the energy through photosynthesis, converting light energy from the sun into chemical energy that is then stored as glucose and other organic compounds. After the producers are consumed, this chemical energy is transferred to herbivores (primary consumers), and it keeps moving up the trophic levels as other organisms are consumed. However, energy is lost during each transfer, primarily as heat energy as a result of metabolic processes and other inefficiencies. Therefore, only about 10% of the energy from one trophic level is typically available for transfer to the next level in the form of chemical energy, making it the primary source of energy in ecological systems.

19. Correct option: (A) Heavy metals, DDT

The correct answer is A) Heavy metals, DDT. In food chains or webs, a process called biomagnification, also known as biological magnification, causes the concentration of some non-biodegradable pollutants, such as heavy metals like mercury and persistent organic pollutants like DDT (dichlorodiphenyltrichloroethane), to rise with each higher trophic level. This occurs because, despite being exposed to these substances in their environment, organisms at lower trophic levels are unable to effectively metabolize or eliminate them. The pollutants present in their prey are accumulated by organisms at the next trophic level when they consume multiple members of the lower level. This process continues as you move up the food chain, leading to a higher concentration of heavy metals and DDT in top predators. Wildlife and people who consume contaminated organisms could suffer as a result, as

20. Correct option: (D) They are made up of non-biodegradable materials

The correct answer is D) They are made up of non-biodegradable materials. Plastics that are not biodegradable, like polyethylene or polypropylene, are typically used to make disposable plastic plates. These materials persist for many years after disposal because they do not easily decompose in the environment. As a result, they significantly contribute to plastic pollution, a global environmental issue. Plastics that are not biodegradable can build up in landfills, oceans, and other ecosystems, endangering wildlife, marine life, and the health of the planet as a whole. There is a growing movement to reduce the use of disposable plastics and support more eco-friendly substitutes that are easily biodegradable or recyclable in order to lessen these negative environmental effects.

21. Correct option: (C) They cause increased agriculture productivity

The correct answer is C) They cause increased agricultural productivity. When it comes to aerosols, this statement is false. Aerosols, in particular specific types of air pollutants and particulate matter, can have a negative impact on the environment and human health. They can aggravate cardiovascular disease, respiratory conditions, and other human health problems. By affecting cloud formation and precipitation patterns, aerosols are also known to change rainfall and monsoon patterns. Aerosols do not, however, result in greater agricultural productivity. In fact, certain aerosols in the atmosphere can have a detrimental effect on agricultural land by degrading water resources, soil quality, and crop health in addition to causing air pollution, which can harm ecosystems and crops. Increased agricultural productivity is generally associated with factors like improved farming practices, technology, and adequate resource management, rather than aerosols.

22. Correct option: (B) Biological Magnification

The correct answer is B) Biological Magnification. A phenomenon in ecosystems known as "biological magnification," or "biomagnification," occurs when the concentration of non-biodegradable pollutants—such as pesticides or specific heavy metals—increases at each higher trophic level of a food chain or web. This happens because, despite being exposed to these substances in their environment, organisms at lower trophic levels are unable to effectively metabolize or eliminate them. The pollutants present in their prey are accumulated by organisms at the next trophic level when they consume multiple members of the lower level. As you move up the food chain, this process keeps going, which causes a higher concentration of the non-biodegradable substance in the top predators. The concentration of these pollutants can increase due to biomagnification, which can have negative effects on wildlife and people who consume contaminated organisms.

23. Correct option: (D) Mortality of fish due to lack of oxygen

The correct answer is D) Mortality of fish due to lack of oxygen. Eutrophication is a process that can occur when a lake becomes abundant in organic waste. In eutrophication, an overgrowth of algae and other aquatic plants in the lake is brought on by excess nutrients, frequently from organic matter like sewage or agricultural runoff. Bacteria break down these plants as they rot and decompose, using up oxygen in the process. The amount of dissolved oxygen in the water decreases as a result of the increased microbial activity. Because fish and other aquatic organisms depend on dissolved oxygen to survive, low oxygen levels can result in fish mortality and a decline in the health of the aquatic ecosystem as a whole. In the end, this process might cause fish deaths and other detrimental ecological effects.

24. Correct option: (B) T_1

The correct answer is B) T1. The different trophic levels in an ecological pyramid correspond to different points along a food chain or web, with T1 representing the primary producers like plants or algae. As you move up the trophic levels, there is less energy available because of energy loss from metabolic processes, heat production, and other inefficiencies. This is represented by the shape of the pyramid. T1 (the primary producers) is the trophic level with the most energy available because they are the highest energy consumers in the pyramid and use photosynthesis to obtain energy from the sun. Energy availability at higher trophic levels, such as T2, T3, and T4, which are made up of herbivores, carnivores, and top predators, decreases as energy moves through the ecosystem and becomes progressively limited.

25. Correct option: (C) Biomagnification

The correct answer is C) Biomagnification. In a food chain, a process known as biomagnification takes place where the concentration of some non-biodegradable pollutants or chemicals, such as pesticides or heavy metals, rises with each higher trophic level. This occurs because, despite being exposed to these substances in their environment, organisms at lower trophic levels are unable to effectively metabolize or eliminate them. The pollutants present in their prey are accumulated by organisms at the next trophic level when they consume multiple members of the lower level. As you move up the food chain, this process keeps going, which causes a higher concentration of the non-biodegradable substance in the top predators. When pollutants are concentrated to toxic levels, biomagnification can have negative effects on wildlife and people who eat contaminated organisms.

26. Correct option: (B) Green plants get their food from organic compounds

The correct answer is B) Polyethylene bags, plastic toys. Waste products classified as non-biodegradable are those that take a long time to naturally decompose in the environment. As examples of non-biodegradable products, polyethylene bags and plastic toys are made of synthetic plastics that can last in the environment for hundreds of years without decomposing. Options A, C, and D, on the other hand, use materials that are more environmentally friendly or biodegradable. Paper straws, plant waste, and used tea bags can all naturally decompose and are typically biodegradable. While non-biodegradable materials like polyethylene bags and plastic toys are persistent and problematic in the environment, so are worn-out clothing and damaged shoes.

27. Correct option: (B) Polyethene bags, plastic toys

The correct answer is B) Polyethene bags, plastic toys. Waste products classified as non-biodegradable are those that take a long time to naturally decompose in the environment. As examples of non-biodegradable products, polyethylene bags and

plastic toys are made of synthetic plastics that can last in the environment for hundreds of years without decomposing. Options A, C, and D, on the other hand, use materials that are more environmentally friendly or biodegradable. Paper straws, plant waste, and used tea bags can all naturally decompose and are typically biodegradable. While non-biodegradable materials like polyethylene bags and plastic toys are persistent and problematic in the environment, so are worn-out clothing and damaged shoes.

28. Correct option: (C) (ii) and (iii)

A food chain is a logical arrangement of living things through which nutrients and energy are transferred. A food chain's constituent organisms belong to various trophic levels. Producers, or living things with the ability to produce their own food, make up the first trophic level. Consumers, or living things that consume other living things, make up the second trophic level. Predators, which are living things that consume other consumers, make up the third trophic level.

Plankton is a producer in option (ii), whereas man is a consumer. As a result, the predator-prey relationship is unclear. There is no herbivore animal to eat grass in option (iii). As a result, the predator-prey relationship is unclear.

Option (i) is a food chain because grass is the producer, rabbit is the consumer, and lion is the predator. Option (iv) is also a food chain because frog is the consumer, snake is the predator, eagle is the predator, and grass and grasshopper are the producers.

29. Correct option: (A) (i) and (iii)

The correct answer is A) (i) and (iii). Humans who are exposed to UV (ultraviolet) rays in excess—primarily from the sun—can develop a number of health problems. I UV radiation can harm the immune system by suppressing it, leaving people more vulnerable to illnesses and infections. (iii) An increased risk of skin cancer, including melanoma, basal cell carcinoma, and squamous cell carcinoma, is one of the most well-known effects of excessive UV exposure. UV rays can alter skin cells' DNA, which can result in the growth of cancerous growths. The provided options I and (iii) are the most closely related to the question, even though excessive UV exposure can have other health effects like eye damage and premature skin ageing.

10. Sustainable Management of Natural Resources

Solutions

LEVEL - I

1. Correct option: (B) 0.02%

The answer is B) 0.02%. This percentage indicates the proportion of freshwater on Earth that is primarily present in lakes, swamps, and rivers. About 71 percent of the Earth's surface is covered by water, but the vast majority of that water—roughly 97.5 percent—is saltwater from the oceans, which is unfit for most people and terrestrial animals to consume. Freshwater makes up the remaining 2.5 percent, but a sizable portion of it is trapped in ice caps and glaciers. As a result, freshwater on Earth is a scarce resource for many ecosystems and human needs for water, making up only a tiny portion of the planet's total freshwater supply, or about 0.02 percent, which is found in liquid form in lakes, swamps, and rivers.

2. Correct option: (A) Conservation of forests and wildlife

The correct answer is A) Conservation of forests and wildlife. Amrita Devi Bishnoi played a significant role in India's history of environmental preservation. She and a group of Bishnoi villagers gave their lives in the 18th century to stop the local Maharaja's men from felling trees so they could build a new palace. The "Khejri tree massacre," also known as the incident, served as a crucial early illustration of environmental activism and the Bishnoi community's steadfast dedication to the preservation of forests and wildlife. Greater awareness and conservation efforts in India were sparked by Amrita Devi Bishnoi's sacrifice and the community's collective action, which ultimately helped the area establish a steadfast tradition of nature and wildlife protection.

3. Correct option: (D) All of the above

The answer is D) All of the above. The term "conservation" refers to a variety of tactics and methods used to ensure the prudent management, preservation, and use of natural resources. It entails careful resource planning and management to avoid overuse and environmental damage while preserving their accessibility for both the present and the future. To balance human needs and the preservation of the environment, conservation efforts can involve managing forests, wildlife, water, energy, and other resources. Therefore, all three options—managing natural resources, protecting natural resources, and using natural resources wisely—collectively capture the comprehensive nature of conservation as a multifaceted strategy to protect the Earth's resources and ecosystems.

4. Correct option: (C) 1.75 - 2%

The answer is C) 1.75 - 2%. The amount of freshwater on Earth that is roughly frozen over the land, primarily in the form of glaciers, ice caps, and ice sheets, is represented by this percentage. Although there is a lot of water on the planet, the vast majority of it—roughly 97.5 percent—is saltwater found in the oceans, making it unfit for most people and land animals to consume. A sizeable portion of the remaining 2.5 percent of freshwater, particularly in the polar and mountainous regions, is trapped in ice formations. Even though it is not easily accessible for many human water needs, this frozen freshwater is essential for maintaining global freshwater resources, controlling sea levels, and supporting various ecosystems.

5. Correct option: (B) Sustainable development meets the current basic human needs and preserves future generations' resources.

The correct answer is B) Sustainable development meets the current basic human needs and preserves future generations' resources. A holistic approach called sustainable development aims to strike a balance between environmental, social, and economic objectives. It aims to ensure that the needs of the current generation are satisfied while not jeopardizing the capacity of future generations to satisfy their own needs. This entails conserving resources, preventing environmental damage, and taking the long view when making social and economic decisions. By addressing the needs of both the present and future populations while safeguarding the planet's natural resources and ecosystems, sustainable development aims to create a more equitable and sustainable future for all. It does this by acknowledging the interdependence of economic growth, environmental conservation, and social equity.

6. Correct option: (A) 2.5 - 2.75%

The answer is A) 2.5 - 2.75%. The hydrosphere of the Earth contains approximately this much fresh water, expressed as a percentage. All of the water on Earth, whether it is present as oceans, glaciers, groundwater, lakes, or rivers, is a part of the hydrosphere. About 71 percent of the Earth's surface is covered by water, but the vast majority of that water—roughly 97.5 percent—is saltwater from the oceans, which is unfit for most people and terrestrial animals to consume. Freshwater, which makes up the remaining 2.5 percent, is necessary for a variety of ecological, animal, and human needs. The specified range (2.5 - 2.75 percent) provides an estimate of the proportion of freshwater within the hydrosphere. This freshwater is distributed across different forms, such as groundwater, surface water in rivers and lakes, and frozen in glaciers and ice caps.

7. Correct option: (C) Reduce, recycle, reuse, recover

The correct answer is C) Reduce, recycle, reuse, recover. These four R's stand for a set of guidelines and methods for preserving natural resources for future use and reducing waste. Reducing waste production and resource consumption are both included in the term "reduce." "Recycle" promotes the gathering and preparation of waste materials for use as components in new products, thereby lowering the demand for virgin resources. "Reuse" encourages reusing objects or materials rather than discarding them after a single use. When waste cannot be reduced, recycled, or reused, "recover" refers to the process of recovering energy or materials from the waste in order to maximize resource efficiency and reduce environmental impact. These ideas work together to promote ecological footprint reduction and more sustainable resource management.

8. Correct option: (D) Afforestation

The answer is D) Afforestation. Excessive water extraction for various uses, including agriculture, industrial processes, and municipal water supply, is the main cause of groundwater depletion. Groundwater depletion can be avoided by reforestation, which entails the planting of trees and the creation of forests. By absorbing rainwater and allowing it to percolate into the soil, trees play a crucial part in groundwater recharge by recharging underground aquifers. They also lessen soil erosion, which can support keeping aquifer recharge areas in good condition. On the other hand, activities like deforestation (Option A), agriculture (Option B), and the operation of thermal power plants (Option C) can raise the risk of groundwater depletion by reducing aquifers' ability to replenish naturally and by increasing the demand for water for a variety of uses.

9. Correct option: (D) All of the above

The best method from an environmental point of view is to implement all of the above options: Reduce, Recycle, and Reuse. The management of resources sustainably and the

reduction of waste are built on these three guiding principles. The most efficient way to prevent environmental effects is to "reduce" resource consumption and minimize waste production at the source. Recyclable materials are gathered and processed so they can be used as the foundation for new products, cutting down on the consumption of virgin resources and waste disposal. "Reuse" encourages reusing objects or materials rather than discarding them after a single use, which further reduces resource consumption and waste production. We can greatly reduce our ecological footprint, protect natural resources, and lessen the effects of resource extraction, production, and waste disposal on the environment by putting all three of these principles into practice.

10. Correct option: (A) Protect biodiversity

The conservation of forests and wildlife is crucial primarily to protect biodiversity. The term "biodiversity" describes the variety and variability of species and life forms within a given ecosystem or across the entire planet. In order to maintain biodiversity, forests and wildlife are essential elements of ecosystems. For a wide variety of plant and animal species, many of which are intricately connected, they provide habitat and resources. We ensure the survival of a variety of species, some of which may be threatened or endangered, and uphold the complex web of life on Earth by conserving forests and wildlife. In turn, this promotes ecological balance, supports the continuation of food chains and ecological processes that sustain life on our planet, and preserves the overall health and resilience of ecosystems.

11. Correct option: (A) Increases groundwater level

The answer is A) Increases groundwater level. Water harvesting is a technique that involves gathering and storing surface runoff or rainwater to restock groundwater sources. Because it enables rainwater to permeate the soil and replenish underground aquifers, it is a useful technique for raising groundwater levels. We can decrease surface runoff and increase the amount of water that percolates into the ground by collecting and storing rainwater in a variety of ways, such as through the use of ponds, check dams, or rainwater harvesting systems. In turn, this aids in replenishing and preserving groundwater levels, which is essential for a sustainable water supply in many areas, particularly in those that experience water scarcity or seasonal variations in precipitation.

12. Correct option: (D) Water

The renewable resource among the options is D) Water. Being naturally replenished by the Earth's water cycle, water is regarded as a renewable resource. Rain and snow fall as precipitation, filling rivers, lakes, and deep aquifers as they reach the Earth's surface. These resources can be used for many purposes, including agriculture, industry, and the production of drinking water. Freshwater is a resource that, when used wisely, is sustainable even though its availability can vary regionally and seasonally because it is constantly replenished by natural processes. As opposed to this, options A, B, and C (petrol, cng, and lpg) are fossil fuels derived from limited, non-renewable sources, and as a result, their supply is constrained and dwindles over time as they are extracted and consumed.

13. Correct option: (B) Petroleum

The resource that needs to be used in a sustainable manner among the options is B) Petroleum. Petroleum is a fossil fuel that is extracted from the Earth's crust and has finite, non-renewable reserves. Petroleum must be used sustainably because it is a vital energy source for heating, transportation, and numerous industrial processes. This will guarantee that petroleum will be available for future generations. The development of alternative renewable energy sources like solar energy (Option A) and biogas (Option D),

as well as the investigation of cleaner and more effective technologies, are all examples of sustainable practices. Sustainable practices also involve reducing energy consumption through energy efficiency measures. In order to lessen the damaging effects on ecosystems and the effects of climate change, sustainable management also involves minimizing the environmental effects of petroleum extraction, transport, and use, such as pollution and greenhouse gas emissions.

14. Correct option: (D) All of above

The correct answer is D) All of the above. Several gases are released into the atmosphere during the burning of coal or gasoline. Carbon dioxide (CO2) and carbon monoxide are two of these gases (CO). A significant greenhouse gas that contributes to climate change and global warming is carbon dioxide. When inhaled in high concentrations, the odorless, colorless gas known as carbon monoxide is toxic. Additionally, other gases that are harmful pollutants that contribute to smog and air quality issues, like nitrogen oxides, may be released during the combustion process. Therefore, option D is the appropriate decision because burning coal or gasoline produces a mixture of gases, including CO2, CO, and occasionally nitrogen compounds.

15. Correct option: (A) Coal

Coal is the main resource used in thermal power plants (Option A). Coal is burned in thermal power plants to create steam, which powers turbines attached to generators to produce electricity. Due to the abundance and relatively low cost of coal, this procedure—known as coal combustion—is frequently used to produce electricity. Coal combustion produces heat energy, which is then transformed into mechanical energy and ultimately electrical energy. Option A, coal, is the preferred option in this situation because kerosene (Option B), charcoal (Option C), and petroleum (Option D) are not frequently used as the primary fuel source in thermal power plants.

16. Correct option: (D) The people who live in urban areas

Urban residents are not directly involved in the management or conservation of forests. They do not directly participate in the processes or decisions relating to forest management and conservation, despite the fact that they may indirectly benefit from the resources and services offered by forests, such as clean air and water. On the other hand, direct stakeholders are those who have a direct impact on forests, such as paper mill owners who depend on timber from forests, officials from the forest department who are in charge of managing and protecting forests, and people or groups actively involved in campaigns for the preservation of forests.

17. Correct option: (C) Biogas

Option C biogas is not a fossil fuel. The long-decomposed remains of ancient plants and animals are the source of fossil fuels, which when burned release carbon dioxide and contribute to climate change. In contrast, biogas is created when organic material like sewage, agricultural waste, or food scraps are digested anaerobically. It is a renewable energy source and less harmful to the environment than fossil fuels like LPG (liquefied petroleum gas), natural gas, and CNG (compressed natural gas), which are all derived from fossilized organic matter and release carbon dioxide when burned. Its main components are methane and carbon dioxide. However, its source is the decomposition of recent organic material.

18. Correct option: (C) Electricity

Option C, electricity, is a human-made form of energy produced from various natural resources rather than being a resource in and of itself. Natural resources are typically components or materials found naturally in the environment and used by people for a variety of activities. Contrarily, the energy used to create electricity comes from sources

like coal, natural gas, water (hydropower), wind, and solar energy. Although these resources, like water, natural gas, and the sun, are all sources of renewable energy, electricity is produced by transforming these resources through a number of processes, such as power generation, transmission, and distribution, into a usable form of energy. As a result, it is not regarded as a primary natural resource in and of itself, but rather as a secondary or derived resource.

19. Correct option: (D) Forests

The world's forests (Option D) are frequently regarded as the natural resource that is depleting the fastest. Forests offer vital ecosystem services like carbon sequestration, wildlife habitat, and control over water cycles. However, the loss of forests has been accelerated globally due to deforestation brought on by logging, the expansion of agriculture, urbanization, and illegal logging. By releasing stored carbon into the atmosphere, this destruction not only directly causes the loss of priceless biodiversity, but it also has a significant negative impact on climate change. Forests are a particularly important natural resource to conserve and manage sustainably since they are depleting at a much faster rate than other resources like water (Option A), soil (Option B), and sunlight (Option C), all of which are also essential.

20. Correct option: (C) Wood, wind, sun

There are no fossil fuels in the term combination "Wood, Wind, Sun" in Option C. Coal, petroleum, and natural gas are examples of resources that are considered fossil fuels because they were created from the remains of extinct plants and animals. In contrast, renewable and non-fossil fuel energy sources are represented by the terms in Option C. When harvested sustainably, wood, a biofuel made from living or recently living organisms like trees, is regarded as renewable. Both wind energy and solar energy, which are both renewable energy sources that are directly derived from natural processes and do not require the burning of fossil fuels, are represented by the wind and the sun, respectively. Coal derivative "coke," which is a fossil fuel, is included in Option A, and "kerosene" is included in Option B.

21. Correct option: (C) Windmills for generating power

Option C, the use of wind turbines to produce electricity, is regarded as environmentally friendly. Wind turbines generate electricity by using the kinetic energy of the wind without emitting any harmful emissions or pollutants. This makes wind power a clean and renewable source of energy that does not contribute to air pollution or greenhouse gas emissions, making it environmentally friendly. In contrast, the other options mentioned have negative environmental impacts: cars for transportation (Option A) typically run on fossil fuels and emit pollutants, polythene bags for shopping (Option B) contribute to plastic pollution, and dyes for coloring clothes (Option D) can involve the use of chemicals harmful to the environment if not managed properly.

22. Correct option: (A) Living and renewable

As shown by Option A, forests and wildlife are both living, renewable resources. Forests contain living trees, plants, and a variety of biodiversity, such as animals, insects, and microorganisms. Wildlife, which refers to the numerous animal species, is made up of living things. Because they can replenish themselves over time through natural processes like reproduction and regrowth, these resources are renewable when they are managed properly and conserved. The cycle of natural renewal is dependent on sustainable practices, though, as overexploitation or habitat destruction can sabotage it. To ensure their sustainability and continued availability for future generations, it is crucial to protect and manage forests and wildlife.

23. Correct option: (C) Carbon monoxide

Carbon monoxide is the lethal gas produced when coal is burned (Option C). Carbon monoxide is produced when coal burns in an environment with insufficient oxygen, such as during incomplete combustion (CO). When inhaled in high concentrations, the colorless, odorless, and highly toxic gas known as carbon monoxide can be dangerous or even fatal. Carbon monoxide poisoning symptoms, such as headache, nausea, and dizziness, are brought on by interference with the body's ability to transport oxygen in the bloodstream. Severe cases can also result in unconsciousness and death. When burning coal or any other fuel, it's crucial to ensure proper ventilation and combustion procedures in order to avoid the formation of carbon monoxide and the associated health risks.

24. Correct option: (D) All of these

Water that enters drains, seeps into the soil, and evaporates is referred to as runoff. When it rains, the water that is not absorbed by the ground or vegetation runs off the surface and gathers in drains or other water bodies. This is one method by which runoff water enters drains. In addition, some of the water that lands on the ground percolates down into the soil and eventually replenishes groundwater sources. Finally, because of the heat from the sun, runoff water can also evaporate and return to the atmosphere. D) All of these is the right response because runoff water can take any of these routes depending on the situation.

25. Correct option: (B) The industrialists

Industrialists (Option B) are often cited as causing the maximum damage to forests. This is due to the fact that industrial processes like logging, mining, and manufacturing frequently lead to extensive deforestation and habitat destruction. Significant natural resources, such as timber and minerals, are needed for industrial processes; these resources are frequently taken from forests, resulting in the destruction of priceless forest ecosystems. Additionally, some industries emit pollutants and other waste products that can degrade the health of forests and worsen air and water pollution. While people who live in or around forests (Option A) can also impact them through activities like subsistence farming and fuelwood collection, industrialists typically have a more extensive and immediate impact due to their large-scale operations and resource-intensive practices. The forest department of the government (Option D) is usually responsible for forest management and conservation, so their role is not typically associated with causing damage to the forests. Wildlife and native enthusiasts (Option C) are often focused on preserving and conserving forests and wildlife, rather than causing harm to them.

26. Correct option: (D) A, B, C and safe disposal of wastes

The answer is D) A, B, C and safe disposal of wastes. To ensure the sustainability of natural resources for future generations, management should be viewed from a long-term perspective. As resource exploitation frequently results in pollution and the decline of ecosystems, it should also address environmental pollution. Additionally, it is essential to distribute resources fairly in order to advance social justice and prevent conflicts. Last but not least, proper waste disposal is necessary to preserve the environment and safeguard public health. Therefore, for effective management of natural resources, a broad strategy that covers all of these factors is required.

27. Correct option: (C) Both A and B

Option C suggests that the generation of hydroelectricity utilizes both the potential energy and the kinetic energy of water. Due to its elevated position, water that is kept at a higher elevation, like in a reservoir behind a dam, has potential energy. This water gains kinetic

energy as it picks up speed and momentum when allowed to flow downhill. These two sources of energy are combined in hydroelectric power plants. As water travels downward through turbines, potential energy is first transformed into kinetic energy. The kinetic energy is then used to drive turbines, which turn a generator mechanically to produce electricity. Water's potential and kinetic energies are therefore essential to the process of producing hydroelectric power.

28. Correct option: (D) Rainwater

Rainwater is the main source of water. When water vapor in the atmosphere condenses and falls to the ground as precipitation, rainwater is created. The replenishment of the water supply on Earth happens naturally. There are many different ways that rainwater is gathered, including direct rainfall into bodies of water like rivers and lakes or infiltration into the ground to become groundwater. It is a crucial source of freshwater for both human consumption and natural ecosystems. In order to support overall food production and to irrigate crops, rainwater is also crucial for agriculture.

29. Correct option: (B) To recharge dried up bore wells

Soak pits primarily serve the purpose of option B, which is to recharge dried up bore wells. The purpose of soak pits is to collect rainwater or surface runoff and allow it to seep into the ground. Rainwater that has been collected in soak pits slowly seeps into the soil to replenish groundwater supplies. Recharging depleted bore wells and regaining access to groundwater resources are two applications of this process that stand out. In order to maintain a sustainable and dependable source of water for a variety of uses, including agricultural, industrial, and domestic use, soak pits are an effective method of groundwater recharge. Their primary purpose is groundwater recharge rather than water storage (Option A) or providing water facilities for animals, even though they indirectly help to keep the soil moist (Option D) by recharging the groundwater (Option C).

30. Correct option: (C) Godavari

The Sri Ram Sagar Project is built on the Godavari River, as indicated by Option C. A significant irrigation and hydroelectric project, this one is situated in the Telangana state of India (formerly part of Andhra Pradesh). It was built over the Godavari River to take advantage of the river's water resources for irrigation, hydropower production, and supplying drinking water to the surrounding areas. The project has a big impact on agriculture in the area because it provides irrigation water, which increases agricultural productivity. It serves as a significant piece of Godavari River water management infrastructure, aiding in regional development and water resource management.

LEVEL – II

1. Correct option: (D) Wetland

The Ramsar Convention is an international agreement that was signed in 1971 with the goal of conserving wetlands. Ecosystems known as "wetlands" include marshes, swamps, lakes, rivers, and estuaries. These areas are essential for biodiversity because they provide homes for many different plant and animal species, purify the water and control flooding, and are essential for maintaining the environment's overall health. The main objective of the Ramsar Convention is to ensure that wetlands are used sustainably and wisely throughout the world, putting special emphasis on their conservation and protection to preserve the crucial ecosystem services they offer to both people and nature.

2. Correct option: (A) Sahara Desert

As shown by Option A, the Noor Complex is situated in the Sahara Desert. The Sahara Desert, which occupies a large portion of North Africa, is the largest hot desert in the world. A significant solar power generation facility called the Noor Complex, or more

specifically, the Noor Ouarzazate Solar Complex, is located in Morocco, a country that is a part of the Sahara Desert. One of the biggest solar power plants in the world, it uses concentrated solar power technology and the Sahara's abundant sunlight to produce renewable, clean energy. The Noor Complex and other large-scale solar energy projects can be found in the Sahara because of its vast area of arid land and strong sunlight.

3. Correct option: (C) 1972

As shown by Option C, India's Wildlife Protection Act was put into effect in 1972. This important piece of legislation was passed in order to legally protect India's wildlife and its natural habitats. The law aimed to protect endangered species, stop poaching and the illegal trade in wildlife, and advance the preservation and sustainable management of India's rich biodiversity. A network of protected areas, including national parks and wildlife sanctuaries, was also established so that wildlife could coexist peacefully. The Wildlife Protection Act of 1972, which reflected the country's dedication to protecting its diverse and distinctive wildlife heritage, was a turning point in India's conservation efforts.

4. Correct option: (B) Kerela

Option B indicates that surangams, also known as suranga or surangam tunnels, are a traditional idea for water harvesting that is mainly found in the Indian state of Kerala. Groundwater is gathered and transported using these conventional underground tunnels to surface wells or other water storage facilities. In areas with rocky terrain or hilly landscapes, surangams are a creative and environmentally friendly way to access groundwater sources. The tunnel is typically built with a slight incline to allow water to flow naturally through it. It aids in storing and channeling groundwater for use in domestic and agricultural irrigation, among other things. As a sustainable solution to the water shortage in the state of Kerala, this traditional water harvesting system has played a significant role in the state's history.

5. Correct option: (C) The environment

We harm the environment when we clear a forest. The balance of ecosystems is critically dependent on forests. They serve as carbon sinks by removing oxygen while removing carbon dioxide from the atmosphere. By affecting rainfall patterns and lessening the effects of natural disasters like floods and landslides, forests also assist in regulating the climate. In addition, forests support biodiversity by serving as habitat for countless species of animals and plants. When we clear forests, we disturb these delicate ecosystems, which causes species extinction and environmental degradation.

6. Correct option: (B) Bamboo

Bamboo (Option B) is often referred to as the "poor man's timber." This moniker refers to bamboo's adaptability, low cost, and quick growth when compared to more conventional timber species like hardwoods. Fast-growing grasses like bamboo can be harvested for a variety of uses, such as building materials, furniture, and handicrafts. Because it matures much more quickly than trees do, it offers a less harmful substitute for timber. This relieves pressure on natural forests. It is accessible to a wide range of people thanks to its affordability, especially in communities where the economy is struggling. Bamboo is known as "poor man's timber" because it offers a valuable resource for communities with limited access to more expensive wood products. Bamboo's strength and durability also make it a suitable replacement for timber in many applications.

7. Correct option: (B) Silviculture

The name given for the replenishment of forests is "Silviculture" (Option B). Silviculture is the cultivation, management, and replenishment of forests using a variety of strategies and methods. It includes tasks like growing new trees, caring for existing ones, and

sustainably harvesting timber. By balancing the economic and ecological aspects of forestry, silviculture aims to maintain the long-term health and productivity of forest ecosystems. In order to promote forest regeneration, increase biodiversity, and lessen the effects of deforestation, it is essential. Silviculture is a crucial part of sustainable forestry methods, helping to preserve and manage forest resources in a responsible manner.

8. Correct option: (B) Make its water pollution-free

The main purpose of the 'Ganga Action Plan' (GAP) project, launched by the Indian government in 1985, was to make the Ganga River water pollution-free, as indicated by Option B. One of India's holiest and most significant rivers, the Ganga (also known as the Ganges), has been severely polluted by the discharge of industrial, domestic, and agricultural waste. By putting policies in place to manage and lessen water pollution in the Ganga, the GAP project sought to address this urgent environmental problem. It included projects like constructing sewage treatment facilities, enhancing industrial waste management, and educating the public about the value of keeping the Ganga clean. The project's objectives were to restore and safeguard the river's ecological stability and guarantee that its water was suitable for a variety of uses, including drinking and bathing.

9. Correct option: (B) Indira Gandhi Canal

The canal that brought about greenery in Rajasthan is the "Indira Gandhi Canal," as indicated by Option B. One of India's largest canal systems is the Indira Gandhi Canal, formerly known as the Rajasthan Canal. In memory of the late prime minister Indira Gandhi, it was renamed. Initiated in the 1950s, this canal was a significant irrigation project used to supply Rajasthan's dry regions with water from the Sutlej River. Large swaths of desert land have been transformed into lush, fertile agricultural areas thanks in large part to the canal. It has made irrigation easier, enabling farmers in Rajasthan to grow crops, plantations, and orchards, boosting agricultural productivity and the state's economy.

10. Correct option: (A) Process of afforestation

The process of afforestation (Option A) does not lead to the depletion of groundwater; in fact, it can help conserve and replenish groundwater. Planting trees and other vegetation where none previously existed is known as afforestation. The planting of trees is essential for groundwater recharge. Their roots aid in soil stabilization, stop soil erosion, and create an ideal environment for rainwater infiltration and groundwater aquifer recharge. Contrarily, depending on factors like water use, land management techniques, and environmental considerations, options like constructing thermal power plants (Option B), growing high-yielding crop varieties (Option C), and deforestation (Option D) can have varying degrees of impact on groundwater depletion.

11. Correct option: (D) all the above

All of the previously mentioned options fall under the main objectives of sustainability, which are (A) the eradication of poverty and hunger, (B) higher standards for healthcare and education, particularly in terms of water quality and sanitation, and (C) the achievement of gender equality. A holistic idea, sustainability aims to protect future generations' prosperity as well as the health of the planet and its inhabitants. It entails tackling social, economic, and environmental issues all at once. Sustainable development must include all of the following: eradicating hunger and poverty; enhancing healthcare; ensuring access to clean water and sanitation; and achieving gender equality. With the environment being protected for the benefit of both the present and future generations, these objectives seek to build a more equitable and just society.

12. Correct option: (B) Monoculture in the area

The practice of clearing vast tracts of forests and cultivating a single species of plant promotes monoculture in the area, as indicated by Option B. The term "monoculture" describes the extensive cultivation of a single crop or plant species. Although this method is frequently employed in industrial agriculture, it may be harmful to ecosystem health and biodiversity. The loss of diverse native plant and animal species, the disruption of natural ecosystems, and increased susceptibility to pests and diseases can all be consequences of clearing forests and replacing them with monoculture crops or plantations. When compared to preserving natural forests or engaging in more varied agricultural practices, it is thought to be a less sustainable land use practice because it decreases overall biodiversity and can cause long-term environmental challenges.

13. Correct option: (A) 200cm

As shown by Option A, tropical rainforests thrive in regions with annual rainfall of at least 200 centimeters (200cm). High and steady precipitation levels are typical throughout the year in these areas, which is necessary for the lush and varied vegetation found in rainforests. Tropical rainforests offer the perfect environment for the growth of a wide range of plant and animal species because of their constant moisture and warm temperatures. In addition to supporting the dense canopy of tall trees, the plentiful rainfall also supports the rich biodiversity and distinctive ecosystems that are characteristic of these areas. One of the most important ecosystems on Earth, rainforests are home to countless species and play a critical role in maintaining global climate patterns.

14. Correct option: (B) A. K. Banerjee

Due to a shift in the Department's strategy and the active participation of neighborhood villagers under the direction of forest officer A.K. Banerjee, the degraded Arabari forests in the Midnapore district underwent a remarkable recovery. A total of 1,272 hectares of severely degraded sal forest were preserved, with the help of the local population. The villagers received opportunities for employment in harvesting and silviculture in exchange for their assistance in preserving the forest. Additionally, they were qualified for a 25% share of the overall harvest and permitted to gather fuelwood and fodder in exchange for a small fee. This community-based approach, with the active and willing participation of the local population, led to the remarkable recovery of the Arabari sal forests. By 1983, a previously worthless forest had gained substantial value, estimated at Rs 12.5 crores. This successful restoration serves as an inspiring example of sustainable forest management and the positive impact of involving local communities in conservation efforts.

15. Correct option: (A) Bundhis

The age-old concept of the water harvesting system in Madhya Pradesh is known as "Bundhis," as indicated by Option A. For the purpose of collecting and storing rainwater, bundhis are conventional earthen embankments or check dams built across small streams or seasonal watercourses. These buildings are purposefully constructed to collect and store rainwater, allowing it to seep into the ground and replenish nearby aquifers. In arid and semi-arid regions, bundhis are a crucial part of rainwater harvesting because they replenish groundwater levels and offer a consistent source of water for drinking, agriculture, and other uses during dry seasons. In Madhya Pradesh and other areas with limited water supplies, they have been instrumental in promoting sustainable agricultural practices and managing water resources.

16. Correct option: (A) It releases harmful gases into the atmosphere when they burn

Option A, "It releases harmful gases into the atmosphere when they burn," explains a significant disadvantage of using fossil fuels as energy sources. When fossil fuels like coal, oil, and natural gas are burned to create energy, they release pollutants like Sulphur dioxide (SO2) and nitrogen oxides as well as greenhouse gases like carbon dioxide (CO2). These emissions play a critical role in climate change, smog formation, and air pollution. Global warming and a variety of environmental and societal problems, such as more frequent and severe weather events, rising sea levels, and disruptions to ecosystems, are caused by an excessive buildup of greenhouse gases in the atmosphere. This drawback emphasizes the urgent need to switch to greener, more sustainable energy sources in order to reduce the negative effects of burning fossil fuels on the environment and human health.

17. Correct option: (D) All of these

The correct option is D) "All of these." Achieving two out of three pillars of sustainable development can still contribute to sustainability in various ways.

A) In order for society to be sustainable on both a social and economic level and ensure that everyone benefits from economic growth, it must address both social well-being and economic prosperity.

B) Social and environmental sustainability (tolerable) refers to actions being taken to ensure that the environment is not unduly burdened or harmed while addressing social issues.

C) Achieving both economic viability and environmental sustainability (viable) entails conducting economic activities while taking into account their potential environmental effects.

These combinations highlight the interconnectedness of social, economic, and environmental aspects in sustainable development, even though achieving all three pillars of sustainable development is the ultimate goal. They also show that sustainability can be achieved even when focusing on just two of the three pillars.

18. Correct option: (C) Sustainable development does not consider the viewpoints of stakeholders

The incorrect statement is C) "Sustainable development does not consider the viewpoints of stakeholders." In fact, the involvement and inclusion of stakeholders' perspectives is fundamental to sustainable development. It places a strong emphasis on a cooperative and participatory approach in which the perspectives of different stakeholders, such as communities, businesses, governments, and civil society, are crucial in decision-making processes. By taking into account the needs and interests of both current and future generations, sustainable development aims to strike a balance between social equity, environmental protection, and economic growth. To make sure that development plans are socially acceptable, environmentally responsible, and economically viable, effective sustainable development practices necessitate active engagement with stakeholders. Therefore, a fundamental principle of sustainable development that encourages inclusive and responsible decision-making is taking into account and incorporating the perspectives of stakeholders.

19. Correct option: (C) “Forests do not conserve soil.”

The wrong statement is C) "Forests do not conserve soil." This is false because forests are essential to soil preservation. The extensive ground cover and plant roots found in forests stabilize the soil and lessen the effects of heavy rainfall, thereby preventing soil erosion. Additionally, the tree canopy serves as a natural barrier to stop raindrops from directly

striking the soil and eroding it. In addition, tree organic matter and leaf litter improve the fertility and health of the soil. Forests are essential to the overall health of the ecosystem and to human well-being because they play a crucial role in preserving the integrity of the soil and halting its degradation.

20. Correct option: (B) Monoculture in the area

The practice of clearing vast tracts of forests and cultivating a single species of plant promotes monoculture in the area, as indicated by Option B. The term "monoculture" describes the extensive cultivation of a single crop or plant species. Although this method is frequently employed in industrial agriculture, it may be harmful to ecosystem health and biodiversity. The loss of diverse native plant and animal species, the disruption of natural ecosystems, and increased susceptibility to pests and diseases can all be consequences of clearing forests and replacing them with monoculture crops or plantations. When compared to preserving natural forests or engaging in more varied agricultural practices, it is thought to be a less sustainable land use practice because it decreases overall biodiversity and can cause long-term environmental challenges.

21. Correct option: (D) Incineration

The name for the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions is "Incineration," as indicated by Option D. A waste management technique called incineration involves the carefully regulated combustion of solid waste. Usually, it takes place in specialized buildings called incinerators. Depending on the technology used and the type of waste, the waste is heated to high temperatures during the incineration process. This method lessens the amount of waste produced, reduces the environmental impact it has, and can even produce energy in the form of heat or electricity. However, incineration can result in air emissions and ash residues that call for careful management to reduce environmental concerns, which makes it a contentious issue in waste management practices.

22. Correct option: (A) To preserve the natural resources for the needs of the future generation

The main significance of sustainable development is to preserve natural resources for the needs of future generations, as indicated by Option A. In order to meet the needs of the present without compromising the capacity of future generations to meet their own needs, sustainable development aims to strike a balance. It acknowledges that the resources on our planet are limited and that overusing them for immediate profit may have unfavorable long-term effects. We want to ensure that economic, social, and environmental well-being are interconnected and sustained over time by practicing sustainability. This strategy encourages prudent resource management, diminished environmental impact, and the development of a more just and equitable society where everyone's needs are met while preserving the environment for future generations.

23. Correct option: (D) All of the above.

The construction of high-rise dams is associated with a range of environmental problems, as encompassed by Option D. First of all, it frequently results in a significant number of human settlements being submerged in the water (Option A), displacing residents and causing them to lose their homes. Additionally, since dam construction frequently necessitates flooding large areas, forests and ecosystems are submerged, which contributes to deforestation and the loss of biodiversity (Option B). Last but not least, building high-rise dams requires significant financial investments (Option C), frequently requiring large sums of money for planning, designing, and building, which can have an effect on the economy. The complex environmental, social, and economic issues raised by high-rise dam projects are highlighted by the interaction of these factors, which calls

for careful planning, assessment, and mitigation measures to lessen their negative effects.

24. Correct option: (B) Planning for safe disposal of wastes after mining

Option B, "Planning for safe disposal of wastes after mining," is an example of sustainable development aimed at conserving natural resources for future generations. Environmental effects from mining operations may include soil and water pollution, habitat destruction, and the production of hazardous waste. Sustainable mining practices must include planning for the safe and ethical disposal of mining wastes. In order to minimize environmental damage, this strategy includes measures like proper waste containment, recycling and reusing mining byproducts, and putting into practice eco-friendly waste management techniques. By taking these precautions, we make sure that mining operations don't permanently harm ecosystems and that natural resources can be preserved for future generations to enjoy.

25. Correct option: (C) It causes severe environmental issues

Burning waste is not an acceptable practice of solid waste management primarily because it causes severe environmental issues, as indicated by Option C. Burning waste contributes to air pollution, smog formation, and climate change by releasing dangerous pollutants and greenhouse gases into the atmosphere. Dioxins, furans, and other toxins found in these emissions, as well as particulate matter, can be harmful to both human and animal health. In addition, waste incineration produces ash byproducts that may contain hazardous materials and need to be disposed of carefully. Environmentally safer and more responsible than waste burning, more sustainable waste management techniques like recycling, composting, and landfilling with proper controls are preferred in light of these health and environmental issues.

26. Correct option: (B) A gift of nature which is very useful to mankind

The most appropriate definition of a natural resource is that it is "a gift of nature which is very useful to mankind," as indicated by Option B. Resources that are naturally occurring in the environment and have intrinsic value or utility for people are known as natural resources. These resources can be both renewable and non-renewable, like minerals and fossil fuels, or both, like water and forests. By providing necessities like food, water, energy, raw materials, and ecosystem services, they play a crucial part in sustaining human life and civilization. Natural resources are the cornerstone of human society and economic growth, and it is imperative to manage them sustainably in order to guarantee their continued availability for both present and future generations.

27. Correct option: (D) Using windmills to generate power for irrigation

The eco-friendly activity among the options listed is "Using windmills to generate power for irrigation," as indicated by Option D. Wind power, which is a clean and renewable energy source, is harnessed by windmills, specifically wind turbines for generating electricity. It is an eco-friendly and sustainable method of farming to generate irrigation power from wind energy because it lessens the need for fossil fuels and the emissions of greenhouse gases. Contrarily, choices like using a car for transportation (Option A) results in air pollution and carbon emissions, while others like using polybags for shopping (Option B) can result in plastic pollution and choosing dyes to color clothing (Option C) can result in the use of potentially hazardous chemicals. As a result, using wind energy to generate electricity is more environmentally friendly.

28. Correct option: (A) Comprehensive programmed to protect all the physical and biological components

A successful forest conservation strategy should indeed involve a comprehensive program to protect all the physical and biological components, as indicated by Option A. Forest

ecosystems are complex and interconnected systems made up of a variety of plant and animal species as well as soil, water, and climatic factors. It is crucial to take into account the overall wellbeing of these ecosystems in order to ensure the long-term health and sustainability of forests. This entails safeguarding not only the forest's trees and other plant life, but also its animal inhabitants, microbial inhabitants, and the physical elements like soil and water. Comprehensive conservation efforts include addressing threats like deforestation and habitat degradation as well as habitat preservation, biodiversity conservation, sustainable management practices, and habitat preservation. Such a strategy acknowledges the significance of preserving the entire ecosystem and the myriad interactions that take place within it, fostering the resilience and vitality of forests for the advantage of both the present and the future.

29. Correct option: (B) (i), (ii) and (iii)

The correct consequences of raising the height of dams, as indicated by Option B, are (i) Terrestrial flora and fauna of the area is destroyed completely, (ii) Dislocation of people and domestic animals living in the area, and (iii) Valuable agricultural land may be permanently lost. Raising the height of dams frequently results in the flooding of sizable tracts of land, which can completely destroy the natural habitat and displace both animal and human populations. A permanent loss to local agriculture could also result from the submersion or destruction of valuable agricultural land. Although the building and modifying of dams can result in economic gains and the production of electricity, they also have significant social and environmental effects that must be carefully considered and mitigated.

30. Correct option: (A) (i) and (iv)

According to Option A, the phrases that best describe the idea of sustainable development are I planned growth that causes the least amount of environmental harm and (iii) growth that is acceptable to all stakeholders. Economic growth and development must be planned and implemented in a way that minimizes environmental damage as part of a balanced strategy for sustainable development (statement i). Additionally, it emphasizes the importance of including all parties involved, addressing their concerns, and ensuring that development is acceptable and advantageous to society as a whole (statement iv). It does not support unrestrained growth regardless of the effects on the environment (statement ii), nor does it require halting all developmental work (statement iii). Instead, sustainable development seeks to find a middle ground that promotes well-being, social equity, and environmental stewardship.